迈向推理时代
长思维链的机理及应用

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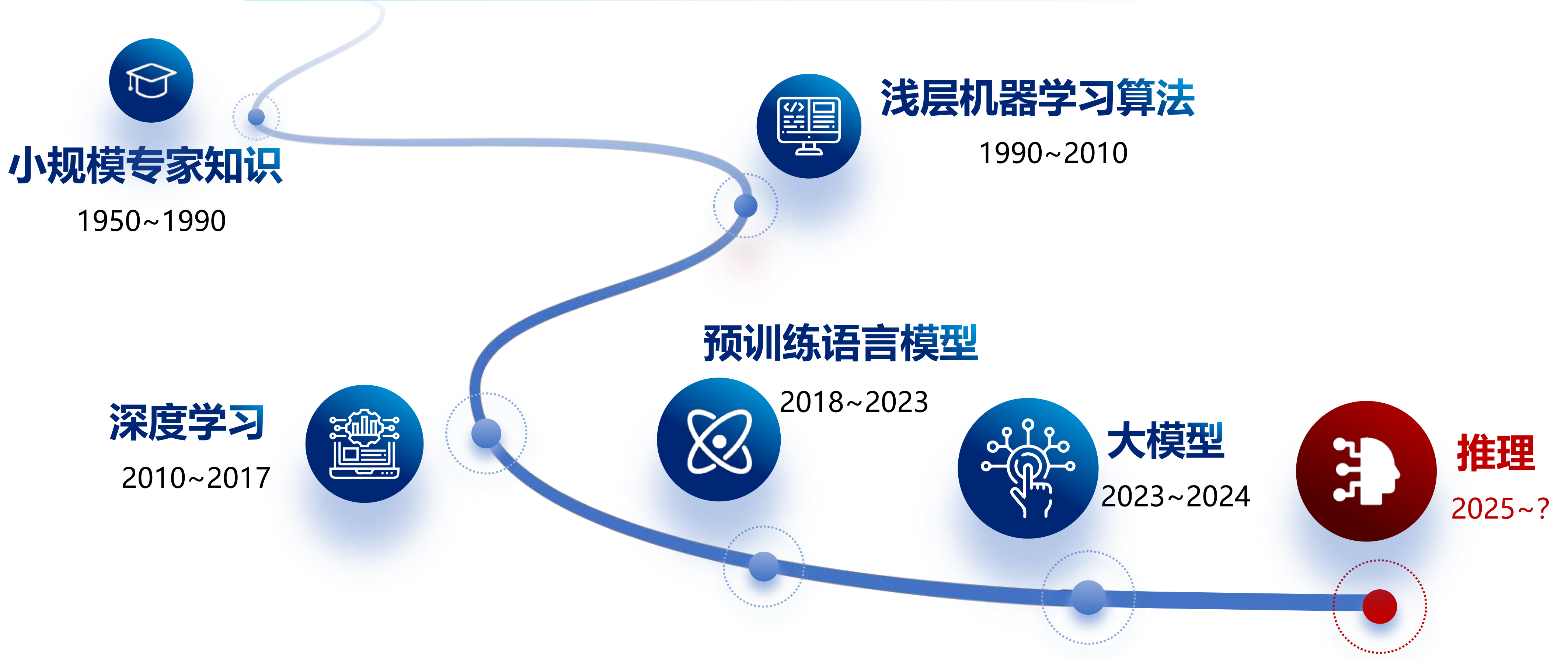
Nature News

2025年1月23日，Nature News报道：“中国的廉价且开源的大型语言模型震撼了科学界！”

由中国研发的DeepSeek-R1大模型是一种既具备**高性价比**又完全**开源**的**“推理”**模型，其性能可与OpenAI的**o1模型媲美**。通过模仿人类推理过程，这些模型能够逐步生成响应，在解决科学问题时表现得比早期大模型更为出色，可能对科研工作产生深远的影响...

人工智能技术范式的变迁

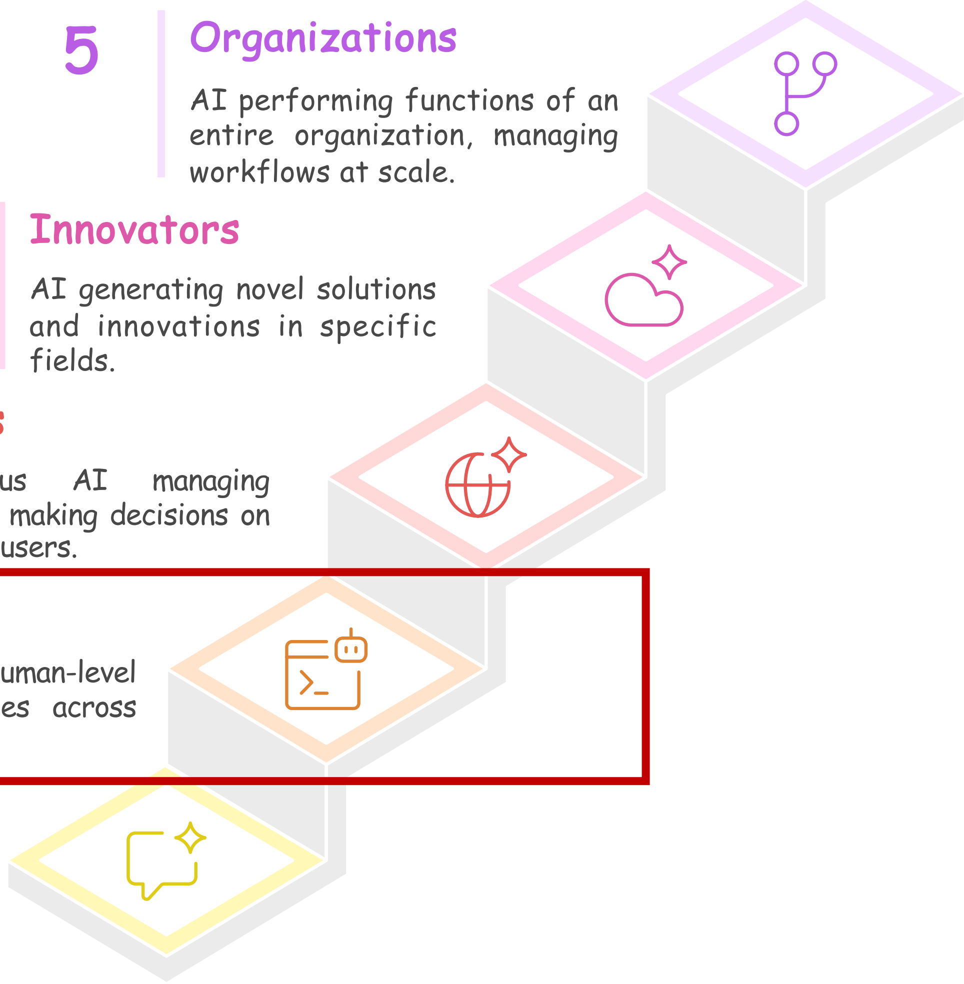
DeepSeek-R1 引发了自然语言处理的 第六次范式变迁



推理（Reasoning）是指根据已知的信息、事实、规则或前提，通过一定的思维过程和方法，推导出新的结论、判断或知识的认知活动。它是人类思维和智能的核心组成部分，也是人工智能、科学研究和日常决策中的关键能力。

OpenAI 提出实现 AGI 的五个阶段

Progression of AGI Levels



5 Organizations

AI performing functions of an entire organization, managing workflows at scale.

4 Innovators

AI generating novel solutions and innovations in specific fields.

3 Agents

Autonomous AI managing tasks and making decisions on behalf of users.

2 Reasoners

AI systems with human-level problem-solving abilities across various domains.

1 Chatbots

Initial stage of conversational AI, understanding and res-ponding to human language.

推理的核心技术"思维链"早已有之

思维链（Chain-of-Thought, COT）

- 一系列中间推理步骤，相当于在求解问题过程中将解题步骤也写出来
- 早期的思维链能力是模型**自发"涌现"**的，形式简单，长度**较短**

(a) 少样例推理

问：小明有5个乒乓球，他又买了2筒乒乓球，每桶有3个。小明现在有多少乒乓球?
答：11个。
问：现在一共有16个球，其中的一半是小华的，小华的球中有一半是蓝色的。小华的球中有多少蓝色?
答:

（输出）8个。❌

(b) 少样例思维链推理 (Wei et al., 2022)

问：小明有5个乒乓球，他又买了2筒乒乓球，每桶有3个。小明现在有多少乒乓球?
答：小明一开始有5个乒乓球，在买了2筒后增加了2乘3等于6个，加起来一共11个。
问：现在一共有16个球，其中的一半是小华的，小华的球中有一半是蓝色的。小华的球中有多少蓝色?
答:

（输出）16的一半是8个，8个的一半是4个。✅

(c) 零样例推理

问：现在一共有16个球，其中的一半是小华的，小华的球中有一半是蓝色的。小华的球中有多少是蓝色的?
答:

（输出）8个。❌

(d) 零样例思维链推理 (Kojima et al., 2022)

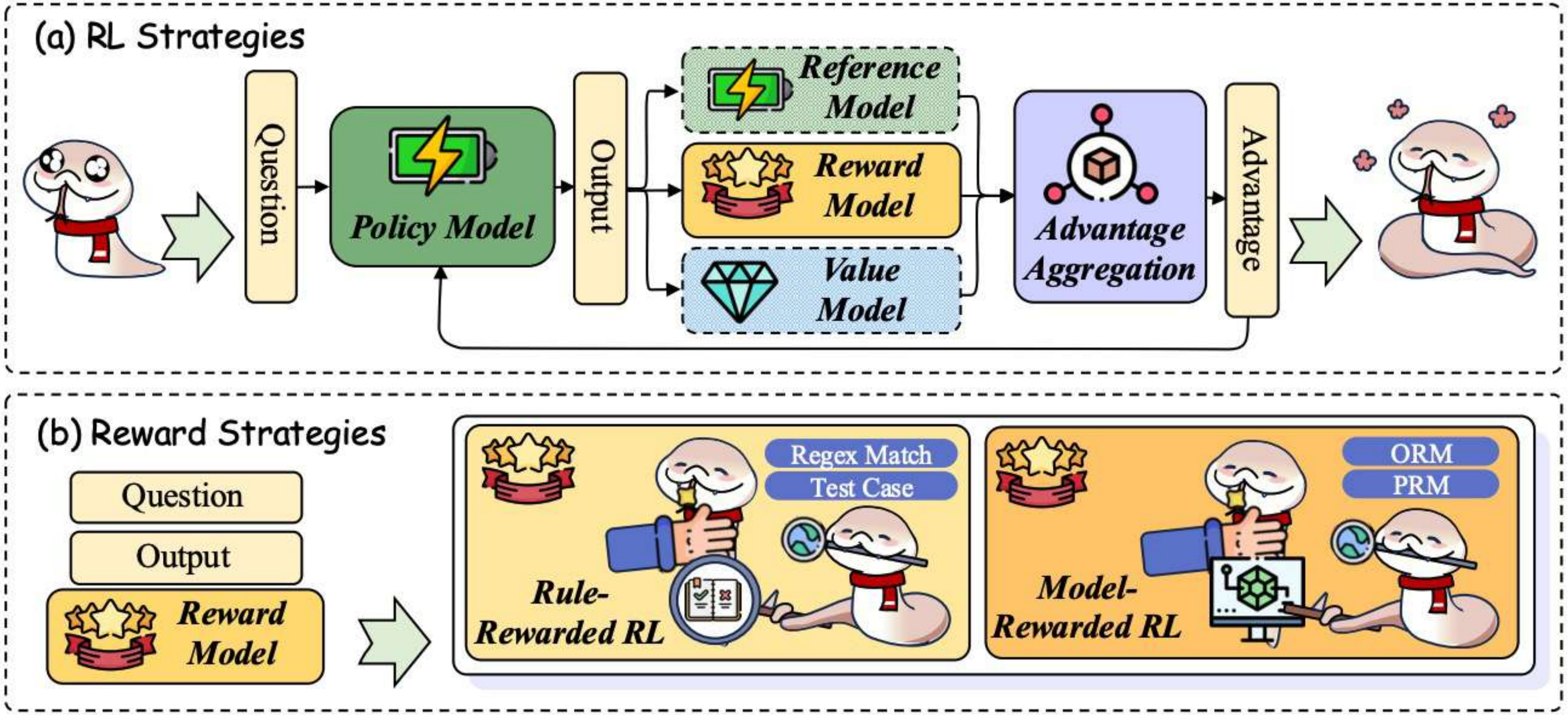
问：现在一共有16个球，其中的一半是小华的，小华的球中有一半是蓝色的。小华的球中有多少是蓝色的?
答：让我们一步一步地思考。

（输出）16的一半是8个，8个的一半是4个。✅

R1引发的强化训练范式

探索式强化学习

- 通过强化学习的方式探索可能的推理路径，更好的学会推理能力



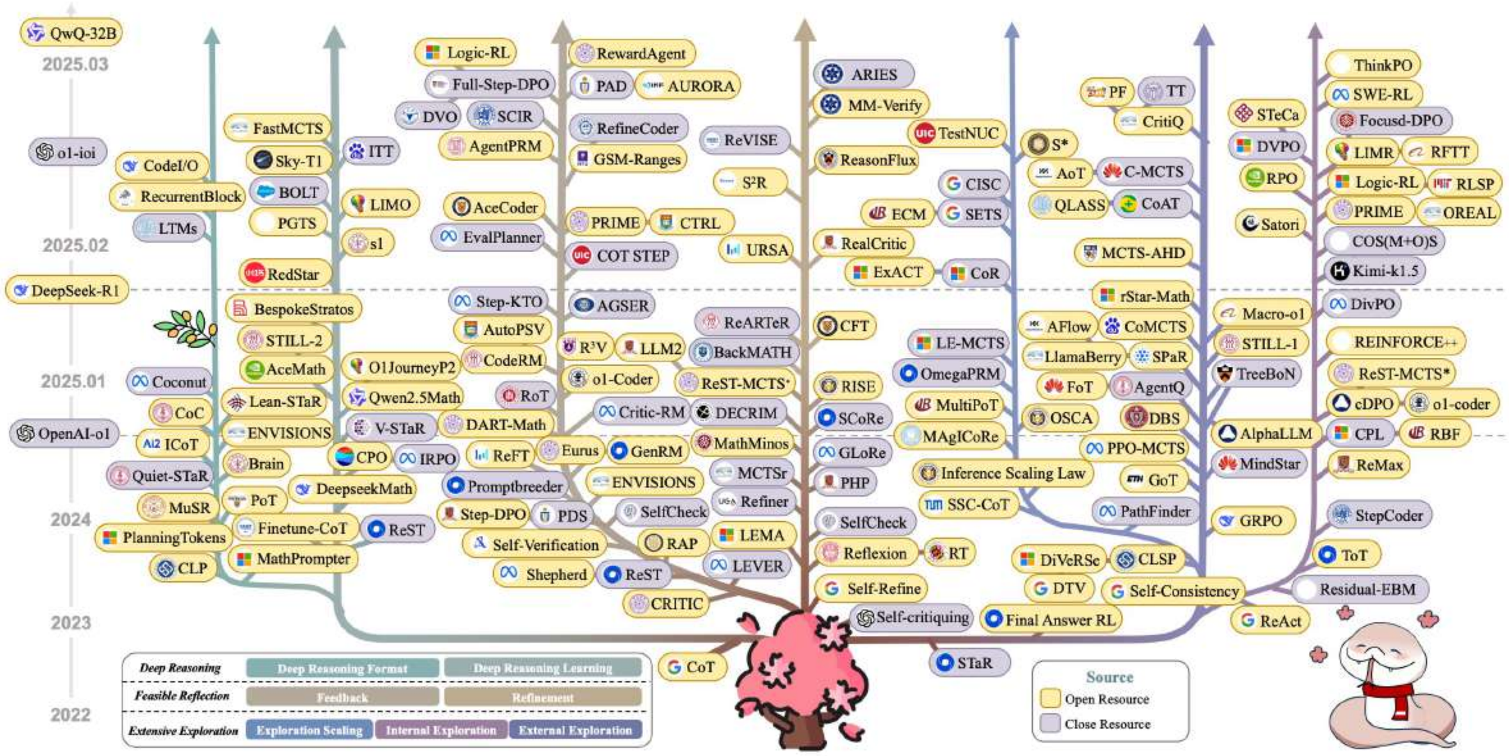
* Chen et al. Towards Reasoning Era: A Survey of Long Chain-of-Thought for Reasoning Large Language Models. Arxiv 2025.

R1引发的新研究

🎯随着OpenAI-o1和Deepseek-R1的成功，开启了长思维链的新时代

✅ 推理能力大幅提升，推理深度远超从前

✅ 研究井喷式爆发 🌋

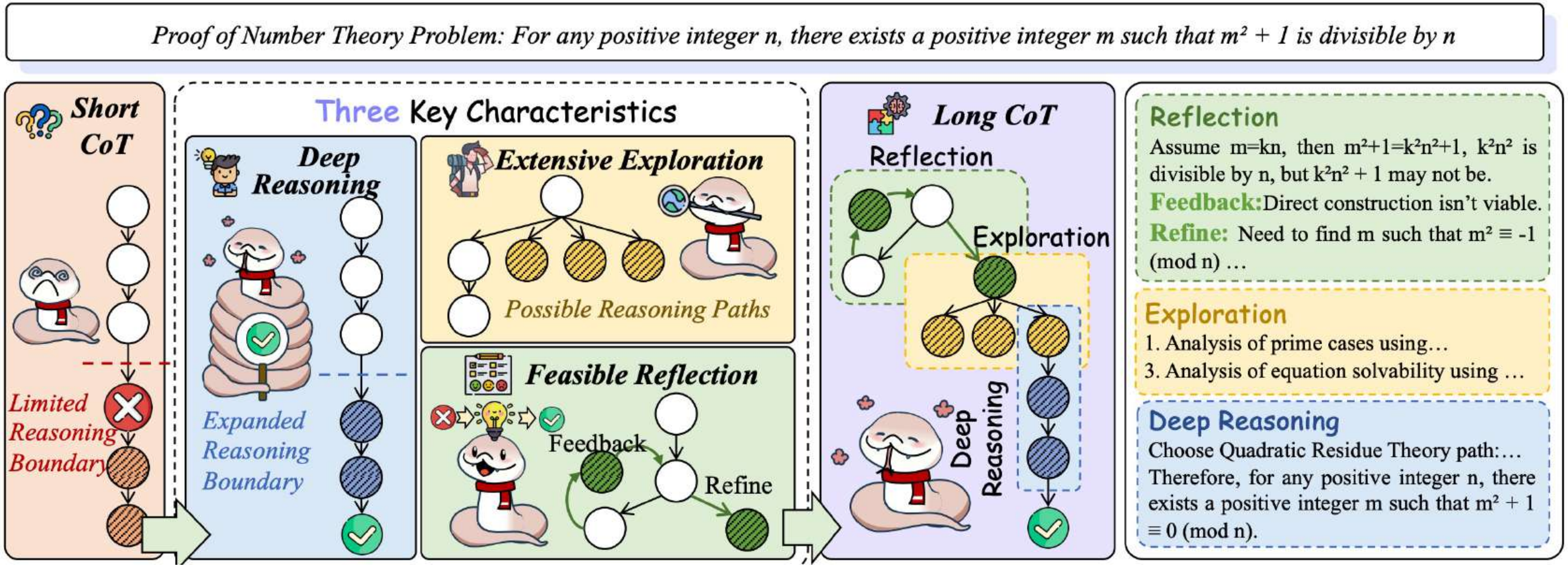


* Chen et al. Towards Reasoning Era: A Survey of Long Chain-of-Thought for Reasoning Large Language Models. Arxiv 2025.

传统思维链与长思维链的本质差别

短思维链

❌ 有限的推理深度　❌ 难以进行反思　❌ 难以进行探索



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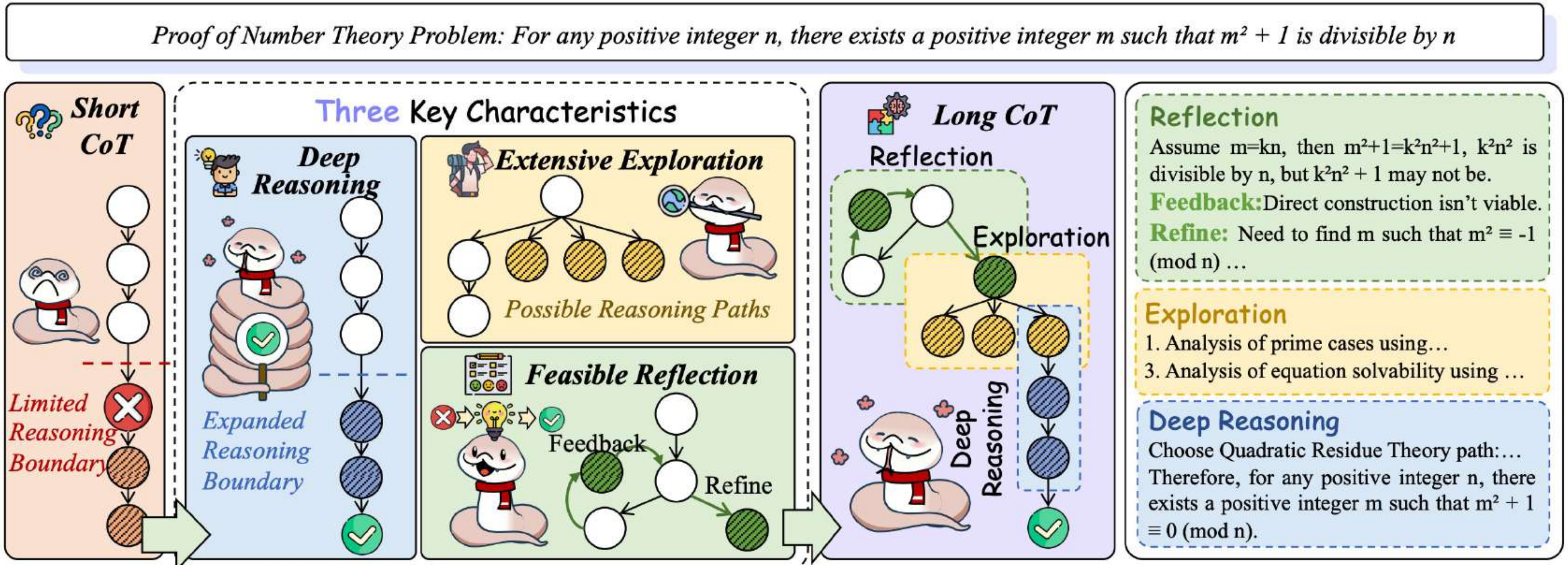
传统思维链与长思维链的本质差别

长思维链

- ✅ 支持更深的思维链
- ✅ 支持适当的反思
- ✅ 支持广泛的探索
- ✅ ***综合上述所有能力特点以完成任务***



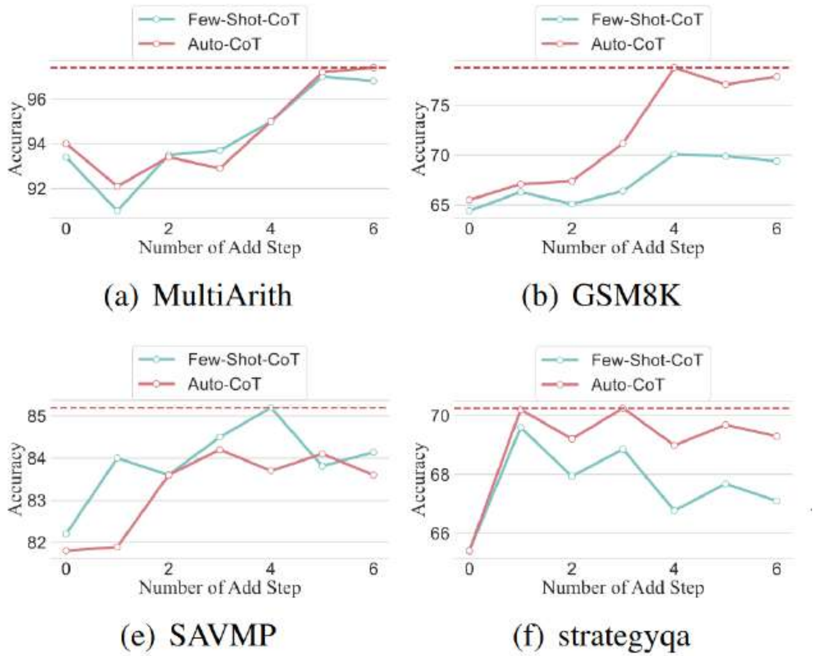
* Chen et al. Towards Reasoning Era: A Survey of Long Chain-of-Thought for Reasoning Large Language Models. Arxiv 2025.

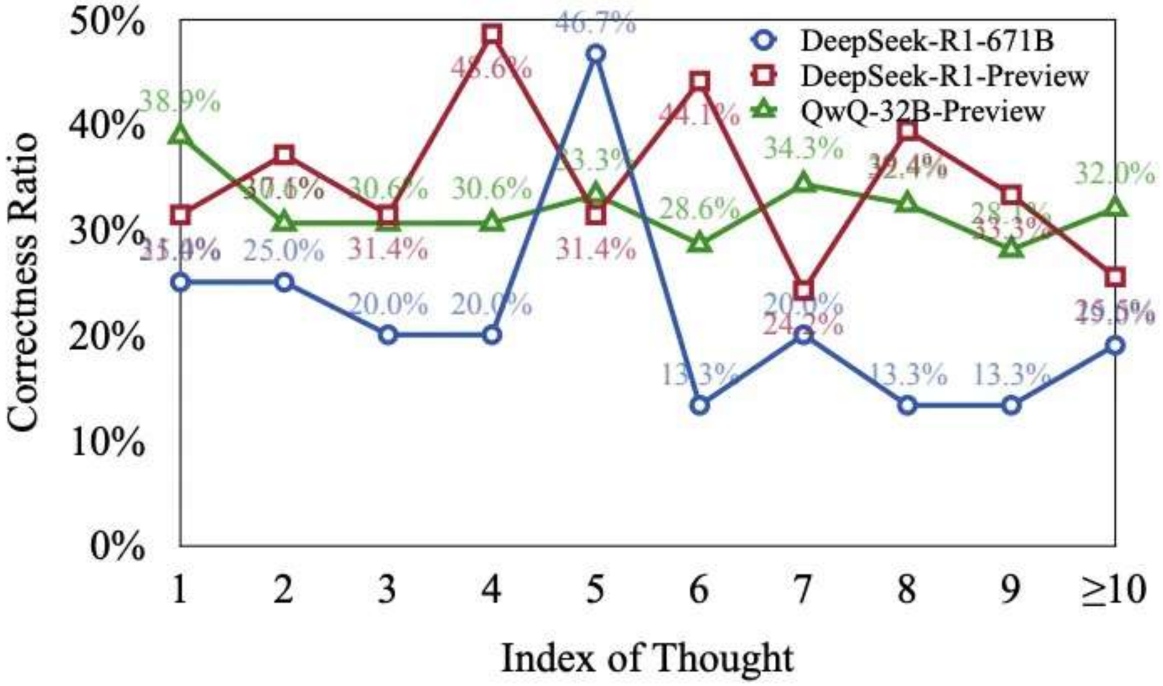
长思维链深度思考能力的边界

推理边界现象

大量工作表明对于特定的任务和模型，在思维链推理过程中，存在一个**推理能力上限**

- 推理的规划长度有上限，规划超过一定长度后性能会下降 （左图）
- **过多次数**的反复规划/思考，超过模型的**反思上限**，也会严重削弱性能（右图）





* Jin et al. The Impact of Reasoning Step Length on Large Language Models. Jin et al., ACL 2024 Findings
* Wang et al. Thoughts Are All Over the Place: On the Underthinking of o1-Like LLMs. Arxiv 2025

推理边界现象——定量化分析

我们首次**定量探索了**现有大模型的推理能力上限，并对相关的思维链优化技术做出指导

具体包括：**最大可行推理边界假设、组合推理边界假设、最短可行思维链分解研究**

基于最短可行思维链的思维链技术研究

推理边界假设 → 组合推理边界假设 → 最短可行思维链分解研究

Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought

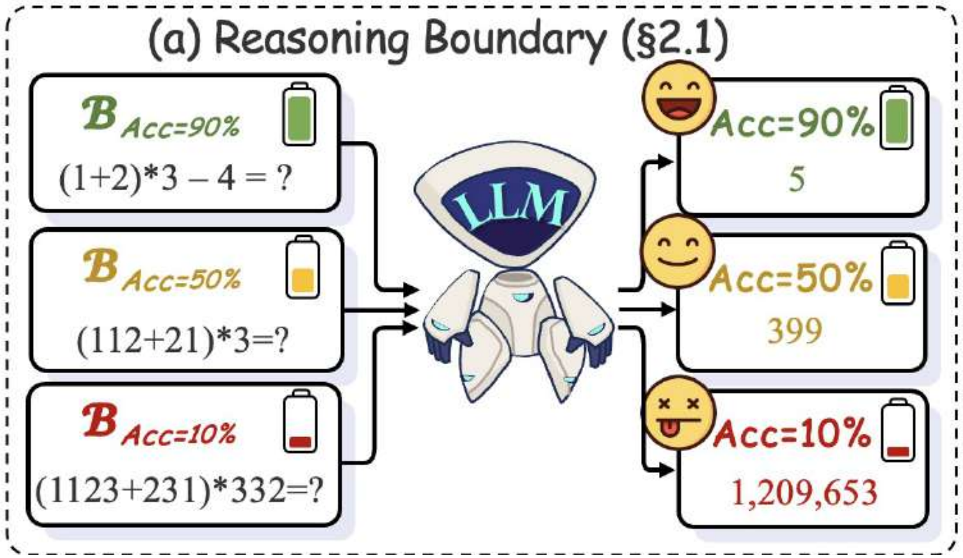
Qiguang Chen† Libo Qin‡* Jiaqi Wang◇ Jinxuan Zhou‡ Wanxiang Che†*

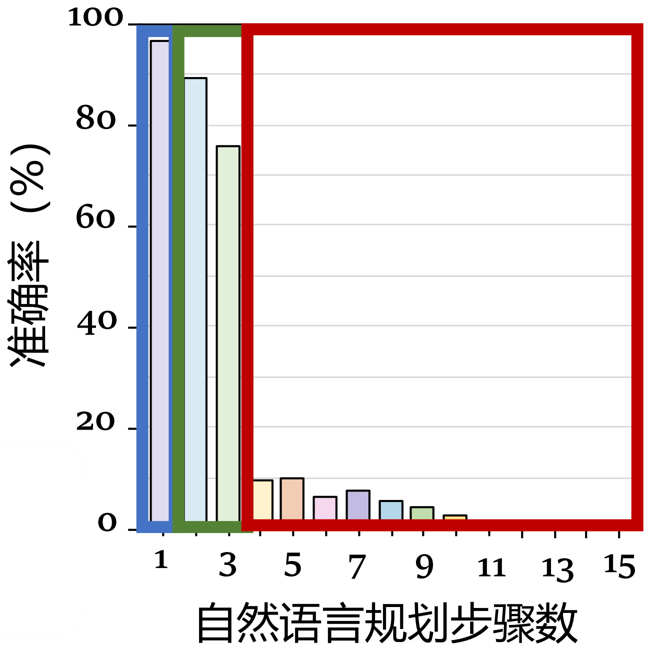
(NeurIPS 2024 Oral)

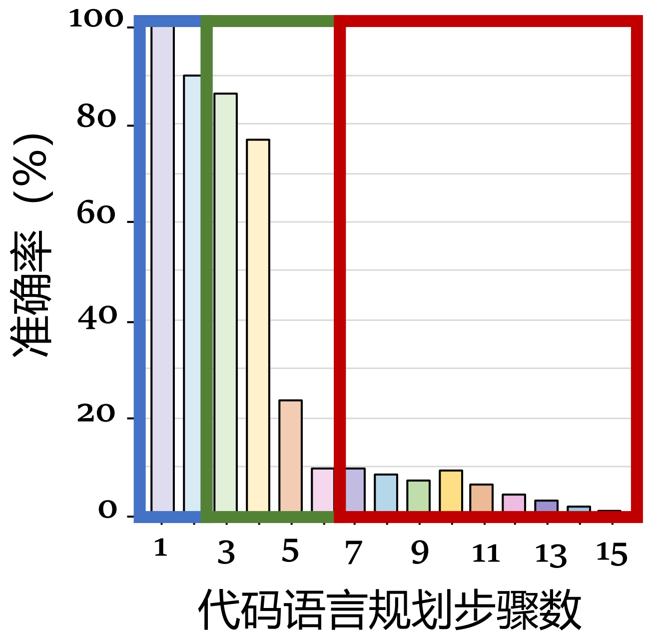
* Chen et al. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. NeurIPS 2024

推理边界现象——推理边界假设

推理边界假设：对于特定的任务和模型，在思维链推理过程中，不同的**推理能力**均有一个上限，即推理边界。超过此边界则无法按预期进行推理





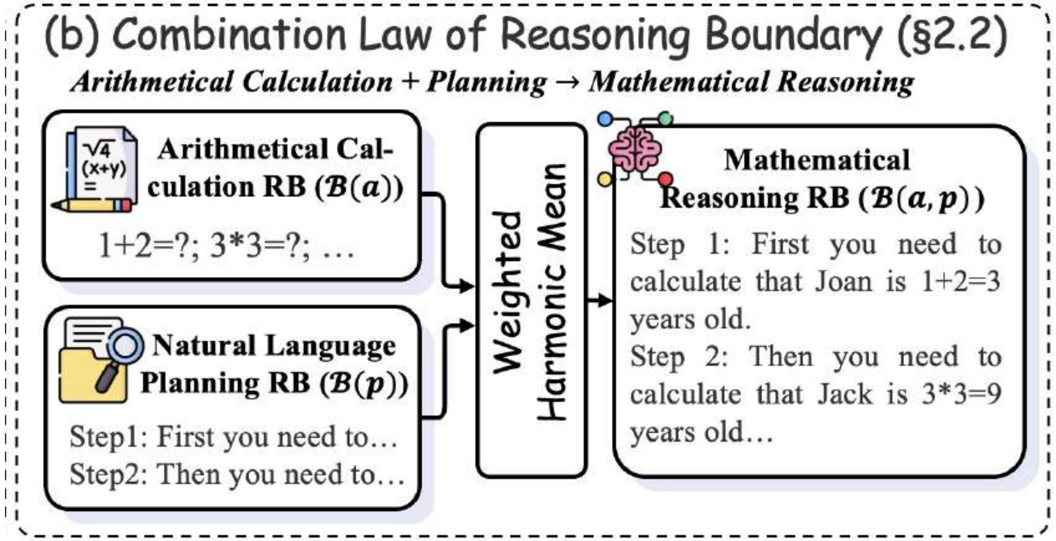


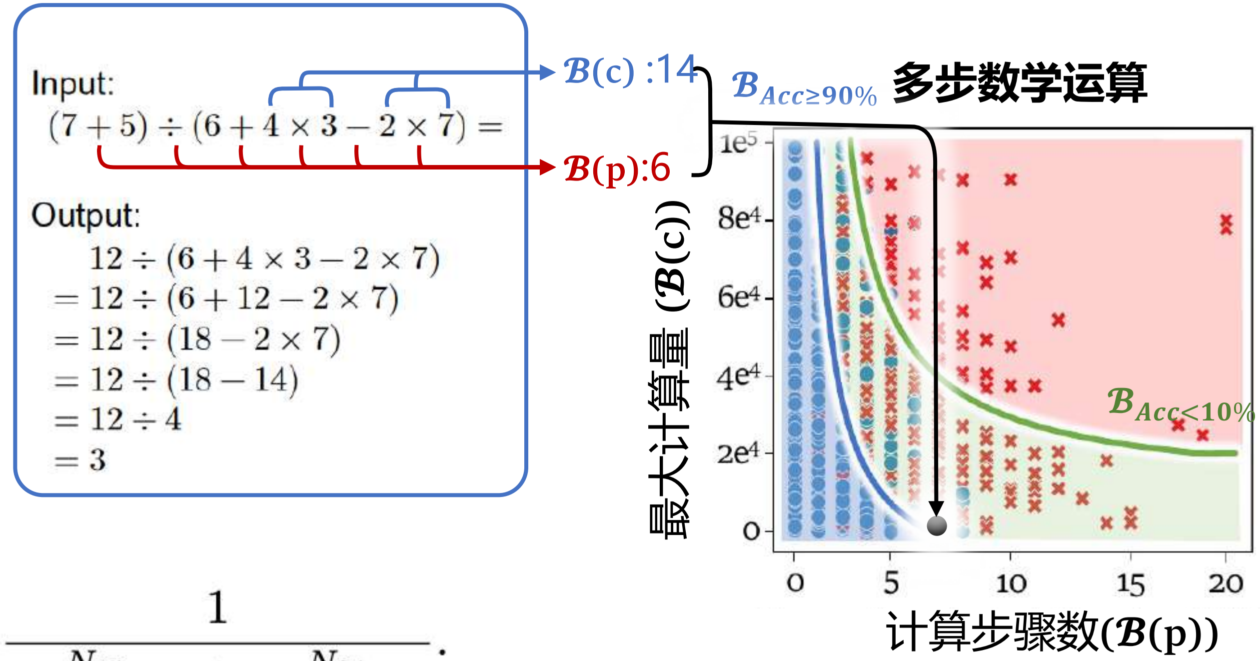
多步自然语言规划准确率分布　**多步代码语言规划准确率分布**

* Chen et al. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. NeurIPS 2024

推理边界现象——组合推理边界假设

组合推理边界假设：针对实际**任务**场景，模型需要利用更加多的不同的基础能力/边界进行组合推理，才能够解决问题。更实际的组合推理边界可以被计算为基础推理边界的**加权调和平均**



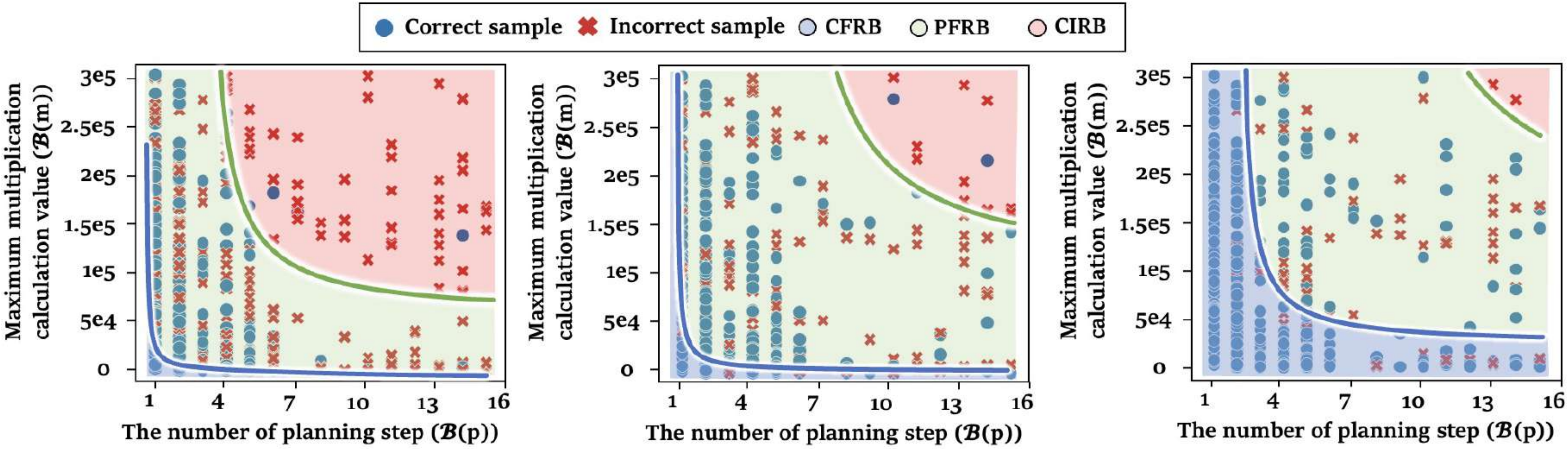


$$\mathcal{B}^{\text{CoT}}(c,p)=\frac{1}{\frac{N_{11}}{(\mathcal{B}(c)-b_1)}+\frac{N_{21}}{(\mathcal{B}(p)-b_2)}}.$$

* Chen et al. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. NeurIPS 2024

推理边界现象——模型的变化趋势

不同的推理边界随着模型优化逐步提升



(a) GPT-3.5-turbo (b) GPT-4o (c) O1-preview

* Chen et al. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. NeurIPS 2024

推理边界现象——组合推理边界优化

最短可行思维链分解： 基于最大可行推理边界，可以将复杂的语言处理任务拆分为步骤更少、更适合模型解决的推理步骤

问： Leo 的作业分为三个部分。他在 25 分钟内完成了作业的第一部分。他花了两倍的时间才完成第二部分。如果他能在 2 小时内完成作业，那么 Leo 需要多少分钟才能完成作业的第三部分？

原始示例样本：

1. Leo 花了 25 x 2 = 50 分钟完成作业的第二部分。
2. Leo 用了 25 + 50 = 75 分钟完成了作业的第一部分和第二部分。
3. 他用 60 x 2 = 120 分钟完成了整个作业。
4. 因此，Leo 花了 120 - 75 = 45 分钟完成作业的第三部分。

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优化后的指令：

要求： … 每个步骤包含尽可能多的基本操作。

限制： … 每步最多包含 5 个基本运算…

优化后的示例样本：

1. Leo 用了 25 + 25 x 2 = 75 分钟完成了作业的第一部分和第二部分。
2. 因此，Leo 花了 2 x 60 - 75 = 45 分钟完成作业的第三部分。

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* Chen et al. Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought. NeurIPS 2024

推理边界现象——组合推理边界优化

最短可行思维链分解：基于最大可行推理边界，可以将复杂的语言处理任务拆分为步骤更少、更适合模型解决的推理步骤

Model	BIGGSM		
	Acc. (↑)	Input Token (↓)	Output Token (↓)
CoT	57.00 $\pm$0.93	780.43	96.76 $\pm$3.22
RG-Optimized Methods			
Tool Usage	71.64 $\pm$0.66	688.43	129.53 $\pm$3.82
PoT	78.25 $\pm$1.09	657.43	78.25 $\pm$1.09
Reasoning-Path-Optimized Methods			
Least-to-most	58.25 $\pm$3.28	679.59	176.09 $\pm$15.22
Complex-CoT	59.78 $\pm$0.60	1111.43	131.82 $\pm$1.91
CoT+MARP	64.37 $\pm$2.24	614.43	95.12 $\pm$0.77
PoT+MARP	**80.55** $\pm$2.40	**576.43**	**76.34** $\pm$2.84

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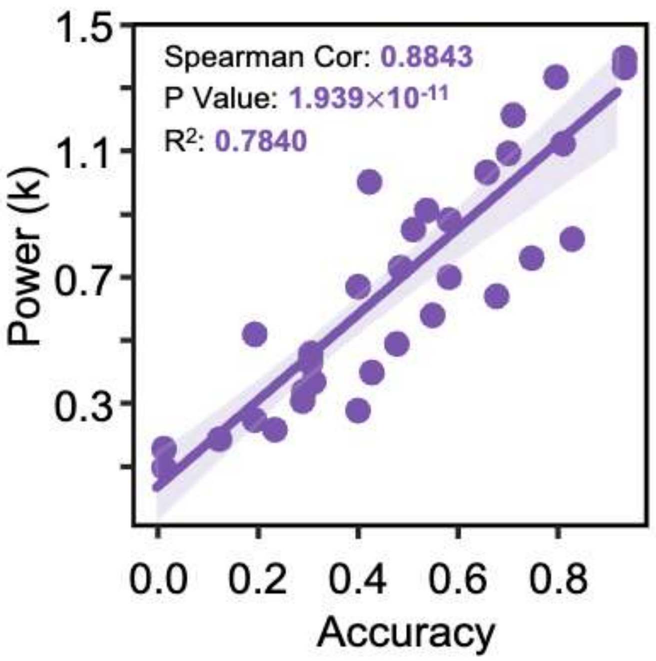
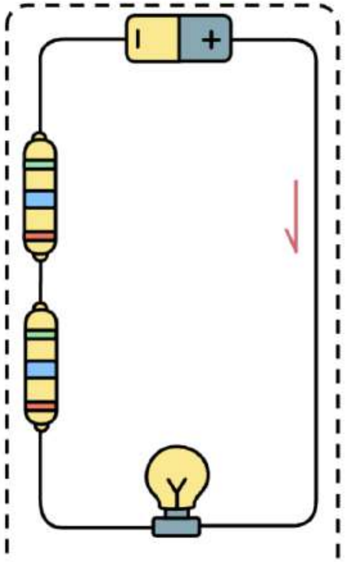
电路猜想

- 长思维链的串联电路现象：推理难度 → 串联电阻【$R = R1 + R2$】
- 逻辑电路输出功率为【$P = \frac{\varepsilon_{model}^2 R_0}{(R_{CoT}+R_0)^2}$】
- 输出功率与模型准确率呈现**强线性相关**（Spearman Cor> 85%）

输入：小华一共有16个球，小明的球数量是小华的两倍。他们一共有有多少球?
Let's think step-by-step!

输出：
- **步骤1：**小华一共有16个球，小明的球数量是小华的两倍，因此小明有16*2=32个球。
- **步骤2：**小明和小华一共有16+32=48个球。





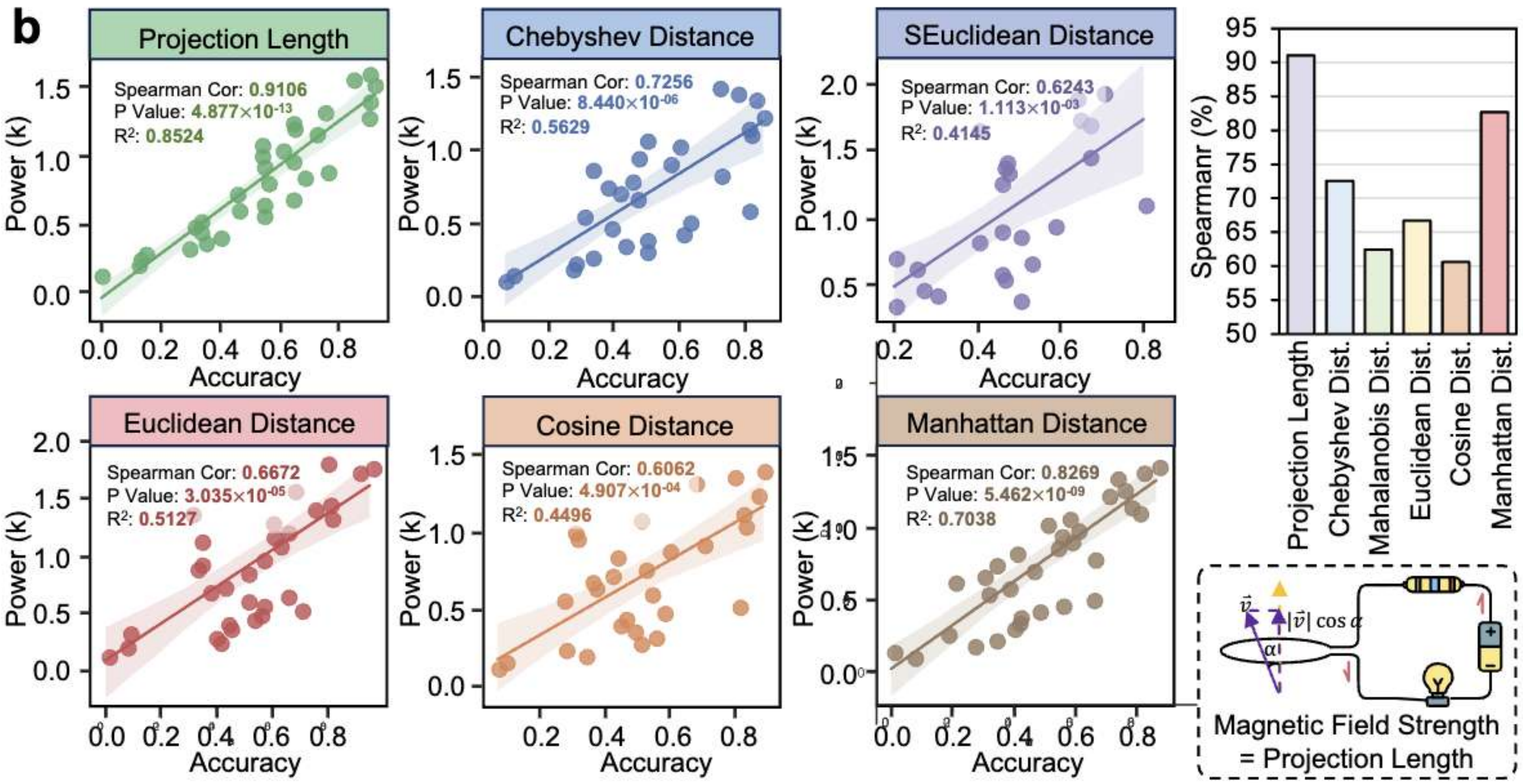
Logic Circuit of Zero-shot CoT

* Chen et al. Electronic Circuit Principles of Large Language Models. Arxiv 2025.

电路猜想

我们认为，语境学习需要依赖于用户输入的**语义信息通量**

- **语境学习的电磁现象：**通过**磁通量（语义信息通量）**的变化使整个**电路（输出过程）**增加更多的电压，从而优化最终的**电灯功率（最终性能）**



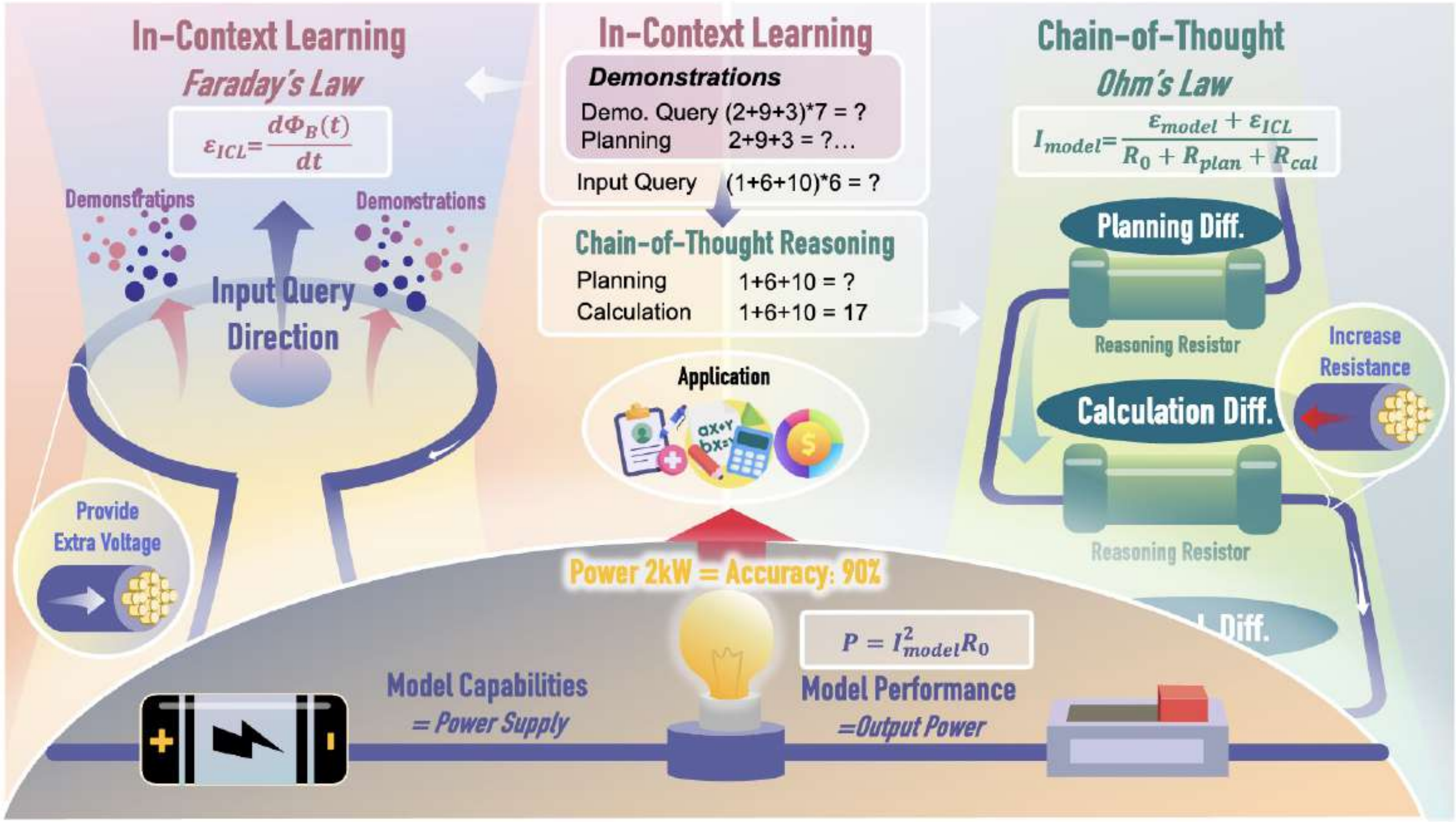
语义通量建模与模型性能的相关性远大于其他语义建模方法

* Chen et al. Electronic Circuit Principles of Large Language Models. Arxiv 2025.

电路猜想

$$P_{\text{out}} = I^2_{model} R_0 = \frac{\left(\mathcal{E}_{\text{model}} + \mathcal{E}_{\text{ICL}}\right)^2 R_0}{(R_{\text{CoT}} + R_0)^2}$$

Circuit Terminology	LLM Terminology	Function & Description
Magnetic Field ($\Phi_B(t)$)	Demonstration in ICL	Modeling a magnetic field change to provide extra voltage (ε_{ICL})
Resistor (R_{CoT})	Difficulty for CoT	Modeling the impact of problem difficulty in CoT
Series Resistors ($R_{plan} + R_{cal}$)	Sub-Difficulty for CoT	Modeling the impact of sub-problem difficulty in CoT
Power Supply	Model Inherent Capabilities	Modeling the capability of LLM itself to provide basic voltage (ε_{model})
Output Resistor (R_0)	Difficulty for Conclusion	Modeling the difficulty for LLM to give the final conclusion
Output Power (P_{out})	Model Performance	Modeling practical performance for the output resistor (R_0)



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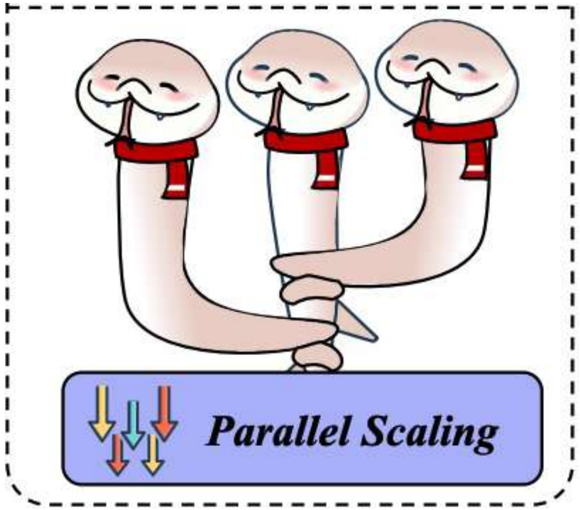
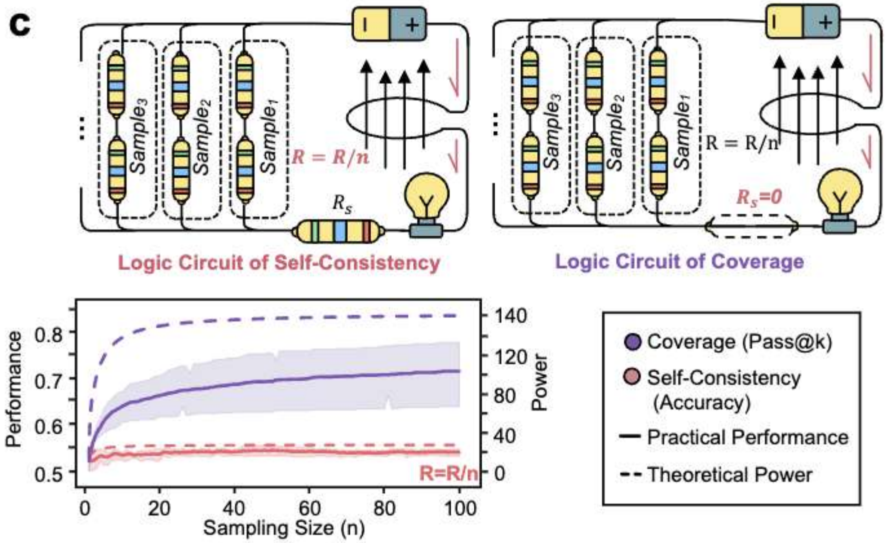
电路猜想

为什么Self-consistency会有效?

- **相当于并行地执行k个推理，则推理电阻变为原来的1/k，可能有额外的集成答案的投票电阻**
- 当有一个无穷大能力的验证器，能够判断哪个答案正确，则得出答案的电阻为0
- 结果趋势一致（**缺陷：把模型采样完全视作独立采样**）





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电路猜想

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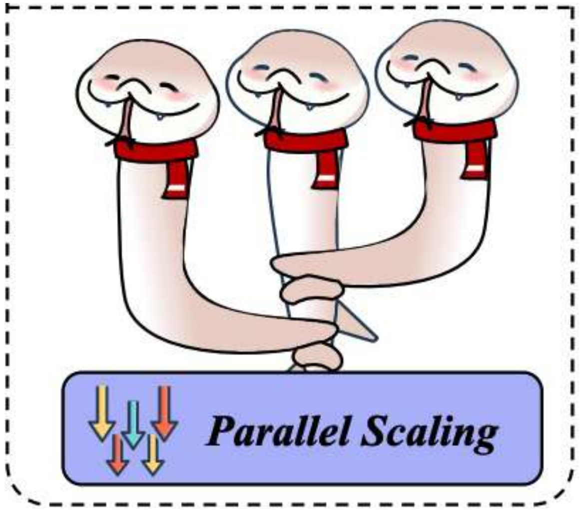
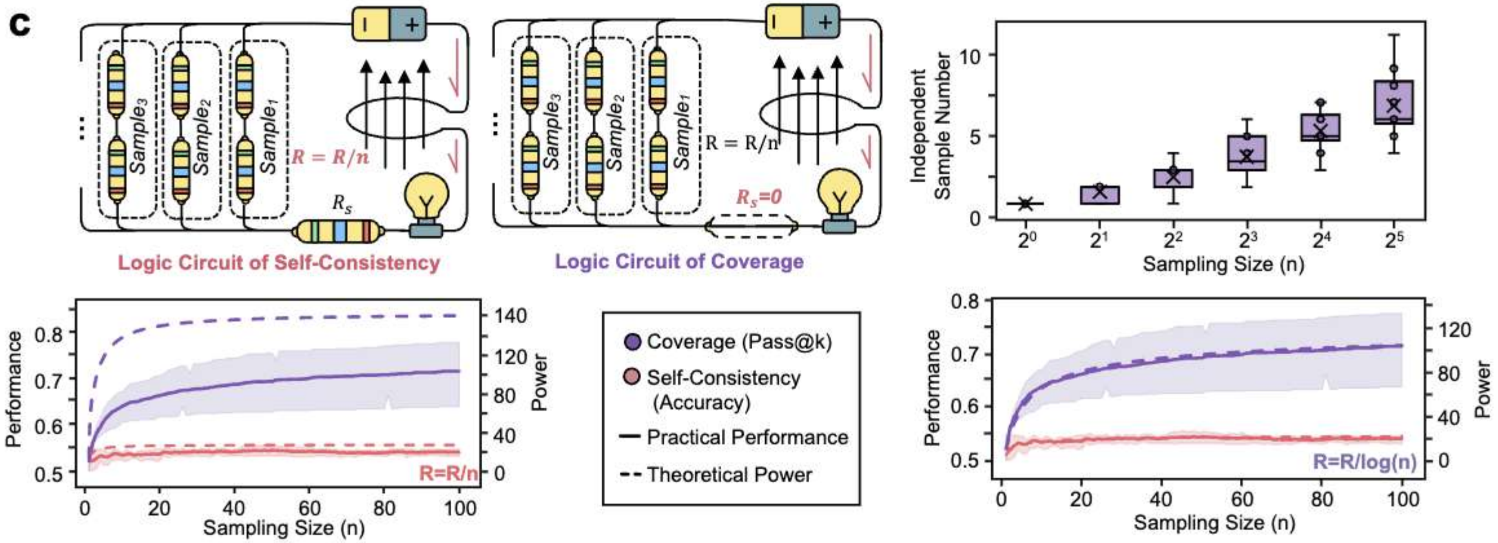
- **相当于并行地执行k个推理，则推理电阻变为原来的1/k，可能有额外的集成答案的投票电阻**
- 实际上模型2^n次才能采出n个独立样本
- **结果完全拟合**





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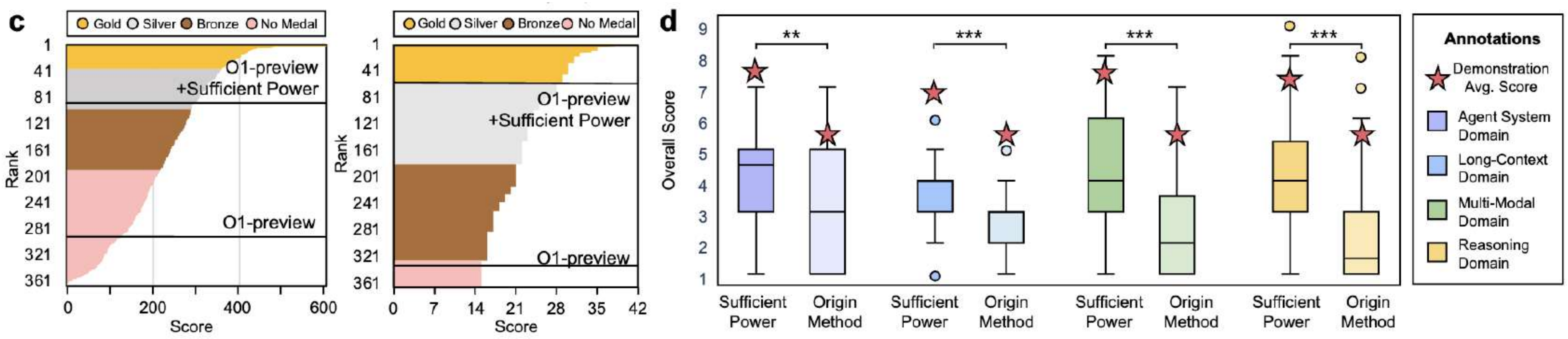
电路猜想

增加足够的功率，能够使大模型在**客观任务（IOI2024，IMO2024）**和**主观任务（论文Idea生成）**上拥有更强的性能表现



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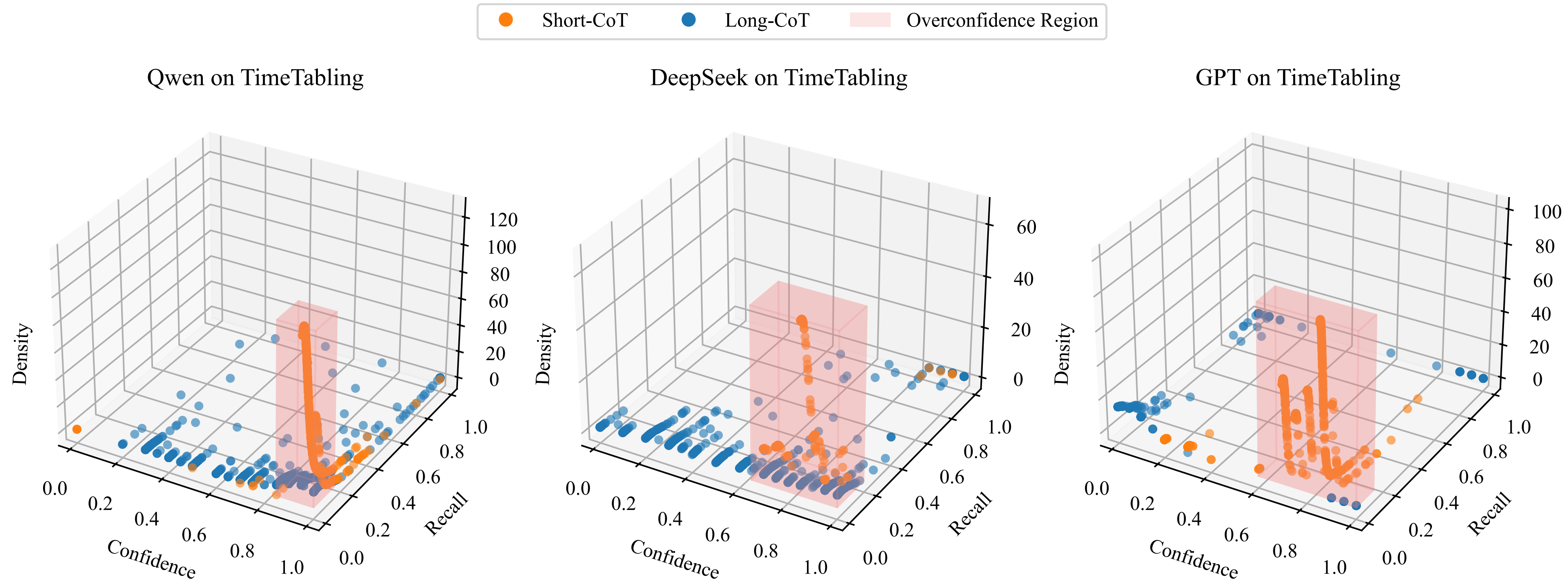
思维链在多解任务上的局限——推理过度自信

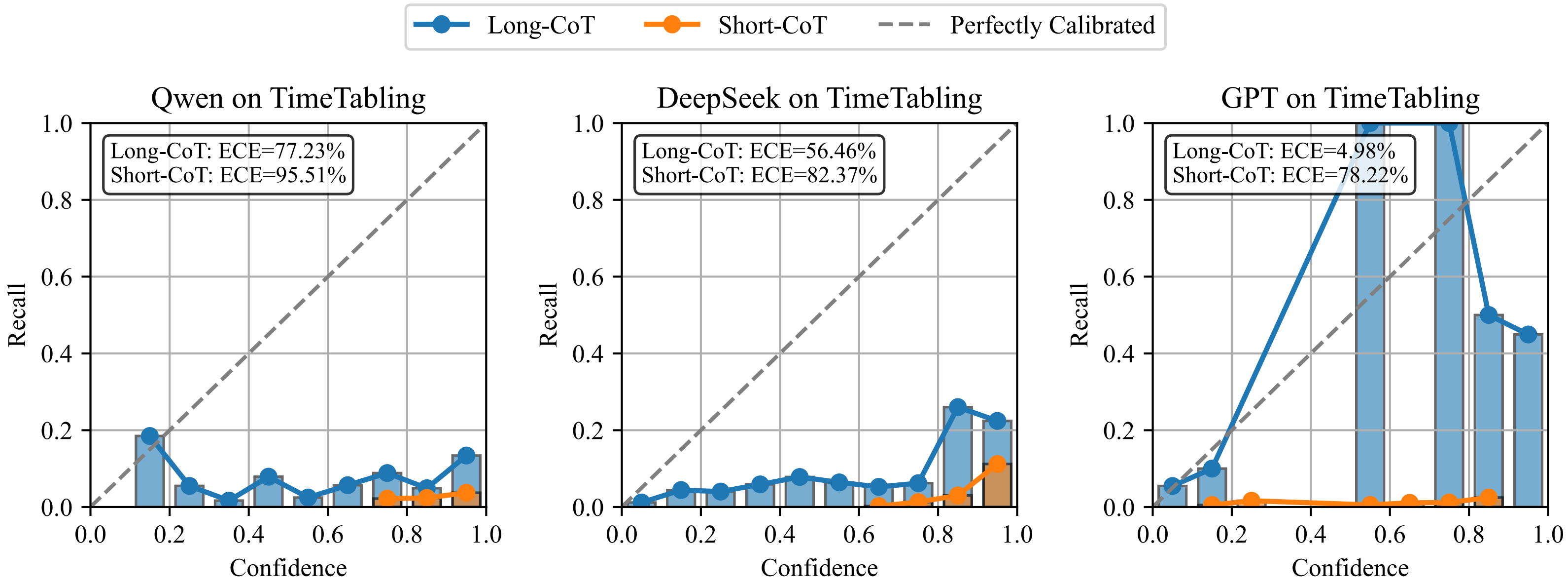
- 多解任务：存在多个可行解；和常规的任务相比，多解任务要求解的**全面性和多样性**
- 推理过度自信（**R**easoning **O**verconfidence，ROC）：使用思维链解决多解任务时，模型对自己答案表现出**过高的置信度，与实际性能严重不匹配**，严重损害幻觉和可靠性评估



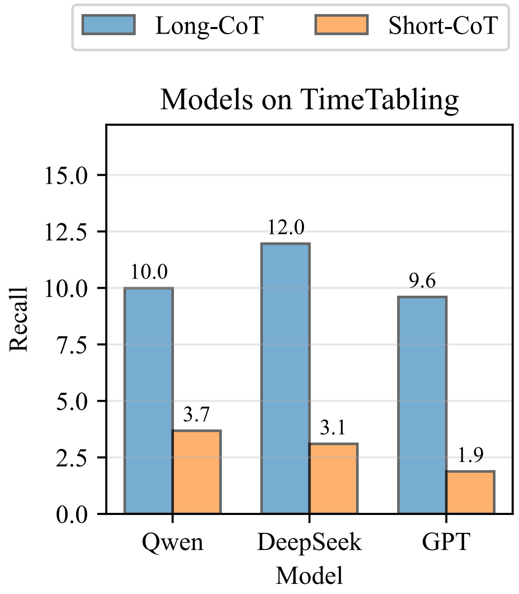
* Guan et al. Beware of Reasoning Overconfidence: Pitfalls in the Reasoning Process for Multi-solution Tasks. AAAI 2026

推理过度自信广泛存在

- 在多种模型中，传统**短思维链的推理过度自信问题广泛存在**
- 长思维链推理范式显著缓解推理过度自信，但仍需改进



(a) Reliability Diagrams of Short-CoT vs. Long-CoT on TimeTabling across different models.

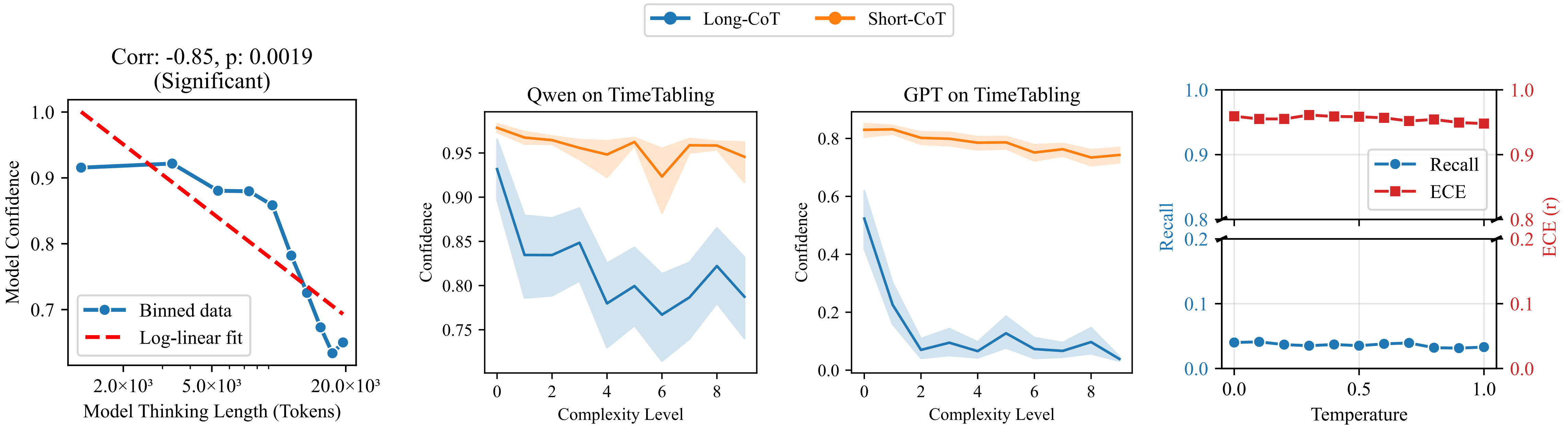


(b) Performance of Short-CoT vs. Long-CoT on TimeTabling.

* Guan et al. Beware of Reasoning Overconfidence: Pitfalls in the Reasoning Process for Multi-solution Tasks. AAAI 2026

推理过度自信的影响因素

- ◆ 更长的**推理链长度**降低了模型估计的置信度，缓解了推理过度自信，符合推理时缩放定律
- ◆ 短思维链的推理过度自信现象在不同**任务复杂度**中均广泛存在
- ◆ 推理过度自信源于模型的内在推理机制，**解码参数**对其几乎无影响

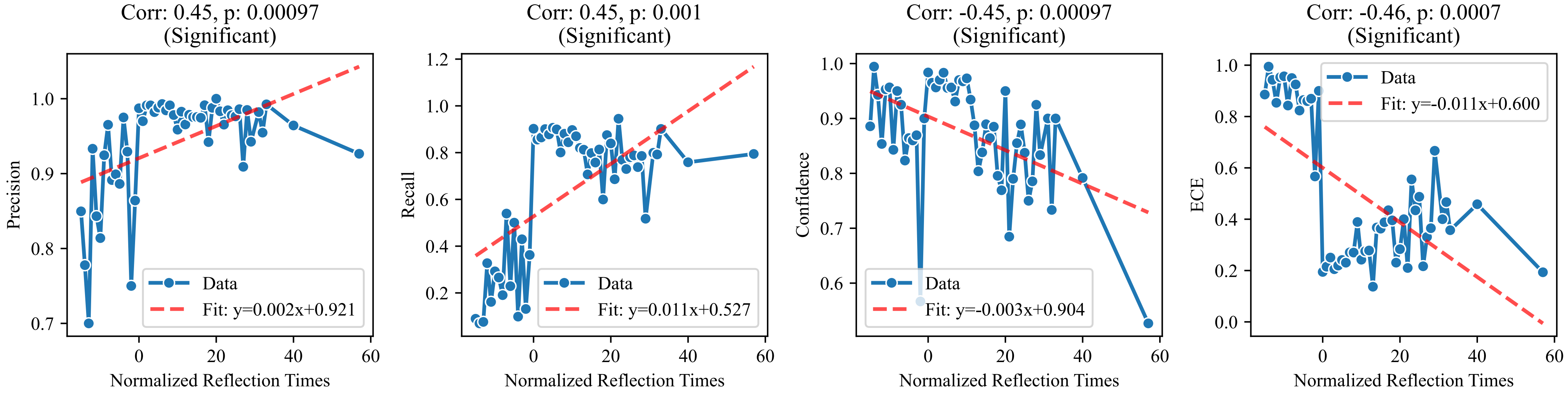


* Guan et al. Beware of Reasoning Overconfidence: Pitfalls in the Reasoning Process for Multi-solution Tasks. AAAI 2026

推理过度自信的缓解

长思维链的**反思**（Reflection）和**探索**（Exploration）是缓解推理过度自信的两大有效策略

◆ 随着长思维链的反思步骤增加，模型性能提升且置信度下降，有效缓解了推理过度自信



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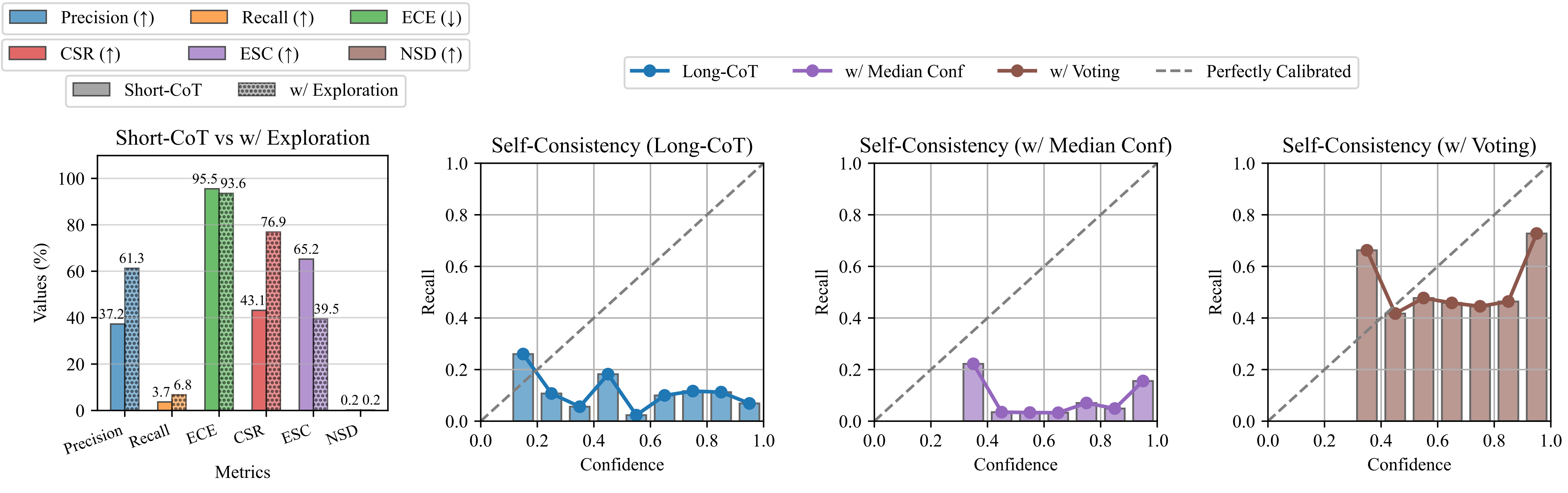
推理过度自信的缓解

长思维链的**反思**（Reflection）和**探索**（Exploration）是缓解推理过度自信的两大有效策略

- ◆ 纵向探索：利用prompt提高推理深度有效缓解了短思维链的推理过度自信
- ◆ 横向探索：基于自一致性策略广泛探索多条推理路径，显著缓解推理过度自信



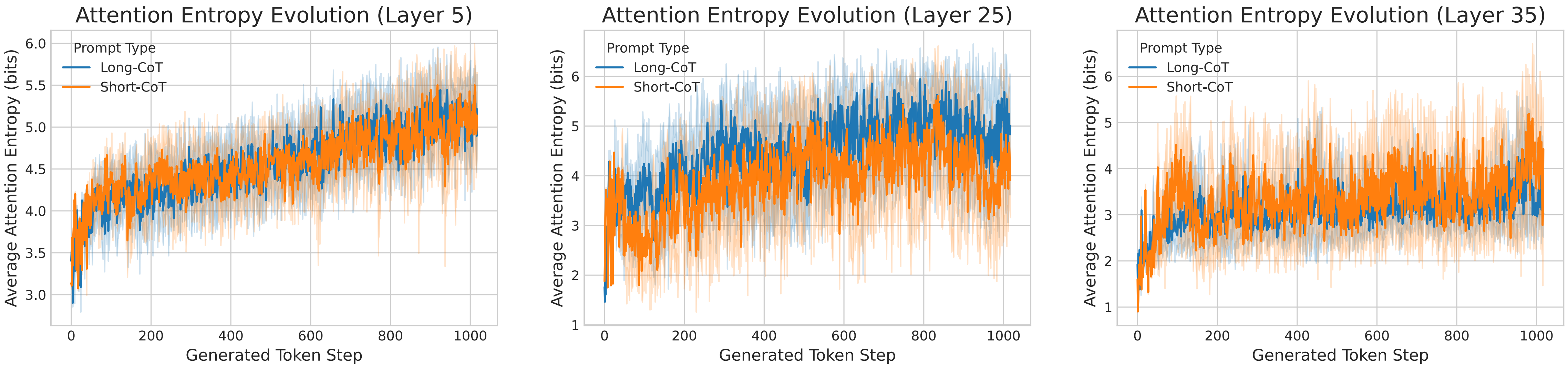
* Guan et al. Beware of Reasoning Overconfidence: Pitfalls in the Reasoning Process for Multi-solution Tasks. AAAI 2026

推理过度自信内部机理初探

认知僵化假设：

◆ **假设：**当模型核心推理层过早锁定单一轨迹，出现认知僵化时，就会出现推理过度自信

◆ **现象：短思维链推理范式核心推理层表现出更低的熵**，导致推理过度自信的出现



* Guan et al. Beware of Reasoning Overconfidence: Pitfalls in the Reasoning Process for Multi-solution Tasks. AAAI 2026

多模态长思维链推理

多模态长思维链

🎯多模态输入 → 文本模态思维输出

◆ 模型基于多模态上下文信息生成文本化的推理步骤，并最终得出答案

◆ **存在问题：性能过高，原因是几乎所有题目都不怎么依赖视觉推理就能解决**

#	Model	Method	Learning	#Size	#P	Link	Date	NAT	SOC	LAN	TXT	IMG	NO	G1-6	G7-12	Avg
*	Human Performance	-	-	-	-	Link	22-09-20	90.23	84.97	87.48	89.60	87.50	88.10	91.59	82.42	**88.40**
*	Random Chance	-	-	-	-	Link	22-09-20	40.28	46.13	29.25	47.45	40.08	33.66	39.35	40.67	**39.83**
1	Mutimodal-T-SciQ_Large 🥇	LLM	Fine-tune	738M	738M	Link	23-05-05	96.89	95.16	95.55	96.53	94.70	96.79	96.44	95.72	**96.18**
2	MC-CoT_F-Large 🥈	VLM	Fine-tune	783M	-	Link	23-11-23	97.47	90.44	93.18	96.97	93.75	94.49	95.30	94.13	**94.88**
3	Honeybee (Vicuna-13B) 🥉	VLM	Fine-tune	13B	-	Link	23-12-11	95.20	96.29	91.18	94.48	93.75	93.17	95.04	93.21	**94.39**
4	Enigma-COT_Large	LLM	Fine-tune	793M	793M	Link	23-07-24	97.51	84.70	94.73	96.68	91.37	95.89	94.46	93.47	**94.11**
5	KAM-CoT	VLM	Fine-tune	280M	280M	Link	24-01-23	94.76	92.24	93.36	94.53	93.16	94.15	94.24	93.21	**93.87**
6	MC-CoT_Large	VLM	Fine-tune	738M	-	Link	23-11-23	95.47	89.99	91.82	95.11	92.66	93.24	94.27	91.76	**93.37**
7	DPMM-CoT_Large	VLM	Fine-tune	738M	738M	Link	23-12-14	95.52	90.33	91.36	95.50	93.26	92.68	93.28	93.47	**93.35**
8	LLaVA (GPT-4 judge)	VLM	Fine-tune	13B	13B	Link	23-04-17	91.56	96.74	91.09	90.62	88.99	93.52	92.73	92.16	**92.53**
9	CoMD (Vicuna-7B)	VLM	Fine-tune	7B	-	Link	23-11-14	91.83	95.95	88.91	90.91	89.94	91.08	92.47	90.97	**91.94**

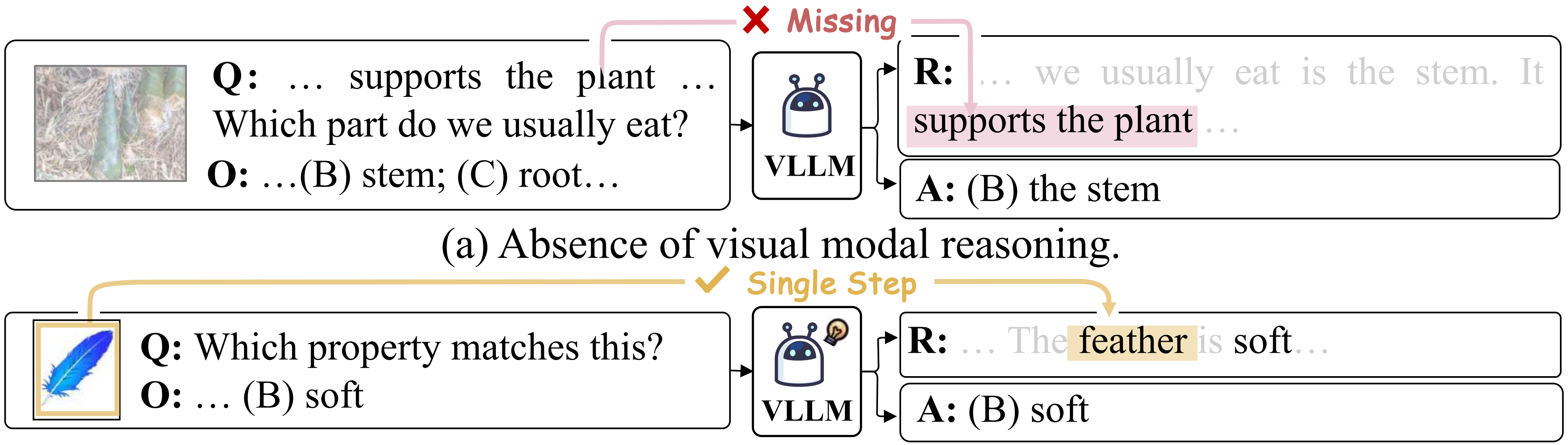
^ ScienceQA （经典多模态思维链数据集）的性能表

* Lu et al. Learn to Explain: Multimodal Reasoning via Thought Chains for Science Question Answering. NeurIPS 2022.

多模态长思维链

现有工作的缺陷

- **缺乏视觉模态推理：**模型能够仅基于文本模态上下文成功生成推理过程并得出答案
- **单步视觉模态推理：**模型仅需一步视觉模态推理即可预测正确的推理过程和答案
- **领域覆盖缺失：**当前在数学与常识推理等领域，尚缺乏高质量的多模态思维链推理数据



(a) Absence of visual modal reasoning.

(b) Single-step visual modal reasoning.

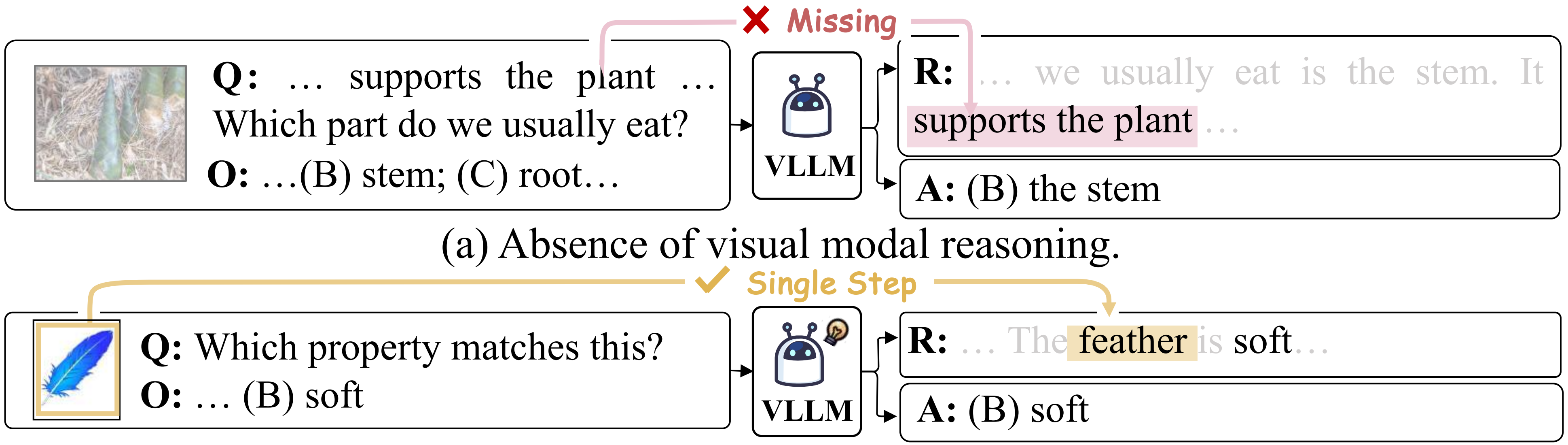
* Chen et al. M3CoT: A Novel Benchmark for Multi-Domain Multi-step Multi-modal Chain-of-Thought. ACL 2024.

多模态长思维链

现有工作的缺陷

- 缺乏视觉模态推理：模型能够仅基于文本模态上下文成功生成推理过程并得出答案。
- 单步视觉模态推理：模型仅需一步视觉模态推理即可预测正确的推理过程和答案。
- 领域覆盖缺失：当前在数学与常识推理等领域，尚缺乏高质量的多模态思维链推理数据。

我们真正期望的多模态思维链推理是什么样的?



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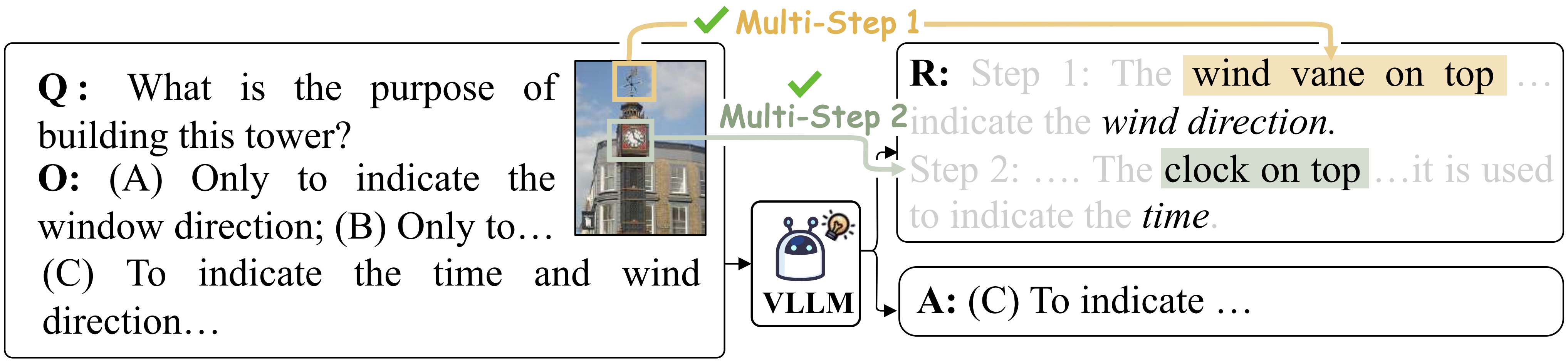
* Chen et al. M3CoT: A Novel Benchmark for Multi-Domain Multi-step Multi-modal Chain-of-Thought. ACL 2024.

多模态长思维链

🎯多模态输入 → 文本模态思维输出 的数据集

◆ **多步多模态思维链推理：**在链式思维推理的多个步骤中，动态地融合视觉线索



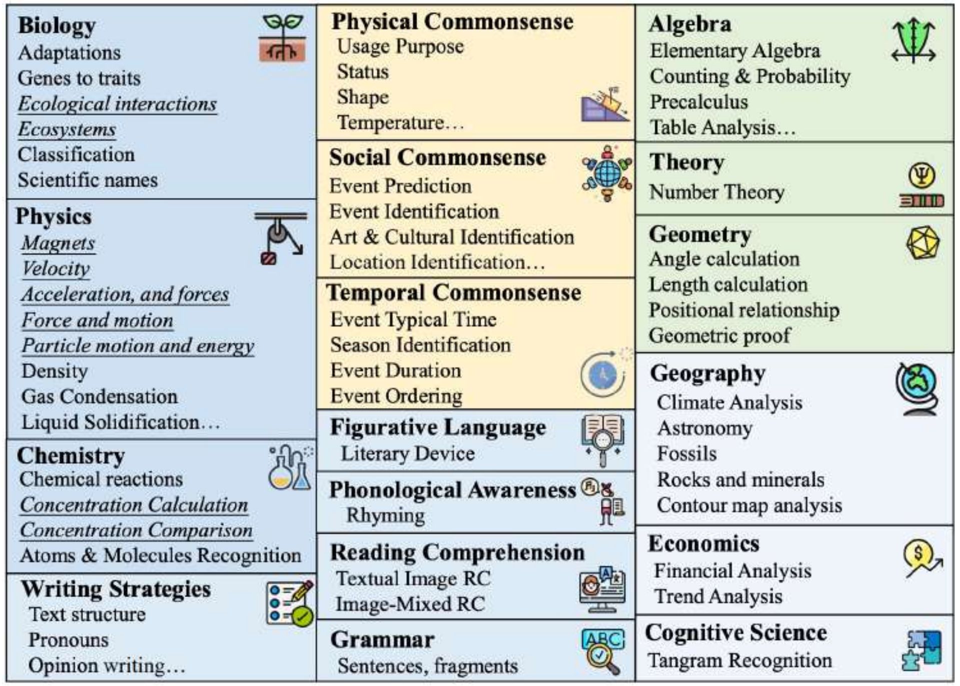
(c) Multi-step visual modal reasoning.

* Chen et al. M3CoT: A Novel Benchmark for Multi-Domain Multi-step Multi-modal Chain-of-Thought. ACL 2024.

多模态长思维链

多模态输入 → 文本模态思维输出 的数据集

- **多步多模态思维链推理：**在链式思维推理的多个步骤中，动态地融合视觉线索
- **多领域多模态思维链推理：**涵盖了链式思维推理的常见领域，如科学、数学和常识推理



Statistic	Number
Total Samples	11,459
Domain classes	3
Topic classes	17
Category classes	263
Science size	7,973
Mathematic size	1,166
Commonsense size	2,163
Train set size	7,973
Dev set size	1,127
Test set size	2,359
Average question length	14.33
Average choice length	3.60
Average context length	19.35
Average rationale length	293.93

○ Science ○ Mathematics ○ Commonsense
○ Natural Science ○ Language Science ○ Social Science

相关工作引用量：80

为InternLM-VL系列的核心科学训练数据

Huggingface数学推理/思维链数据集下载量前5

Huggingface几何数据集下载量前3

加入DL4GPS （面向几何问题的深度学习）必读list

* Chen et al. M3CoT: A Novel Benchmark for Multi-Domain Multi-step Multi-modal Chain-of-Thought. ACL 2024.

多模态长思维链

🎯多模态输入 → 文本模态思维输出 的性能

◆ 开源多模态大模型 << GPT-4V << 人类性能

◆ 多模态思维链需要更加视觉理解能力与逻辑能力更强的大视觉语言模型

Model	Science			Commonsense			Mathematics			Total
	Lang	Natural	Social	Physical	Social	Temporal	Algebra	Geometry	Theory	
Random	32.70	30.62	26.71	32.97	22.22	20.33	35.71	27.50	23.81	28.56
InstructBLIP-13B (Dai et al., 2023)										
Direct (Dai et al., 2023)	38.39	30.52	26.27	76.67	70.66	35.77	30.00	22.50	19.05	35.94
CoT (Kojima et al., 2022)	38.39	30.01	27.55	80.00	70.25	33.33	30.71	21.25	19.05	36.07
Desp-CoT (Wu et al., 2023b)	16.59	27.84	22.77	54.44	52.89	30.08	27.86	28.75	28.57	29.25
CCoT (Mitra et al., 2023)	13.27	26.95	24.84	62.22	67.36	41.46	25.00	25.00	23.81	31.28
LLava-V1.5-13B (Liu et al., 2023)										
Direct (Liu et al., 2023)	36.97	27.46	20.22	52.22	23.55	27.64	22.86	45.00	4.76	27.05
CoT (Kojima et al., 2022)	46.45	38.31	27.87	67.78	64.05	49.59	26.43	30.00	23.81	39.52
Desp-CoT (Wu et al., 2023b)	47.87	29.25	27.23	68.89	59.92	47.15	26.43	36.25	9.52	35.98
CCoT (Mitra et al., 2023)	38.86	31.55	28.18	72.22	61.57	39.84	29.29	36.25	28.57	36.45
CogVLM-17B (Wang et al., 2023c)										
Direct (Wang et al., 2023c)	52.61	37.42	26.91	55.56	54.13	29.27	29.29	32.50	23.81	37.19
CoT (Kojima et al., 2022)	51.18	43.81	29.30	54.44	39.26	31.71	35.71	33.75	33.33	38.91
Desp-CoT (Wu et al., 2023b)	46.92	35.63	25.80	48.89	47.52	38.21	27.14	31.25	19.05	35.07
CCoT (Mitra et al., 2023)	47.39	34.99	25.80	62.22	46.28	35.77	30.71	37.50	23.81	35.63
GPT4V (OpenAI, 2023)										
Direct (OpenAI, 2023)	80.09	54.66	43.95	87.78	67.77	82.11	42.14	43.75	**42.86**	56.95
CoT (Kojima et al., 2022)	**90.52**	**63.09**	**46.97**	83.33	**75.21**	**82.93**	**45.71**	**50.00**	38.10	**62.60**
Desp-CoT (Wu et al., 2023b)	79.62	54.66	36.94	**88.89**	74.38	73.98	20.71	32.50	33.33	53.54
CCoT (Mitra et al., 2023)	84.83	55.30	39.81	80.00	65.70	81.30	32.86	21.25	28.57	54.44
Human	97.63	91.70	87.92	97.80	94.24	91.87	85.71	90.00	76.19	91.17

Model	Science			Commonsense			Mathematics			Total
	Lang	Natural	Social	Physical	Social	Temporal	Algebra	Geometry	Theory	
Random	32.70	30.62	26.71	32.97	22.22	20.33	35.71	27.50	23.81	28.56
Kosmos-2-2B (Peng et al., 2023)										
Direct (Peng et al., 2023)	10.43	28.61	21.18	33.33	17.77	28.46	21.43	21.25	14.29	23.17
CoT (Kojima et al., 2022)	18.48	24.14	14.65	30.00	14.46	9.76	17.14	18.75	0.00	18.68
Desp-CoT (Wu et al., 2023b)	0.00	0.00	0.00	1.11	0.00	0.00	0.00	0.00	0.00	0.04
CCoT (Mitra et al., 2023)	0.00	0.00	0.16	2.22	7.44	1.63	0.00	0.00	0.00	0.99
InstructBLIP-7B (Dai et al., 2023)										
Direct (Dai et al., 2023)	30.81	32.31	27.55	60.00	66.94	39.02	35.71	31.25	33.33	36.11
CoT (Kojima et al., 2022)	38.39	30.01	26.43	80.00	70.25	33.33	30.71	21.25	19.05	35.76
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Direct (Dai et al., 2023)	38.39	30.52	26.27	76.67	70.66	35.77	30.00	22.50	19.05	35.94
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LLava-V1.5-7B (Liu et al., 2023)										
Direct (Liu et al., 2023)	43.13	37.16	26.43	66.67	58.26	30.89	22.14	35.00	14.29	36.63
CoT (Kojima et al., 2022)	38.86	33.59	25.48	71.11	65.29	39.02	29.29	16.25	4.76	35.81
Desp-CoT (Wu et al., 2023b)	34.12	32.18	25.32	63.56	57.83	41.46	24.29	31.25	28.57	34.43
CCoT (Mitra et al., 2023)	26.54	35.50	28.66	62.22	55.79	44.72	29.29	31.25	9.52	35.72

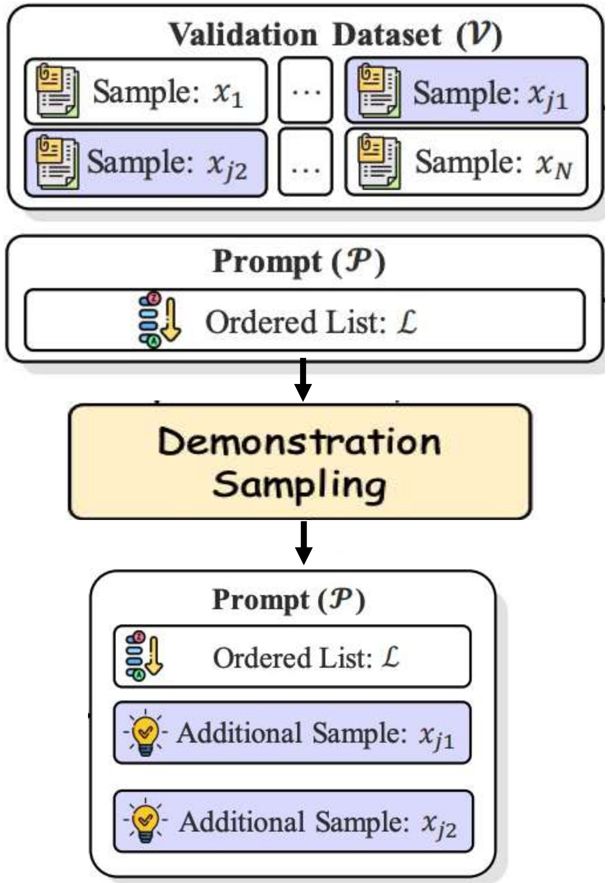
加了思维链反而性能下降

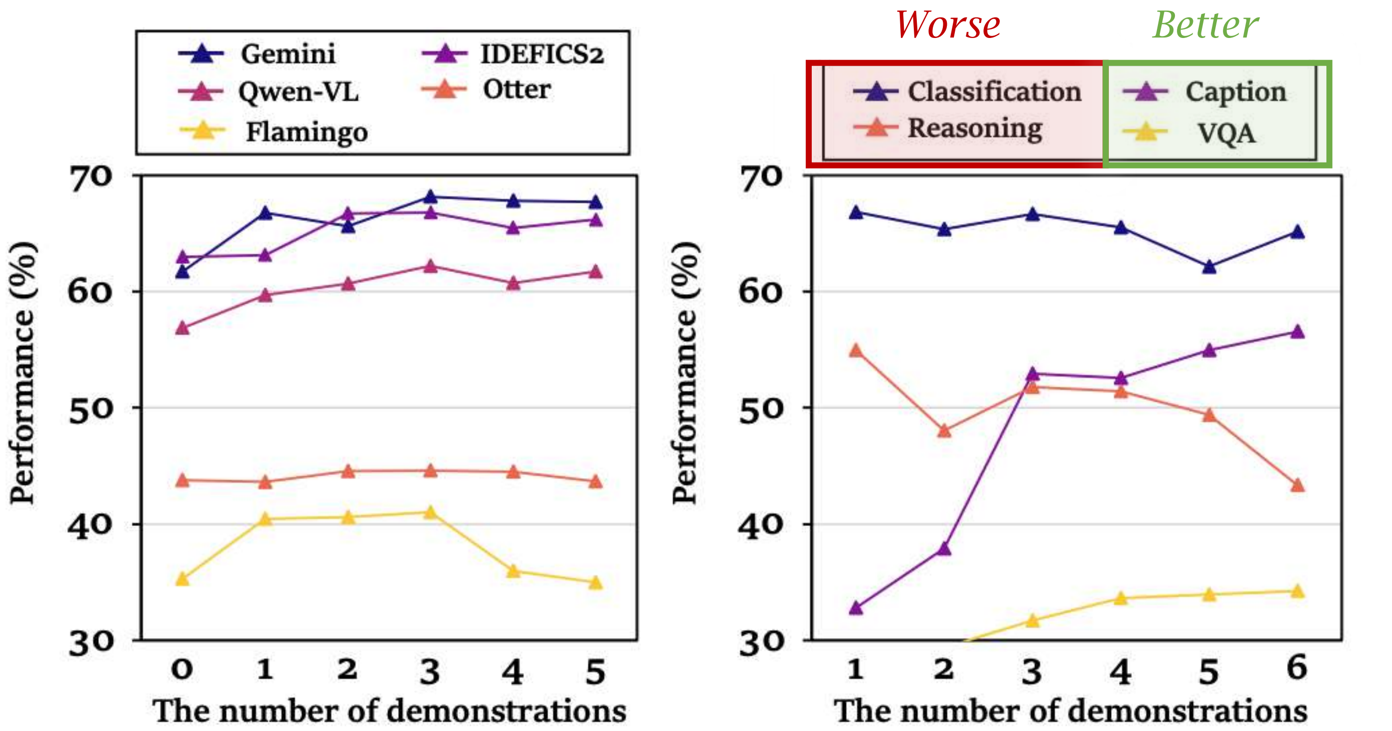
* Chen et al. M3CoT: A Novel Benchmark for Multi-Domain Multi-step Multi-modal Chain-of-Thought. ACL 2024.

多模态长思维链

😟多模态输入 → 文本模态思维输出缺陷：在多模态场景下，文本模态的经验会失效

◆ 增加推理示例往往难以带来更好的推理性能





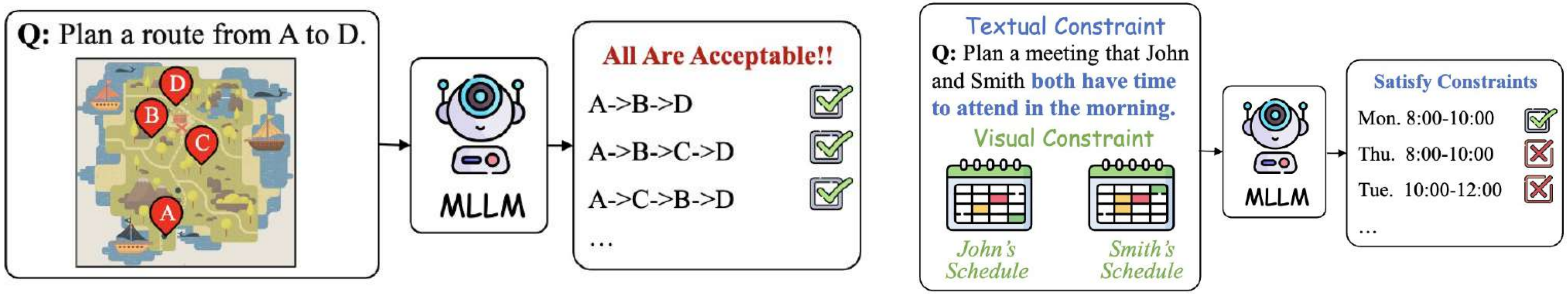
(a) The impact of the number of demonstrations on average score performance.

(b) The impact of the number of demonstrations on different task performance.

* Qin et al. What Factors Affect Multi-Modal In-Context Learning? An In-Depth Exploration. NeurIPS 2024.

长思维链在解空间中的探索能力

- 多解任务：存在多个可行解；和常规的任务相比，多解任务要求解的**全面性和多样性**
- 现有工作尚未考虑**多模态的在条件解空间中探索的多解规划任务**
- 我们提出了第一个**多模态有条件规划基准**



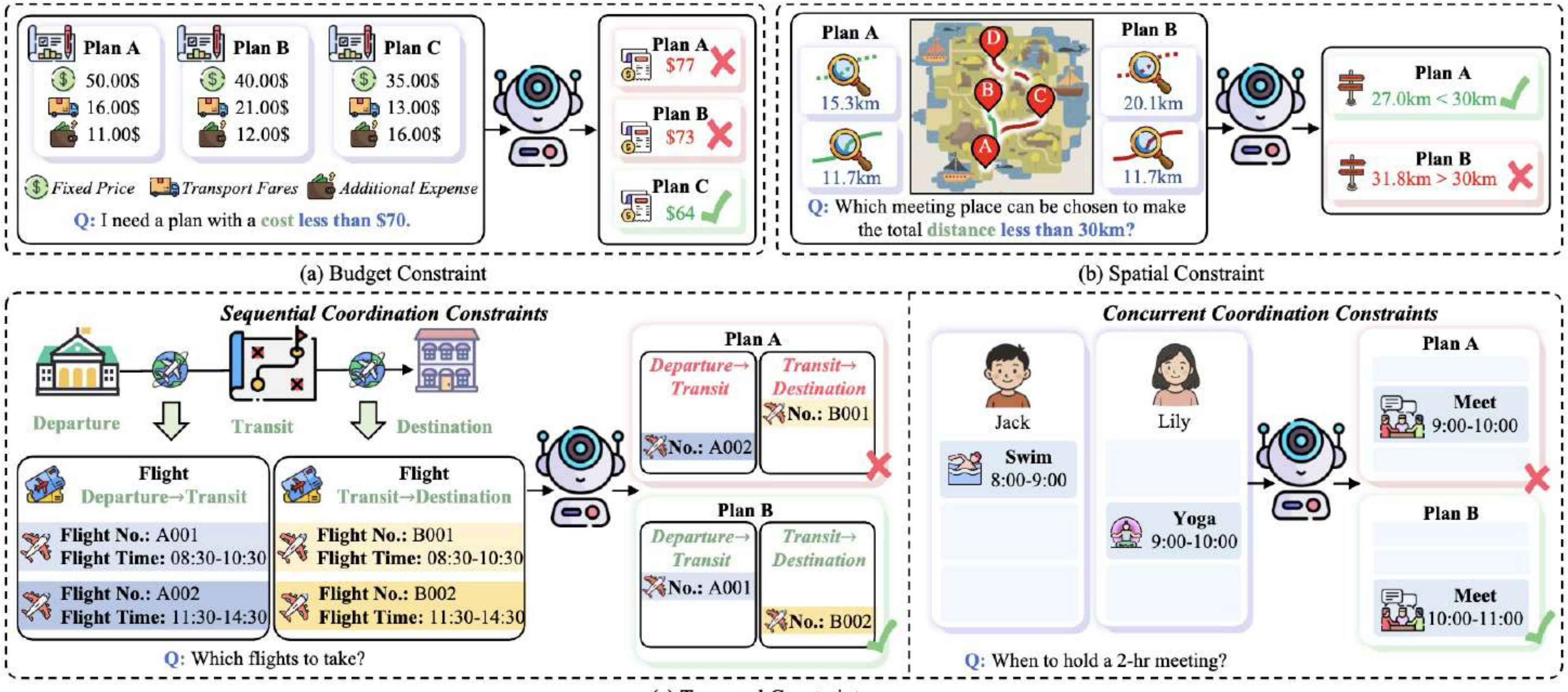
(a) Multimodal planning task without any constraint. **(b) Multimodal planning task with multimodal constraints.**

Dataset	Task Type	#Q	#I	Real-world	Graded Difficulties	Multi-modal Cons.	Composite Cons.
MMMU [42]	Understanding	11,550	11,264	✗	✗	✗	✗
SEED-Bench [18]	Generation	19,242	-	✗	✗	✗	✗
MLLM-CompBench [13]	Reasoning	39,800	79,600	✗	✗	✗	✗
EgoPlan-Bench [6]	Planning	4,939	-	✓	✗	✗	✗
MPCC(ours)	Planning	2,700	6,300	✓	✓	✓	✓

* Ji et al. MPCC: A Novel Benchmark for Multimodal Planning with Complex Constraints in Multimodal Large Language Models. ACM MM 2025

长思维链在解空间中的探索能力

- 我们提出了第一个**多模态有条件规划基准**
- 提出三类跨模态条件限制：**预算限制，空间限制，时间限制**
- 按照可探索的条件解空间大小，定义了多样的探索难度



* Ji et al. MPCC: A Novel Benchmark for Multimodal Planning with Complex Constraints in Multimodal Large Language Models. ACM MM 2025

长思维链在解空间中的探索能力

◆ 模型仅仅能够生成较好的简单的条件解空间较小的可行解

Model	Flight Planning			Calendar Planning			Meeting Planning			Average
	EASY	MEDIUM	HARD	EASY	MEDIUM	HARD	EASY	MEDIUM	HARD	
Empirical Max	-	-	-	11.7/10.0	**11.3/10.3**	**7.0/5.3**	12.0/8.0	**8.0**/5.3	**7.0/6.3**	-
Closed-source MLLMs										
GPT-4o	65.0/33.0	16.7/1.7	2.7/0.0	24.0/21.0	9.0/7.7	2.0/1.0	17.7/16.0	7.7/5.3	4.0/3.3	16.7/9.9
Gemini-2.0-Flash	53.3/27.0	11.7/1.0	0.7/0.0	21.0/12.0	5.7/1.7	3.3/1.0	25.0/23.0	6.7/4.3	4.0/2.7	14.6/8.1
Claude-3.5V-Sonnet	**75.3/46.7**	**34.0/4.7**	0.0/0.0	**29.0/23.3**	9.3/2.0	2.0/1.7	**37.7/34.3**	2.7/1.7	2.0/1.7	**21.3/12.9**
Open-source MLLMs										
InternVL-4B	18.0/3.3	0.0/0.0	0.0/0.0	4.3/3.3	7.0/0.3	2.0/2.0	9.0/5.3	6.3/5.0	4.3/1.0	5.7/2.2
InternVL-8B	32.0/11.0	0.0/0.0	0.0/0.0	10.7/7.7	0.3/0.0	4.0/0.0	5.3/4.0	1.3/1.3	1.0/0.7	6.1/2.7
InternVL-26B	31.7/10.0	1.3/0.3	0.3/0.0	13.7/7.3	5.0/0.7	5.7/2.0	11.3/10.0	**8.0/7.0**	4.3/4.0	9.0/4.6
InternVL-38B	35.7/13.0	0.7/0.3	0.0/0.0	14.0/9.0	5.7/0.0	3.3/3.0	6.3/5.3	**8.0**/6.7	3.0/1.7	8.5/4.3
InternVL-78B	37.3/15.0	1.7/0.0	0.3/**0.3**	7.3/3.3	5.7/1.0	3.0/3.0	7.3/6.3	7.7/6.7	3.0/2.7	8.1/4.3
LLaVa-OV-0.5B	0.0/0.0	0.0/0.0	0.0/0.0	3.7/3.0	1.7/0.0	1.3/1.3	6.3/3.0	0.3/0.3	1.0/0.7	1.6/0.9
LLaVa-OV-7B	6.7/2.7	0.0/0.0	0.0/0.0	9.0/6.3	5.7/0.0	2.3/0.0	7.7/4.0	7.3/4.0	2.3/1.3	4.6/2.0
Qwen2-VL-2B	5.3/1.0	0.0/0.0	0.0/0.0	4.7/4.3	1.7/0.0	0.7/0.0	5.7/2.7	3.0/1.3	1.3/0.7	2.5/1.1
Qwen2-VL-7B	32.0/8.7	0.0/0.0	0.0/0.0	12.7/9.0	7.3/0.0	6.3/0.3	9.0/4.0	7.0/3.7	4.7/3.0	8.8/3.2
Qwen2-VL-72B	46.0/17.7	12.7/1.0	**9.3/0.3**	8.3/6.7	4.7/0.0	2.3/2.3	10.0/8.7	3.3/2.7	1.7/1.7	10.9/4.6

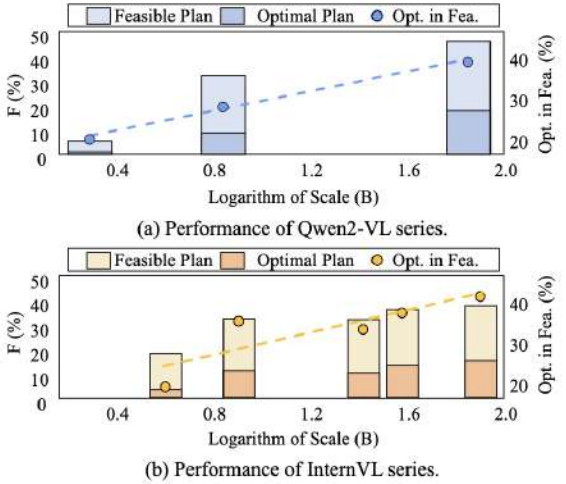


Figure 5: Selected open source model scales and their performance on Flight Planning EASY. Opt. in Fea.: Ratio of *optimal plan* in *feasible plan*.

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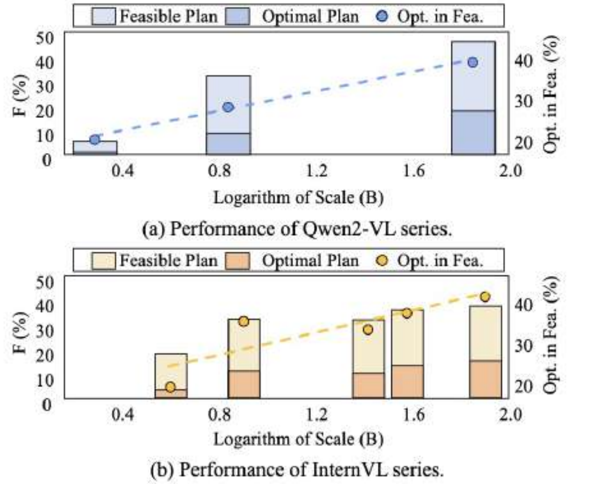
长思维链在解空间中的探索能力

- 模型仅仅能够生成较好的简单的条件解空间较小的可行解
- 模型几乎无法生成最优解

Model	Flight Planning EASY	Flight Planning MEDIUM	Flight Planning HARD	Calendar Planning EASY	Calendar Planning MEDIUM	Calendar Planning HARD	Meeting Planning EASY	Meeting Planning MEDIUM	Meeting Planning HARD	Average
Empirical Max	-	-	-	11.7/10.0	**11.3/10.3**	**7.0/5.3**	12.0/8.0	**8.0**/5.3	**7.0/6.3**	-
Closed-source MLLMs										
GPT-4o	65.0/33.0	16.7/1.7	2.7/0.0	24.0/21.0	9.0/7.7	2.0/1.0	17.7/16.0	7.7/5.3	4.0/3.3	16.7/9.9
Gemini-2.0-Flash	53.3/27.0	11.7/1.0	0.7/0.0	21.0/12.0	5.7/1.7	3.3/1.0	25.0/23.0	6.7/4.3	4.0/2.7	14.6/8.1
Claude-3.5V-Sonnet	**75.3/46.7**	**34.0/4.7**	0.0/0.0	**29.0/23.3**	9.3/2.0	2.0/1.7	**37.7/34.3**	2.7/1.7	2.0/1.7	**21.3/12.9**
Open-source MLLMs										
InternVL-4B	18.0/3.3	0.0/0.0	0.0/0.0	4.3/3.3	7.0/0.3	2.0/2.0	9.0/5.3	6.3/5.0	4.3/1.0	5.7/2.2
InternVL-8B	32.0/11.0	0.0/0.0	0.0/0.0	10.7/7.7	0.3/0.0	4.0/0.0	5.3/4.0	1.3/1.3	1.0/0.7	6.1/2.7
InternVL-26B	31.7/10.0	1.3/0.3	0.3/0.0	13.7/7.3	5.0/0.7	5.7/2.0	11.3/10.0	**8.0/7.0**	4.3/4.0	9.0/4.6
InternVL-38B	35.7/13.0	0.7/0.3	0.0/0.0	14.0/9.0	5.7/0.0	3.3/3.0	6.3/5.3	**8.0**/6.7	3.0/1.7	8.5/4.3
InternVL-78B	37.3/15.0	1.7/0.0	0.3/**0.3**	7.3/3.3	5.7/1.0	3.0/3.0	7.3/6.3	7.7/6.7	3.0/2.7	8.1/4.3
LLaVa-OV-0.5B	0.0/0.0	0.0/0.0	0.0/0.0	3.7/3.0	1.7/0.0	1.3/1.3	6.3/3.0	0.3/0.3	1.0/0.7	1.6/0.9
LLaVa-OV-7B	6.7/2.7	0.0/0.0	0.0/0.0	9.0/6.3	5.7/0.0	2.3/0.0	7.7/4.0	7.3/4.0	2.3/1.3	4.6/2.0
Qwen2-VL-2B	5.3/1.0	0.0/0.0	0.0/0.0	4.7/4.3	1.7/0.0	0.7/0.0	5.7/2.7	3.0/1.3	1.3/0.7	2.5/1.1
Qwen2-VL-7B	32.0/8.7	0.0/0.0	0.0/0.0	12.7/9.0	7.3/0.0	6.3/0.3	9.0/4.0	7.0/3.7	4.7/3.0	8.8/3.2
Qwen2-VL-72B	46.0/17.7	12.7/1.0	**9.3/0.3**	8.3/6.7	4.7/0.0	2.3/2.3	10.0/8.7	3.3/2.7	1.7/1.7	10.9/4.6



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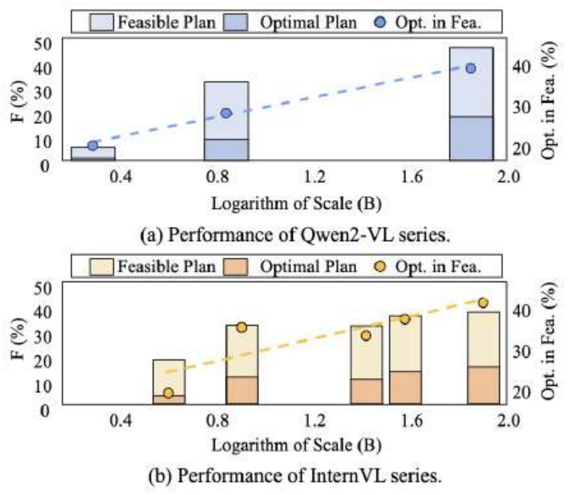
- 模型几乎无法生成最优解
- 基准性能满足模型Scaling Law，模型的参数扩展仍然有效

Model	Flight Planning			Calendar Planning			Meeting Planning			Average
	EASY	MEDIUM	HARD	EASY	MEDIUM	HARD	EASY	MEDIUM	HARD	
Empirical Max	-	-	-	11.7/10.0	**11.3/10.3**	**7.0/5.3**	12.0/8.0	**8.0**/5.3	**7.0/6.3**	-
Closed-source MLLMs										
GPT-4o	65.0/33.0	16.7/1.7	2.7/0.0	24.0/21.0	9.0/7.7	2.0/1.0	17.7/16.0	7.7/5.3	4.0/3.3	16.7/9.9
Gemini-2.0-Flash	53.3/27.0	11.7/1.0	0.7/0.0	21.0/12.0	5.7/1.7	3.3/1.0	25.0/23.0	6.7/4.3	4.0/2.7	14.6/8.1
Claude-3.5V-Sonnet	**75.3/46.7**	**34.0/4.7**	0.0/0.0	**29.0/23.3**	9.3/2.0	2.0/1.7	**37.7/34.3**	2.7/1.7	2.0/1.7	**21.3/12.9**
Open-source MLLMs										
InternVL-4B	18.0/3.3	0.0/0.0	0.0/0.0	4.3/3.3	7.0/0.3	2.0/2.0	9.0/5.3	6.3/5.0	4.3/1.0	5.7/2.2
InternVL-8B	32.0/11.0	0.0/0.0	0.0/0.0	10.7/7.7	0.3/0.0	4.0/0.0	5.3/4.0	1.3/1.3	1.0/0.7	6.1/2.7
InternVL-26B	31.7/10.0	1.3/0.3	0.3/0.0	13.7/7.3	5.0/0.7	5.7/2.0	11.3/10.0	**8.0/7.0**	4.3/4.0	9.0/4.6
InternVL-38B	35.7/13.0	0.7/0.3	0.0/0.0	14.0/9.0	5.7/0.0	3.3/3.0	6.3/5.3	**8.0**/6.7	3.0/1.7	8.5/4.3
InternVL-78B	37.3/15.0	1.7/0.0	0.3/**0.3**	7.3/3.3	5.7/1.0	3.0/3.0	7.3/6.3	7.7/6.7	3.0/2.7	8.1/4.3
LLaVa-OV-0.5B	0.0/0.0	0.0/0.0	0.0/0.0	3.7/3.0	1.7/0.0	1.3/1.3	6.3/3.0	0.3/0.3	1.0/0.7	1.6/0.9
LLaVa-OV-7B	6.7/2.7	0.0/0.0	0.0/0.0	9.0/6.3	5.7/0.0	2.3/0.0	7.7/4.0	7.3/4.0	2.3/1.3	4.6/2.0
Qwen2-VL-2B	5.3/1.0	0.0/0.0	0.0/0.0	4.7/4.3	1.7/0.0	0.7/0.0	5.7/2.7	3.0/1.3	1.3/0.7	2.5/1.1
Qwen2-VL-7B	32.0/8.7	0.0/0.0	0.0/0.0	12.7/9.0	7.3/0.0	6.3/0.3	9.0/4.0	7.0/3.7	4.7/3.0	8.8/3.2
Qwen2-VL-72B	46.0/17.7	12.7/1.0	**9.3/0.3**	8.3/6.7	4.7/0.0	2.3/2.3	10.0/8.7	3.3/2.7	1.7/1.7	10.9/4.6



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长思维链在解空间中的探索能力

◆ 模型超过30%以上的生成解原因是由于**多模态条件的理解错误**

◆ 模型超过30%以上的生成解原因是由于**与多模态条件的矛盾**

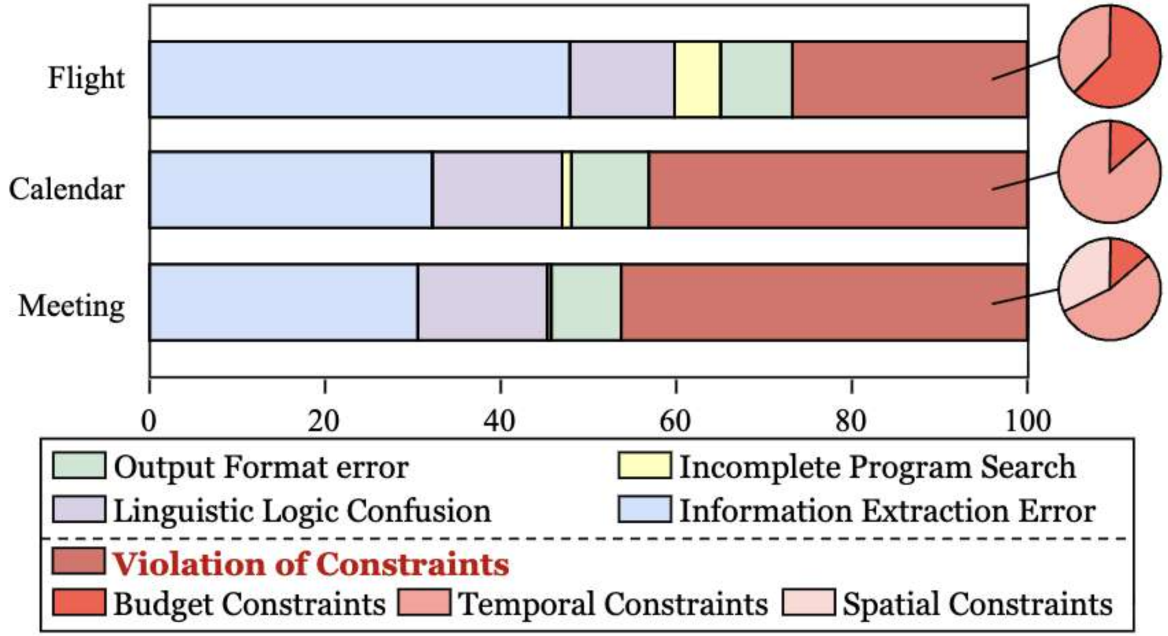

Figure 10: Manual analysis of incorrect responses. Over 40% of incorrect responses were due to failure to meet constraints.

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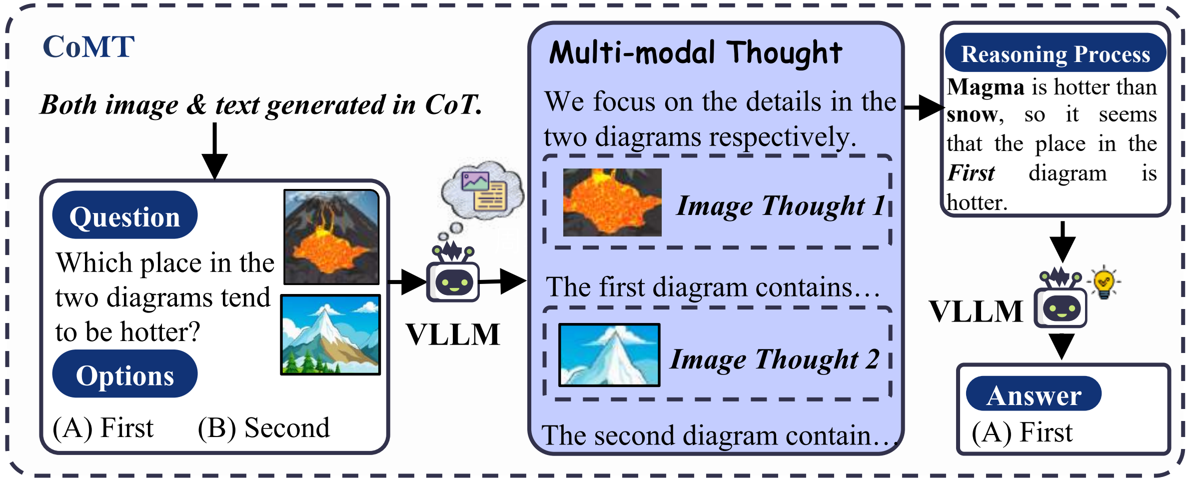
多模态长思维链

多模态输入 → 多模态思维输出

- 推理过程中生成多模态输出（文本+图像）
- 更接近人类思考方式
- 便于执行视觉操作
- 便于描述空间信息



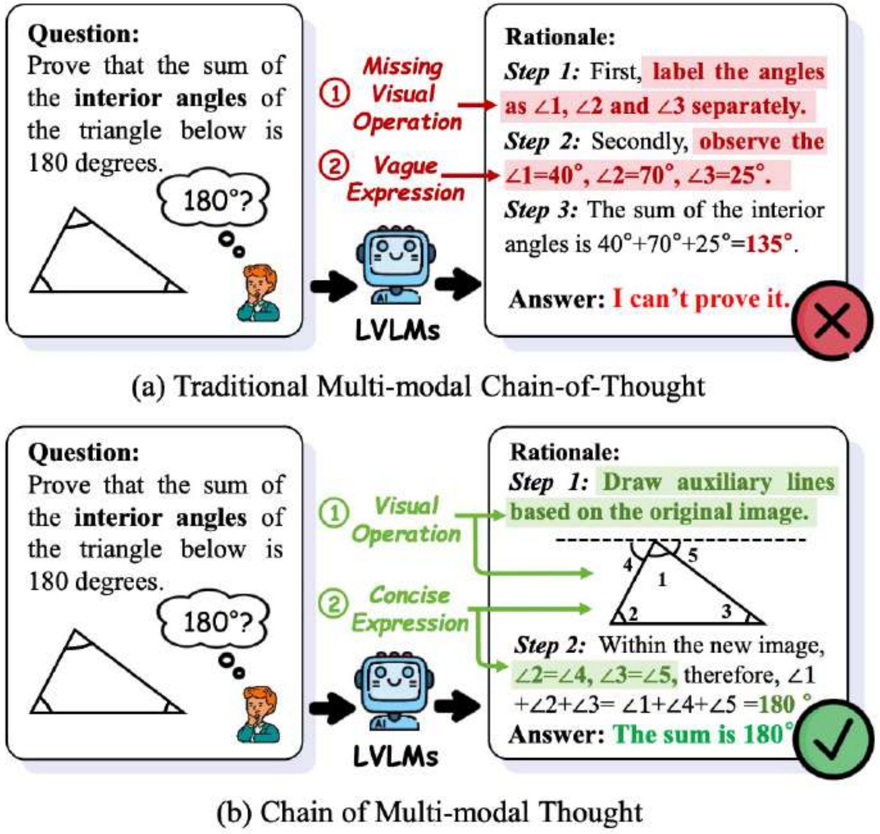


Figure 1: Comparison between (a) traditional multi-modal CoT and (b) chain of multi-modal thought, where images in rationales are needed to be generated from LVLMs to assist textual reasoning in rationale.

* Cheng et al. CoMT: A Novel Benchmark for Chain of Multi-modal Thought on Large Vision-Language Models. AAAI 2025

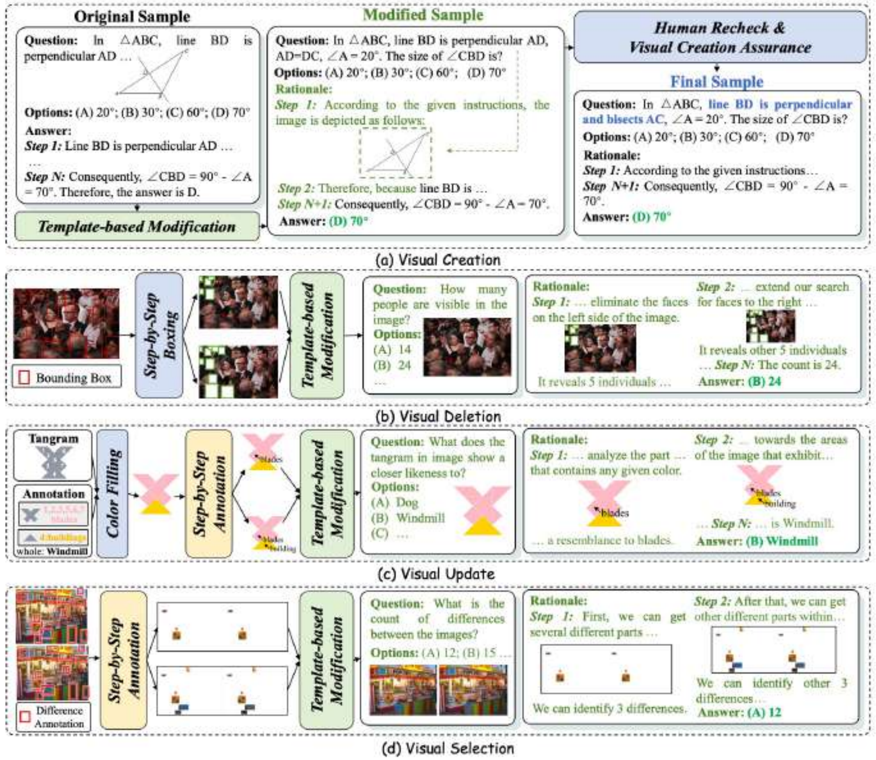
多模态长思维链

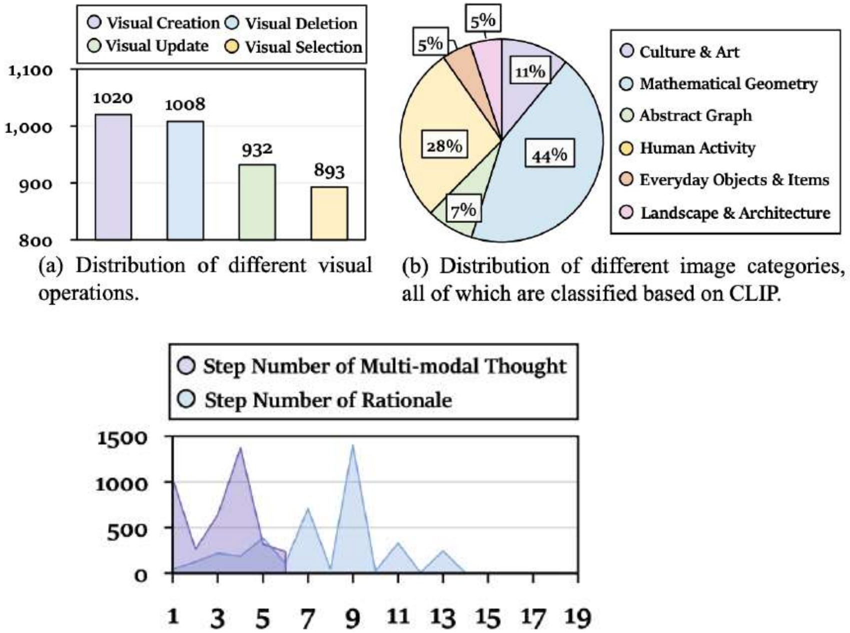
多模态输入 → 多模态思维输出 的数据集

- 包含四种视觉任务，全面评估模型的多模态思维链能力
- 结合多个领域数据集内容，确保推理任务的多样性和复杂性

Huggingface思维链数据集下载量第 1

Huggingface Image-to-Image 数据集下载量第 2





(a) Distribution of different visual operations.

(b) Distribution of different image categories, all of which are classified based on CLIP.

(c) Distribution of *the number of steps in multi-modal thought* and *the number of steps in complete rationale*.

* Cheng et al. CoMT: A Novel Benchmark for Chain of Multi-modal Thought on Large Vision-Language Models. AAAI 2025

多模态长思维链

多模态思维性能缺陷

- 传统的**纯文本思维链（CoT）**因无法执行视觉逻辑表达而对性能提升几乎**没有帮助**
- 大型视觉语言模型**无法**通过文本**准确表达视觉推理**

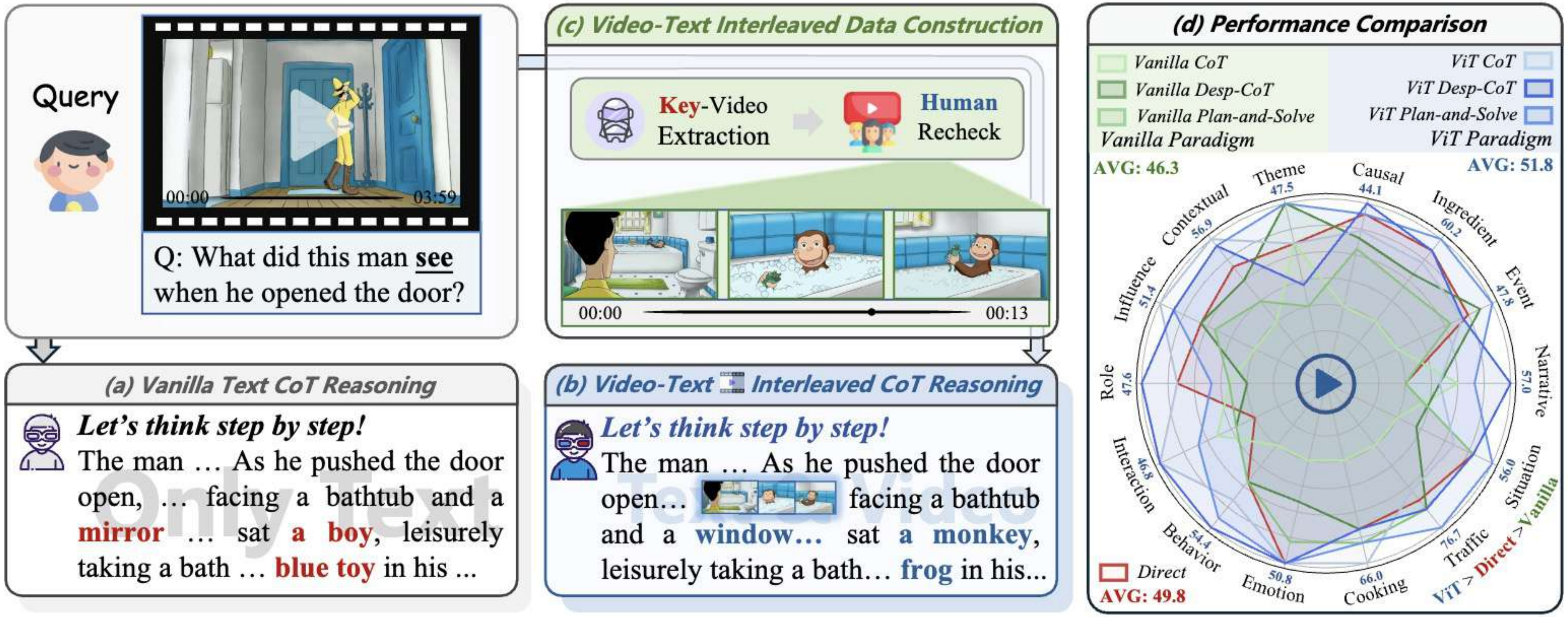
Model	Visual Creation		Visual Deletion		Visual Update		Visual Selection		Average	
	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1	Acc	Macro-F1
Random	27.10	26.75	25.17	25.15	24.06	24.05	25.59	25.55	25.48	25.37
Qwen-VL-7B (Bai et al. 2023)										
Direct (Bai et al. 2023)	21.49	12.78	26.35	18.29	37.64	30.34	22.08	13.80	26.89	18.80
CoT (Kojima et al. 2022)	23.96	19.22	12.63	11.81	33.62	26.13	23.22	18.00	23.26	18.79
Desp-CoT (Wu et al. 2023)	19.90	13.23	20.94	7.73	30.59	23.85	26.05	10.48	24.37	13.82
VoT (Wu et al. 2024b)	22.08	17.51	14.43	11.71	28.52	21.02	22.08	12.47	21.78	15.68
LLaVA-NeXT-13B (Liu et al. 2024a)										
Direct (Liu et al. 2024a)	26.34	19.72	20.64	20.06	35.47	34.26	22.76	19.60	26.30	23.41
CoT (Kojima et al. 2022)	22.18	12.33	21.44	15.21	26.36	18.99	24.92	19.91	23.73	16.61
Desp-CoT (Wu et al. 2023)	19.90	12.82	23.45	17.47	27.01	18.82	25.59	20.77	23.99	17.47
VoT (Wu et al. 2024b)	20.79	15.58	25.55	18.55	27.55	18.95	26.61	17.23	25.13	17.58
Gemini (Team et al. 2023)										
Direct (Team et al. 2023)	28.91	25.43	**30.86**	22.28	**46.36**	**46.26**	**27.63**	20.69	**33.44**	**28.67**
CoT (Kojima et al. 2022)	27.92	23.07	28.76	**22.73**	40.24	40.02	27.39	**23.60**	31.08	27.36
Desp-CoT (Wu et al. 2023)	18.04	14.61	29.36	21.43	31.05	23.20	25.14	11.32	25.90	17.64
VoT (Wu et al. 2024b)	**33.27**	**26.48**	27.05	20.79	35.36	27.83	24.92	19.38	30.15	23.62

* Cheng et al. CoMT: A Novel Benchmark for Chain of Multi-modal Thought on Large Vision-Language Models. AAAI 2025

多模态长思维链——缺陷解决方案

多模态思维性能缺陷解决方案

- 在视频思维链理解过程中，通过插入与**推理逻辑相关的关键视频帧**，添加额外的跨模态思维



* Zhang et al. ViTCoT: Video-Text Interleaved Chain-of-Thought for Boosting Video Understanding in Large Language Models. ACM MM 2025

多模态长思维链——缺陷解决方案

多模态思维性能缺陷解决方案

在视频思维链理解过程中，通过插入与推理逻辑相关的关键视频帧，添加额外的跨模态思维，**进一步显著地提升模型性能**

Table 1: The experimental results of Acc. (%) on MLLMs. area: vanilla CoT reasoning. area: video-text interleaved CoT reasoning. The underline indicates better performance in the MLLM. The Red represents better AVG performance in the MLLM.

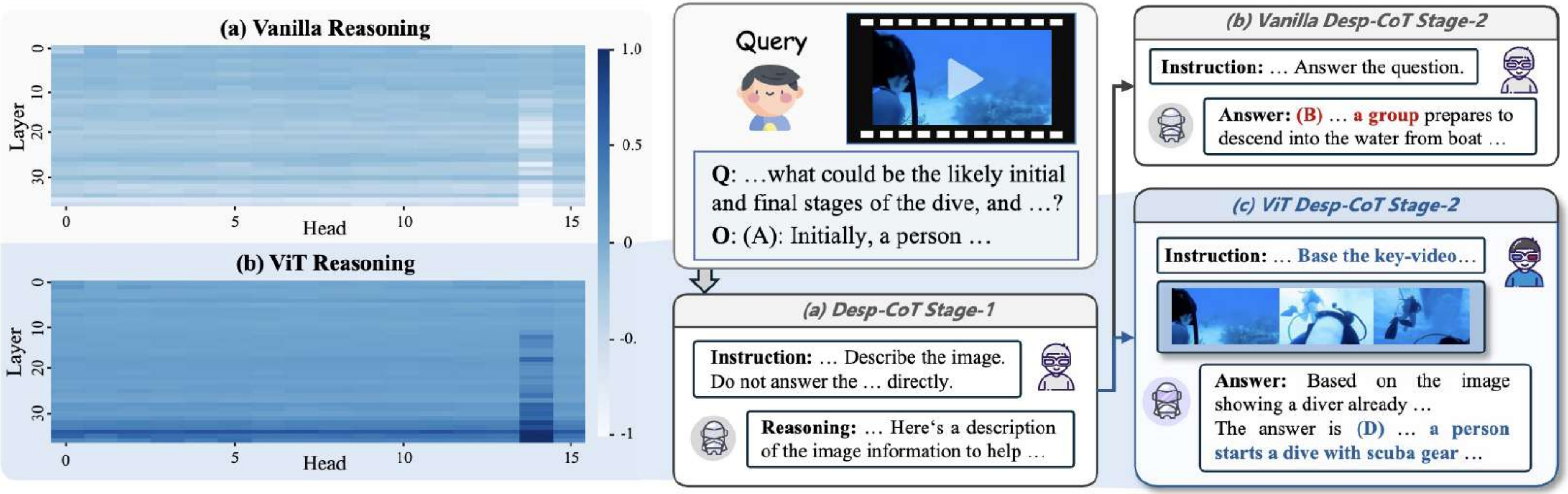
Methods	Narra.	Event	Ingre.	Causal	Theme	Conte.	Influ.	Role	Inter.	Behav.	Emoti.	Cook.	Traff.	Situa.	AVG
Qwen2.5-VL-3B-Instruct [47]															
Direct	38.0	36.9	46.9	28.4	34.4	39.4	41.7	31.7	35.5	22.8	32.3	41.5	53.3	38.0	37.2
Vanilla CoT	34.2	29.3	37.8	26.1	36.1	39.4	36.1	36.5	30.6	19.3	32.3	32.1	46.7	30.0	33.3
Vanilla Desp-CoT	44.3	37.6	50.0	35.0	34.4	43.1	51.4	33.3	41.9	29.8	40.0	39.6	56.7	44.0	41.5
Vanilla Plan-and-Solve	49.4	40.8	58.2	42.3	47.5	48.6	48.6	39.7	50.0	38.6	41.5	50.9	63.3	50.0	47.8
ViT CoT	43.0	31.2	40.8	30.5	37.7	39.4	41.7	33.3	33.9	21.1	32.3	32.1	43.3	34.0	35.3
ViT Desp-CoT	50.6	37.6	53.1	34.3	41.0	45.0	51.4	34.9	46.8	33.3	43.1	47.2	56.7	46.0	44.4
ViT Plan-and-Solve	51.9	41.4	62.2	43.0	49.2	53.2	52.8	41.3	54.8	40.4	43.1	54.7	66.7	52.0	50.5
Qwen2.5-VL-7B-Instruct [47]															
Direct	49.4	45.2	58.2	43.4	44.3	53.2	47.2	44.4	38.7	45.6	50.8	56.6	66.7	54.0	49.8
Vanilla CoT	53.2	37.6	45.9	37.3	44.3	45.9	41.7	39.7	41.9	38.6	43.1	37.7	46.7	46.0	42.8
Vanilla Desp-CoT	51.9	46.5	54.1	41.8	47.5	51.4	47.2	38.1	40.3	45.6	46.2	56.6	70.0	48.0	48.9
Vanilla Plan-and-Solve	49.4	43.3	52.0	41.1	39.3	48.6	45.8	39.7	40.3	45.6	49.2	60.4	70.0	54.0	48.5
ViT CoT	51.9	44.6	56.1	42.3	44.3	56.9	51.4	44.4	46.8	49.1	50.8	66.0	63.3	52.0	51.4
ViT Desp-CoT	54.4	47.8	60.2	43.4	47.5	56.0	50.0	41.3	43.5	52.6	50.8	60.4	76.7	56.0	52.9
ViT Plan-and-Solve	57.0	44.6	58.2	44.1	41.0	56.0	50.0	47.6	46.8	54.4	50.8	56.6	70.0	54.0	52.2

* Zhang et al. ViTCoT: Video-Text Interleaved Chain-of-Thought for Boosting Video Understanding in Large Language Models. ACM MM 2025

多模态长思维链——缺陷解决方案

多模态思维性能缺陷解决方案

- 跨模态思维能够使模型**推理激活的更加活跃**
- 跨模态信息会**传递到大模型更深的模型层**中



(a) Comparison of activation parameters for MLLMs between Vanilla reasoning and ViT reasoning.

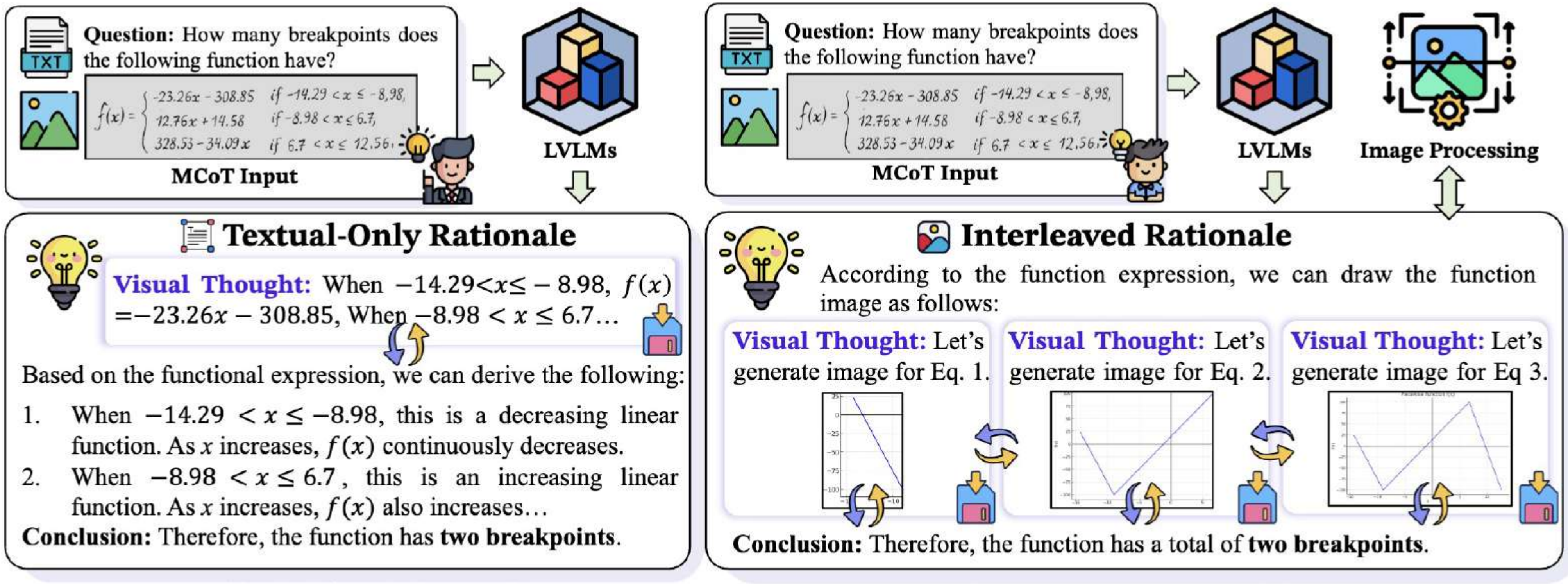
(b) Vanilla Desp-CoT selects an incorrect answer under the text-based reasoning. ViT Desp-CoT selects the correct answer in *Video-Text* interleaved reasoning.

* Zhang et al. ViTCoT: Video-Text Interleaved Chain-of-Thought for Boosting Video Understanding in Large Language Models. ACM MM 2025

多模态长思维链——性能提升的核心机理

多模态思维性能提升的核心机理

所有的多模态思维链能够有效都是因为有效的传递了“视觉思维（Visual Thoughts）”



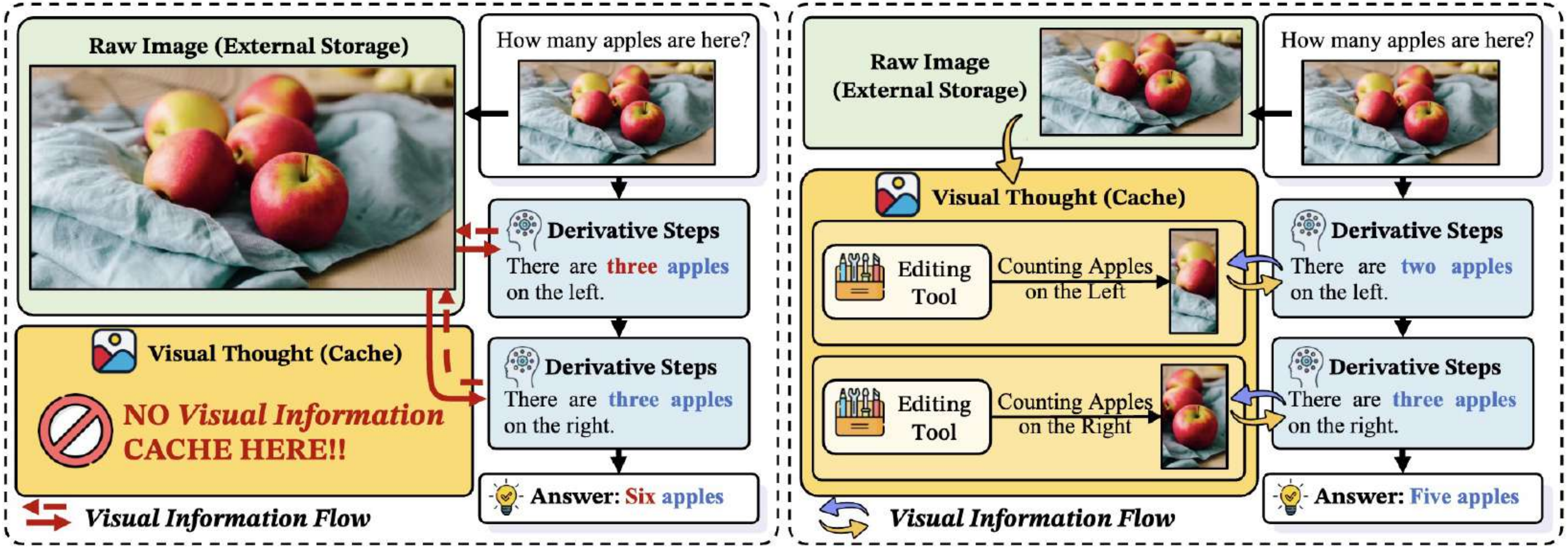
(a) Textual-MCoT (T-MCoT)　　(b) Interleaved-MCoT (I-MCoT)

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——性能提升的核心机理

多模态思维性能提升的核心机理

视觉思维（Visual Thoughts）相当于一个**寄存器**，模型每次进行下一步进行推理时，都**优先**从之前的视觉思维中进行存取



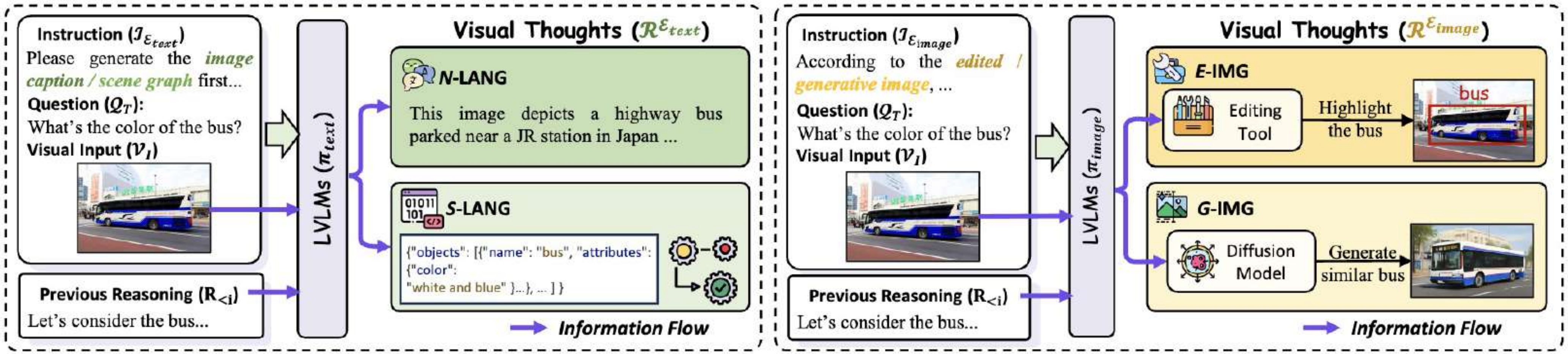
(a) Multimodal Reasoning without visual thoughts (b) Multimodal Reasoning with visual thoughts

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——性能提升的核心机理

详细的视觉思维分类

- **语言形式的视觉思维：** 自然语言表述& 结构化语言表述
- **视觉形式的视觉思维：** 编辑图像表述& 生成图像表述



(a) Visual Thoughts with Textual Expression

(b) Visual Thoughts with Visual Expression

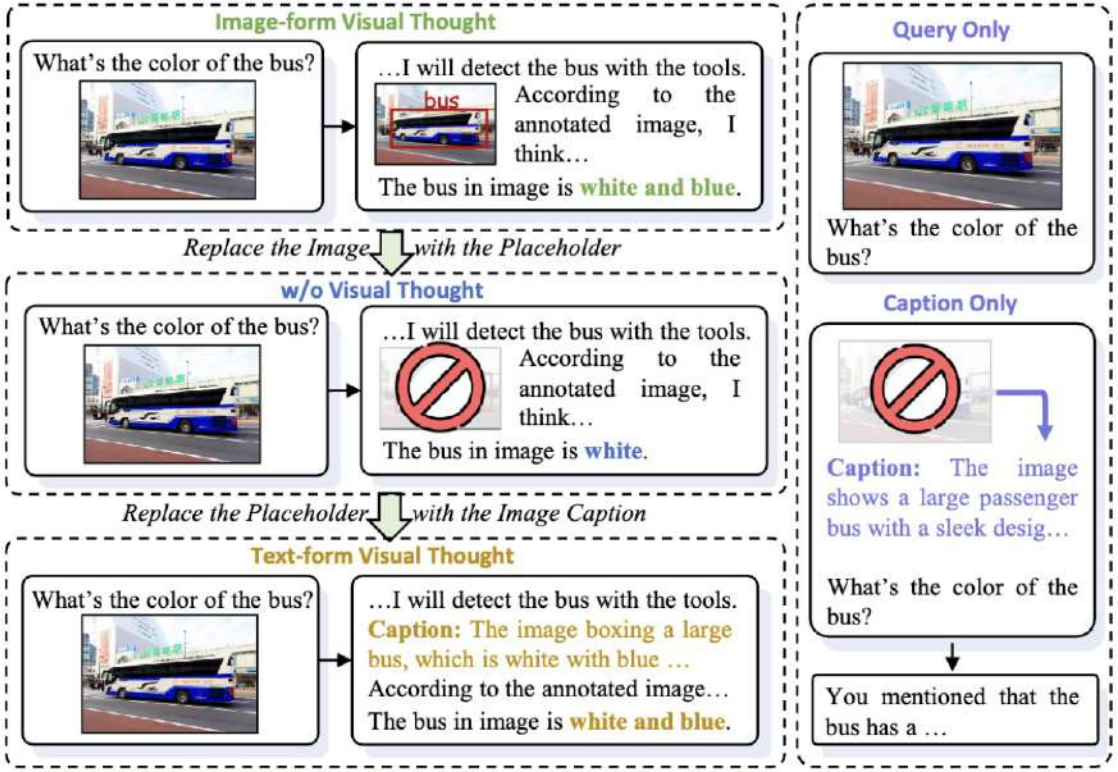
Figure 3: Visual Thoughts in textual expression (a) and visual expression (b). Specifically, the textual expression includes N-LANG and S-LANG, while the visual expression includes E-IMG and G-IMG.

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

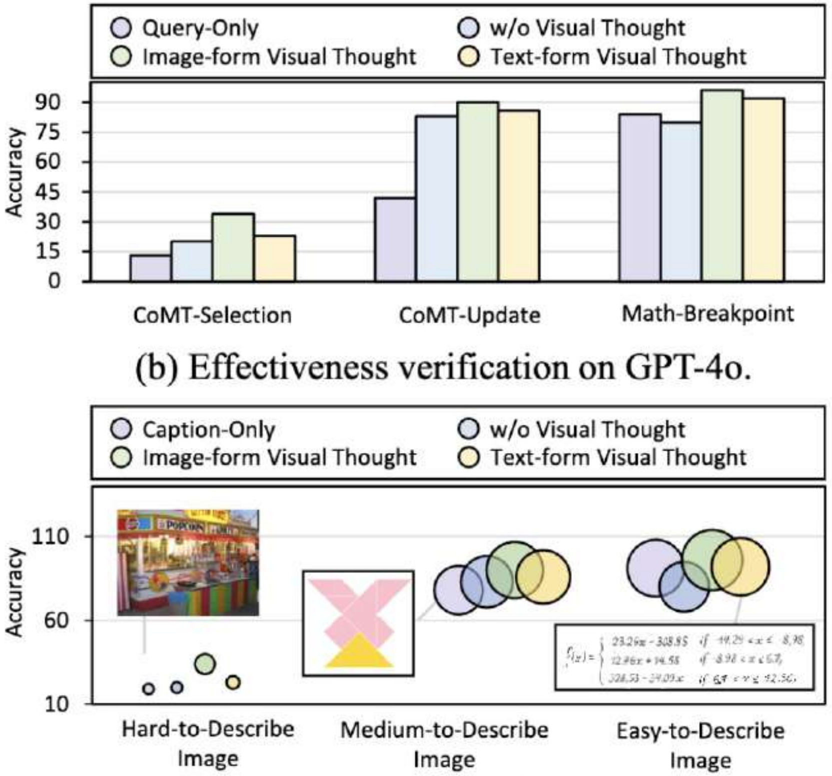
多模态长思维链——性能提升的核心机理

机理合理性验证

- 移除模型的视觉思维，性能甚至会比移除图像的性能还要差
- 重新补充文本视觉思维（原始图像视觉思维的标题），性能会恢复原始多模态思维链的性能



(a) Workflow of effectiveness verification.



(b) Effectiveness verification on GPT-4o.

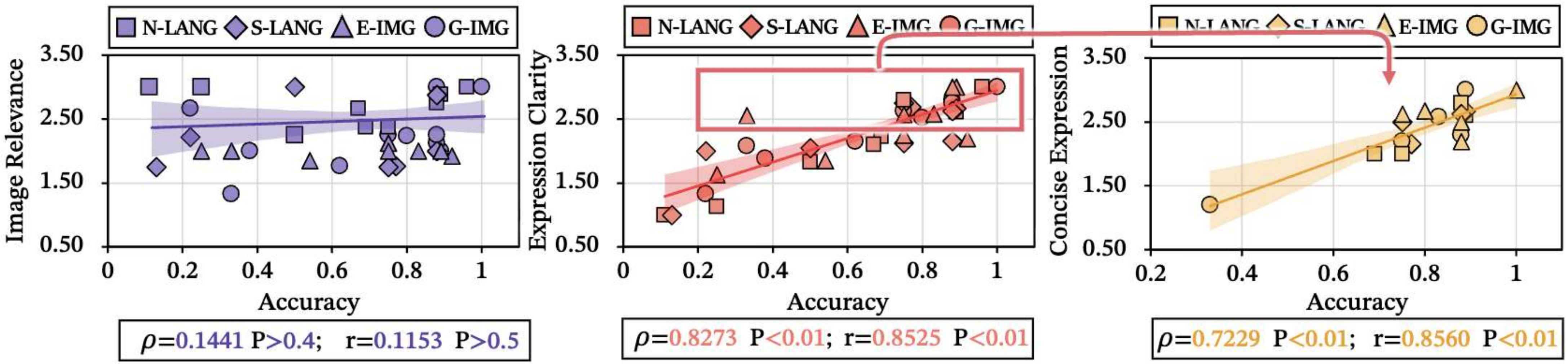
(c) Visual Thoughts v.s. Caption-Only.

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——性能提升的核心机理

究竟是视觉思维的哪些因素影响了模型性能?

- 视觉思维并**不是**原始图像的简单**复制**【多模态思维链性能和视觉思维与原图相关性不大】
- 模型性能取决于视觉思维的表述质量包括表述的**清晰性**和**简洁性**



(a) The correlation between input image relevance of visual thought and accuracy.

(b) The correlation between the clarity of visual thought expression and its accuracy.

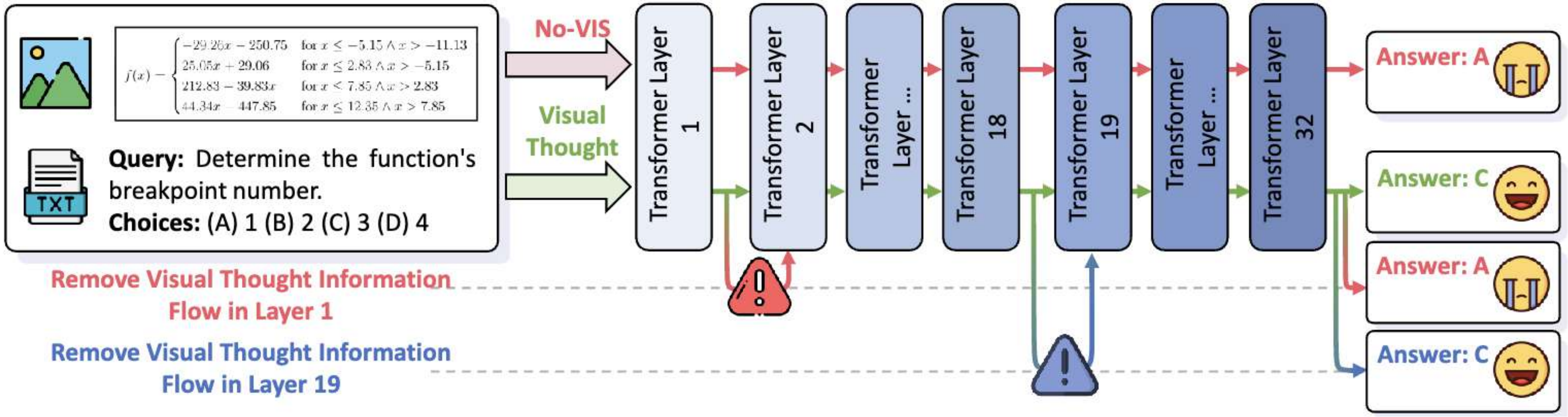
(c) The correlation between the conciseness of visual thought expression and its accuracy.

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——性能提升的核心机理

视觉思维的内部机理

- 移除大模型**底层的输入问题关于视觉思维的Attention**，会让模型生成答案出现显著变化
- **输入问题在模型中通过查询视觉思维寄存器，来获取视觉信息**



* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——性能提升的核心机理

视觉思维的内部机理

- 模型会通过视觉思维将信息流传递到**更深的**模型层中
- 模型会把**原始图像的注意力**转移到**视觉思维的注意力**上，起到了寄存器的作用

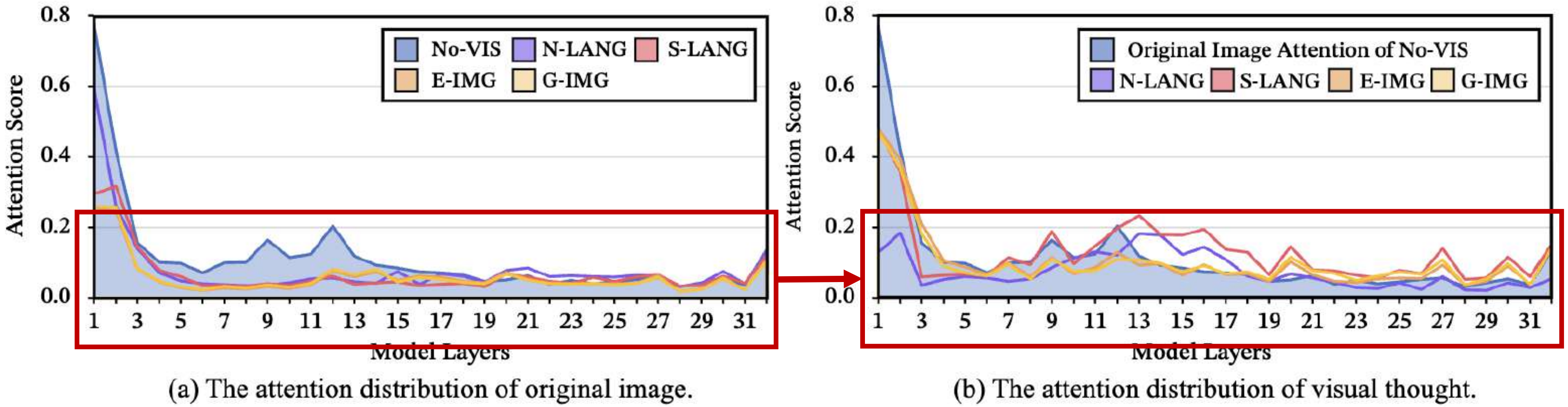


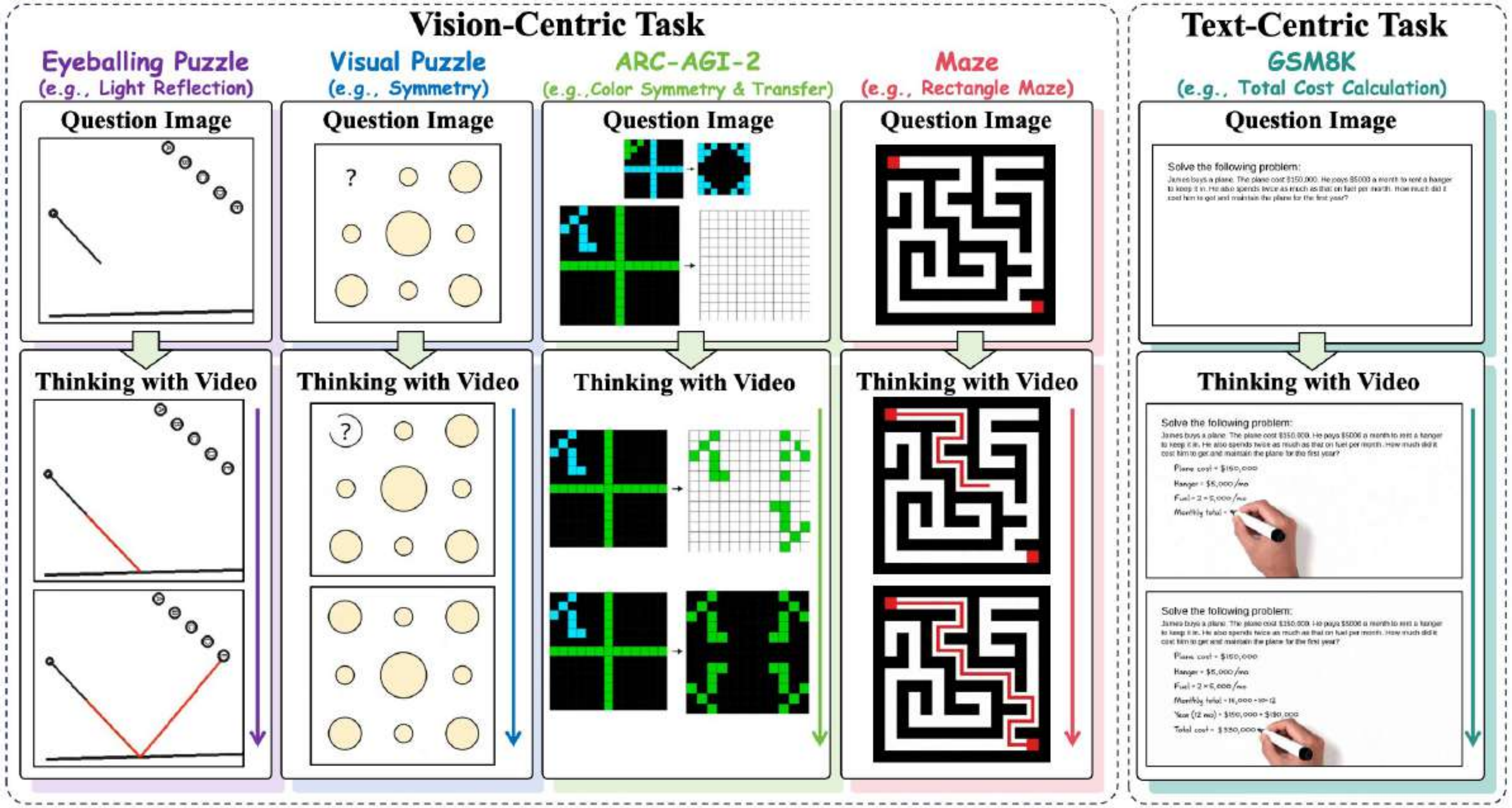
Figure 7: Attention distribution of Visual Thought and Image in MCoT within LLaVA model.

* Cheng et al. Visual Thoughts: A Unified Perspective of Understanding Multimodal Chain-of-Thought. NeurIPS 2025.

多模态长思维链——Video推理的未来

视觉思维的内部机理

- **动态推理优势**：Sora-2可以进行连续推理，如物理模拟，在目测谜题上超过GPT5 10%
- **模态融合前景**：首次发现视频生成模型有潜力解决文本推理问题，进而统一多模态理解与生成



* Tong et al. Thinking with Video: Video Generation as a Promising Multimodal Reasoning Paradigm. Arxiv 2025.

多模态长思维链——Audio推理的未来

畅想一下，Auto推理是什么样的?

长思维链的推理加速

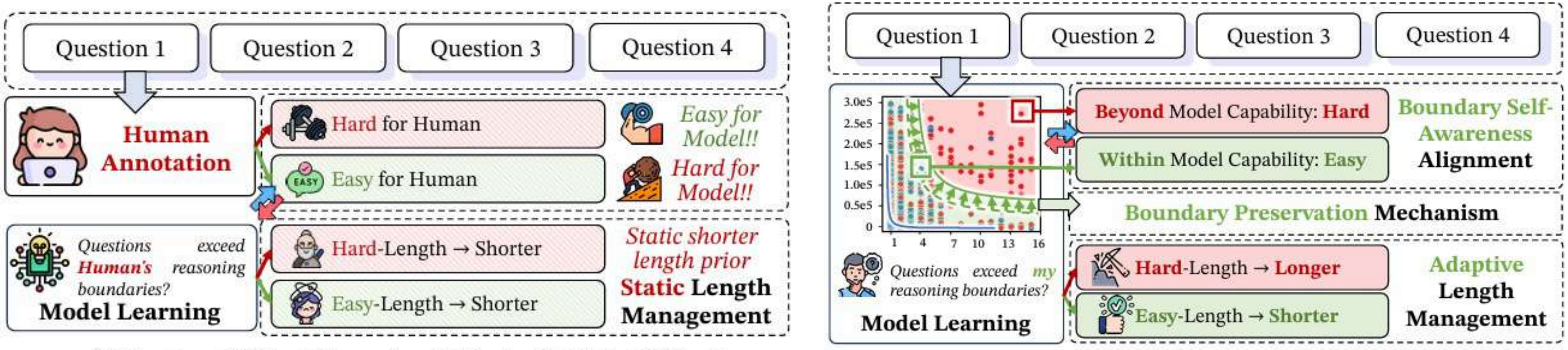
长思维链的推理加速——基于强化学习的思维链动态压缩

静态压缩技术——存在问题

☹ **现有的思维链压缩方法往往需要在训练开始前判定题目的难度，是一种静态压缩**

◆ **忽略了模型自身推理能力的动态变化：** 当前模型的推理压缩往往基于人为规定题目的静态难度，没有与模型动态能力进行对齐（**模型认为的难度 ≠ 人类认为的难度**）

针对上述问题，我们进一步探索在动态难度感知下的思维链压缩



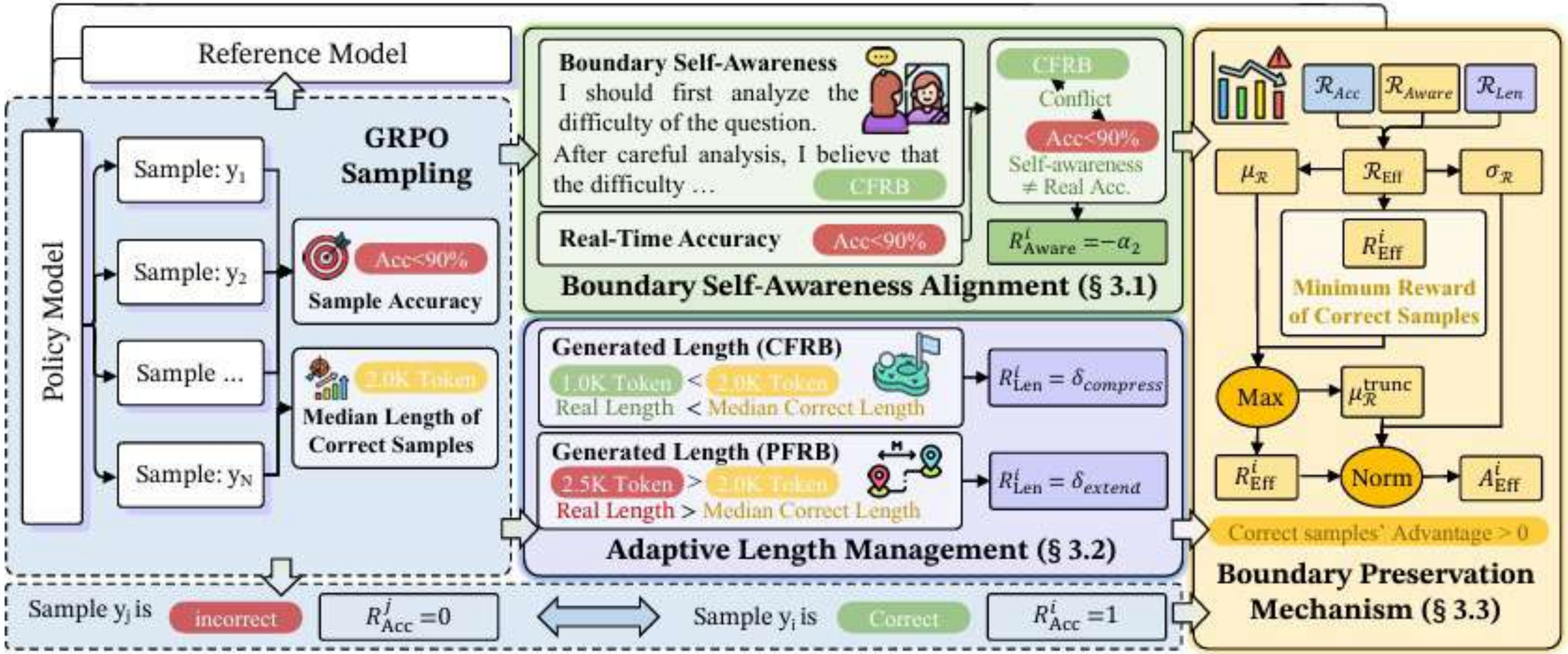
(a) Previous Efficient Reasoning Method with Static Difficulty.　(b) Dynamic Reasoning Boundary Self-Awareness Framework (DR. SAF).

* Chen et al. Aware First, Think Less: Dynamic Boundary Self-Awareness Drives Significant Gains in Reasoning Efficiency in Large Language Models. AAAI 2026.

动态压缩技术

😄 **我们使用模型强化学习中的真实准确率作为客观难度判定，与模型实时能力对齐**

- **根据题目难度进行自我难度感知训练：**要求模型输出的难度判定与客观难度判定对齐
- **根据题目难度进行动态压缩：**对简单问题进行压缩，对复杂问题进行反向思维拓展
- **使用推理边界保持技术保证模型推理能力：**保证GRPO中的正确回答获得的强化优势始终正



* Chen et al. Aware First, Think Less: Dynamic Boundary Self-Awareness Drives Significant Gains in Reasoning Efficiency in Large Language Models. AAAI 2026.

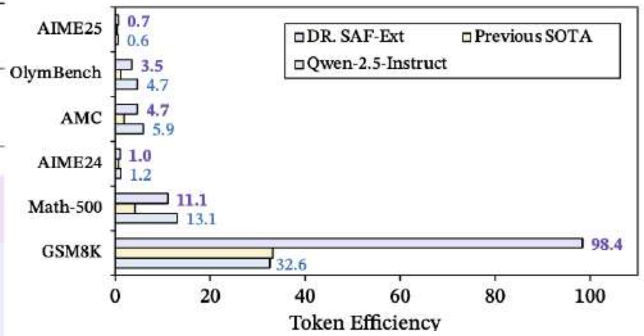
动态压缩技术

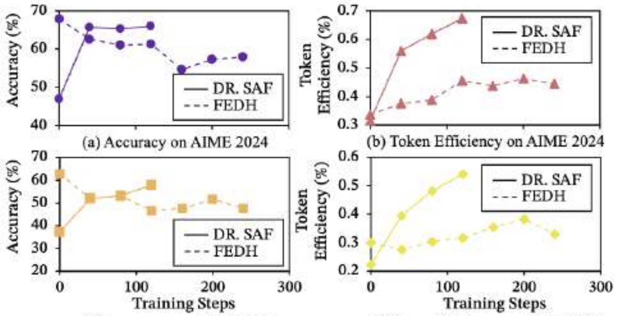
取得效果

- 保证推理能力的同时大幅压缩思维链，所有基准上token效率均取得SOTA水平
- Reasoning模型的极限Token效率甚至可以超过超越Instruct模型
- 在相同token压缩效率下，训练效率提升5-6倍

Model Name	GSM8K			MATH500			AIME24			AMC			OlymBench			AIME25		
	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF
Qwen2.5-7B-Ins	90.9	279	32.58	74.2	567	13.09	12.0	1016	1.18	47.5	801	5.93	39.2	827	4.74	7.6	1240	0.61
Qwen2.5-7B-Math	93.2	439	21.23	63.4	740	8.57	19.0	1429	1.33	62.5	1022	6.12	31.5	1037	3.04	4.0	2562	0.16
Qwen2.5-7B-Math-Ins	95.2	323	29.47	81.4	670	12.15	10.3	1363	0.76	60.0	1029	5.83	38.9	1027	3.79	9.3	2087	0.45
R1-Distill-Qwen2.5-7B	92.4	1833	5.04	90.8	3854	2.36	49.2	10200	0.48	90.0	6476	1.39	66.1	7789	0.85	35.0	10518	0.33
+ ThinkSwitcher	**92.5**	1389	6.66	91.3	3495	2.61	48.3	7936	0.61	–	–	–	57.0	5147	1.11	37.5	6955	0.54
+ Dynasor-CoT	89.6	1285	6.97	89.4	2661	3.36	46.7	12695	0.37	85.0	5980	1.42	–	–	–	–	–	–
+ DEER	90.6	917	9.88	89.8	2143	4.19	49.2	9839	0.50	85.0	4451	1.91	–	–	–	–	–	–
+ OverThink	91.4	879	10.39	**92.9**	2405	3.86	50.0	9603	0.52	–	–	–	–	–	–	–	–	–
+ Spirit	87.2	687	12.68	90.8	1765	5.14	38.3	6926	0.55	–	–	–	–	–	–	–	–	–
+ ConCISE-SimPO	92.1	715	12.88	91.0	1945	4.68	48.3	7745	0.62	–	–	–	–	–	–	–	–	–
+ DAST	86.7	459	18.89	89.6	2162	4.14	45.6	7578	0.60	–	–	–	–	–	–	–	–	–
+ AdaptThink	91.0	309	29.45	92.0	1875	4.91	**55.6**	8599	0.65	85.0*	4265*	1.99*	58.4*	5988*	0.98*	**38.3***	10380*	0.37*
+ Length-Penalty	87.2	263	33.16	89.1	2121	4.20	51.9	7464	0.70	82.5	4411	1.87	**59.8**	4919	1.22	33.3	8902	0.37
+ FEDH	90.1	218	41.33	88.5	1306	6.50	42.3	7242	0.58	–	–	–	–	–	–	–	–	–
+ DR. SAF	88.1	**162**	**54.38**	88.3	**1061**	**8.32**	50.6	**6288**	**0.80**	**90.0**	**3096**	**2.91**	59.4	**3259**	**1.82**	38.2	**6764**	**0.56**
R1-Distill-Qwen3-8B	94.2	2135	4.41	90.6	7051	1.28	**67.9**	20155	0.34	83.5	11931	0.70	60.1	12895	0.47	**62.9**	20992	0.30
+ FEDH*	**94.4**	2014	4.69	92.6	6761	1.37	61.3	13463	0.42	94.7	11928	0.79	63.5	12353	0.51	46.7	14730	0.32
+ Length-Penalty*	93.3	604	15.45	92.4	2581	3.58	63.7	12303	0.52	89.2	6166	1.45	68.4	7383	0.93	54.7	12446	0.44
+ DR. SAF	92.3	**521**	**17.72**	**93.3**	**2168**	4.30	66.0	**9807**	**0.67**	**95.6**	**4003**	**2.39**	**71.3**	**5766**	**1.24**	57.9	**10692**	**0.54**

Model	GSM8K			MATH500			AIME24			AMC			OlymBench			AIME25		
	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF	ACC	LEN	EFF
DR. SAF ext	86.6	88	98.41	84.1	759	11.08	41.3	4,002	1.03	75.0	1,581	4.74	55.10	1,576	3.50	26.2	3,998	0.66





(a) Accuracy on AIME 2024
(b) Token Efficiency on AIME 2024
(c) Accuracy on AIME 2025
(d) Token Efficiency on AIME 2025

* Chen et al. Aware First, Think Less: Dynamic Boundary Self-Awareness Drives Significant Gains in Reasoning Efficiency in Large Language Models. AAAI 2026.

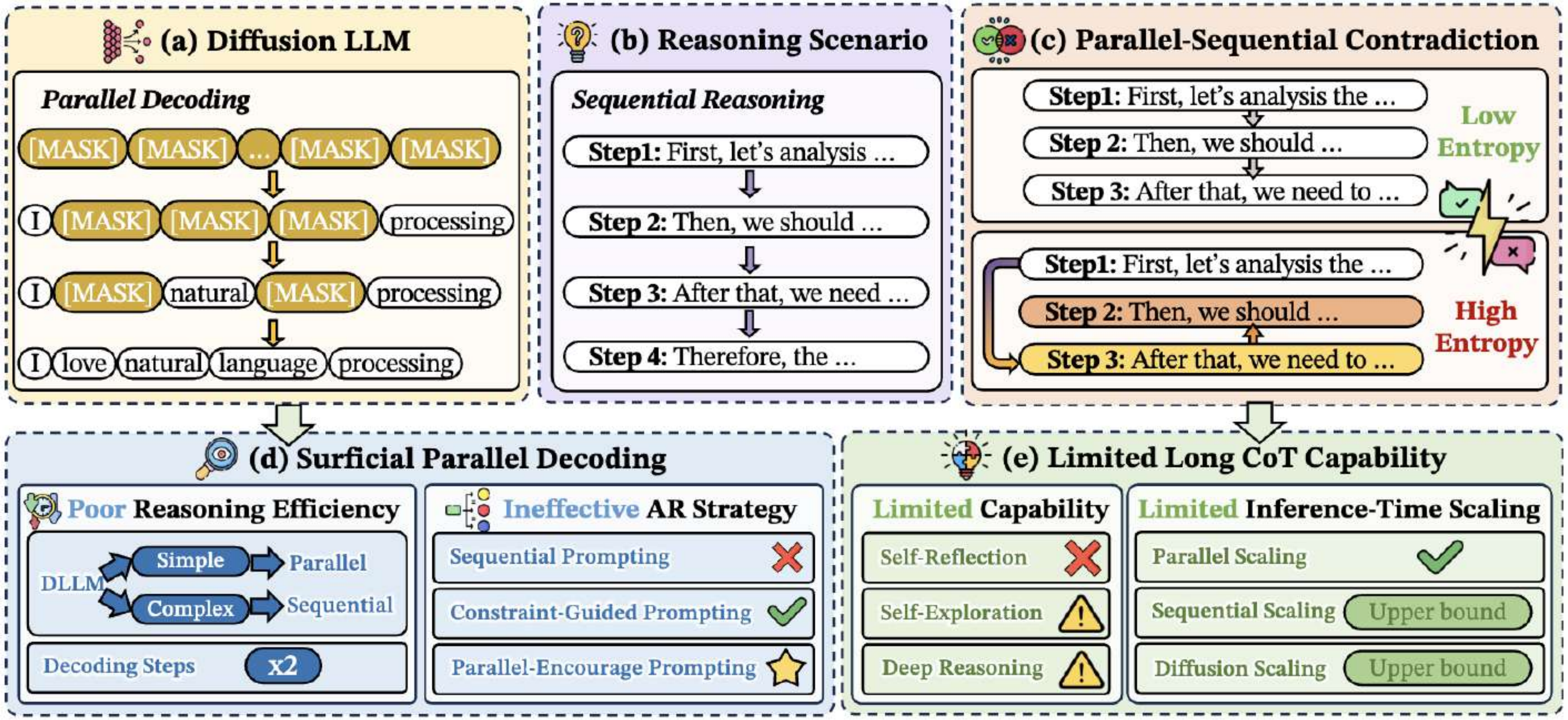
长思维链的推理加速——
基于扩散语言模型的并行推理

扩散语言模型的并行推理——概述

扩散语言模型的并行推理探索

- 扩散语言模型是并行生成的
- 推理步骤是串行依赖的

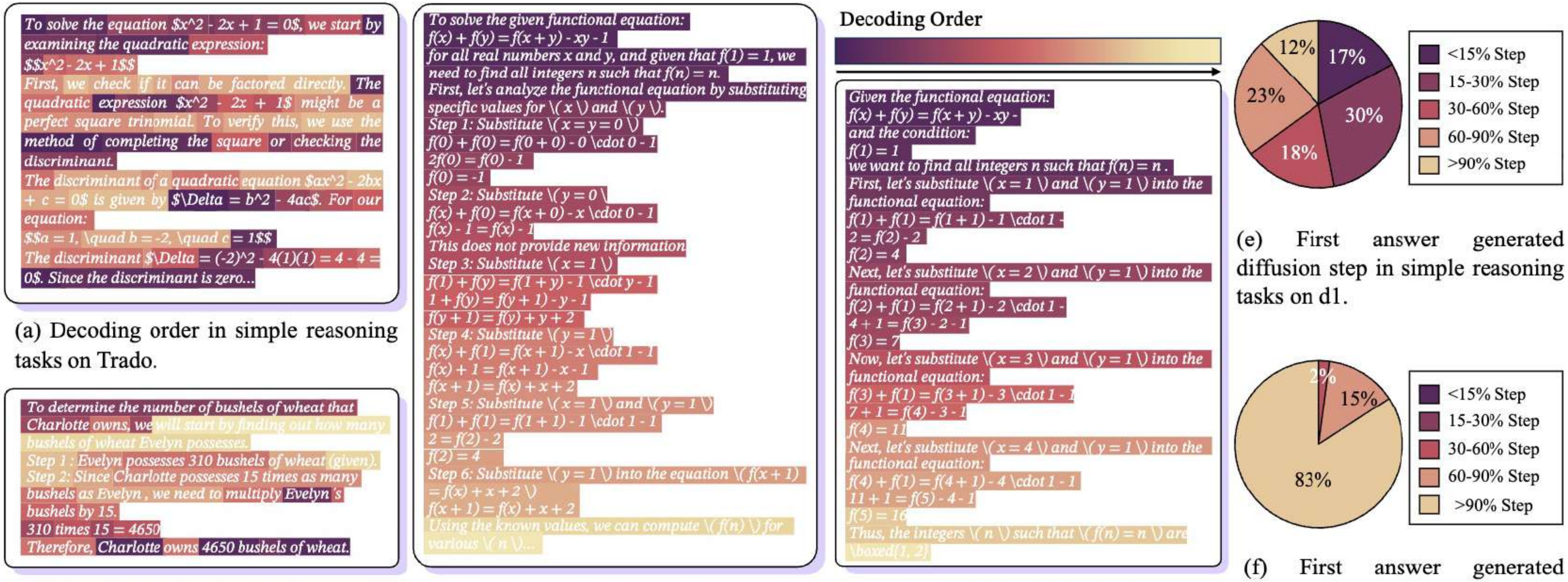
存在“并行-串行矛盾”，基于Step 1，并行先生成Step 3比串行生成Step 2的熵要高得多



* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

扩散语言模型的并行推理——矛盾导致"虚假"并行推理

- 对于可以直接解答的问题，是**在装模作样的并行推理**
- 对于无法直接回答的问题，会**退化为自回归推理**



(a) Decoding order in simple reasoning tasks on Trado.

(b) Decoding order in simple reasoning tasks on d1.

(c) Decoding order in complex reasoning tasks on Trado.

(d) Decoding order in complex reasoning tasks on d1.

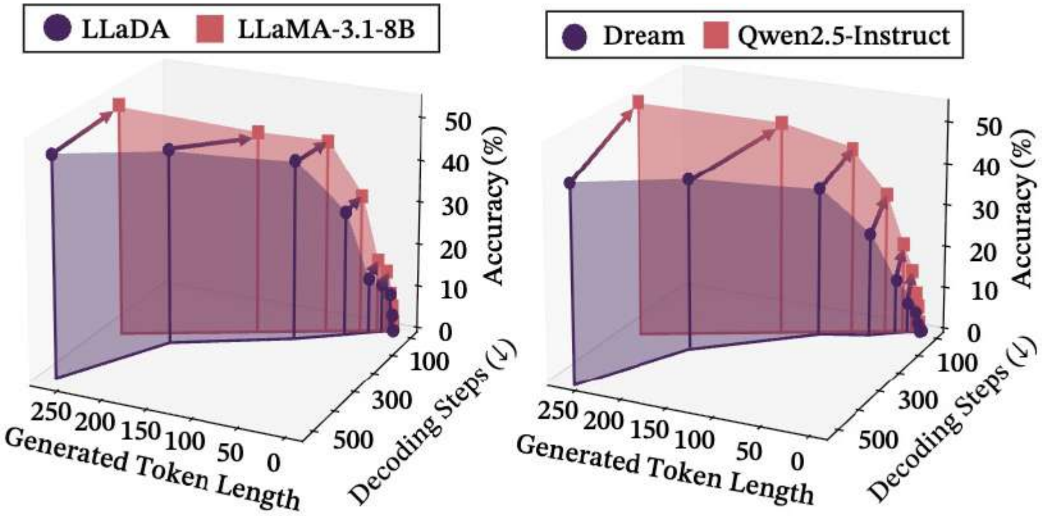
(e) First answer generated diffusion step in simple reasoning tasks on d1.

(f) First answer generated diffusion step in complex reasoning tasks on d1.

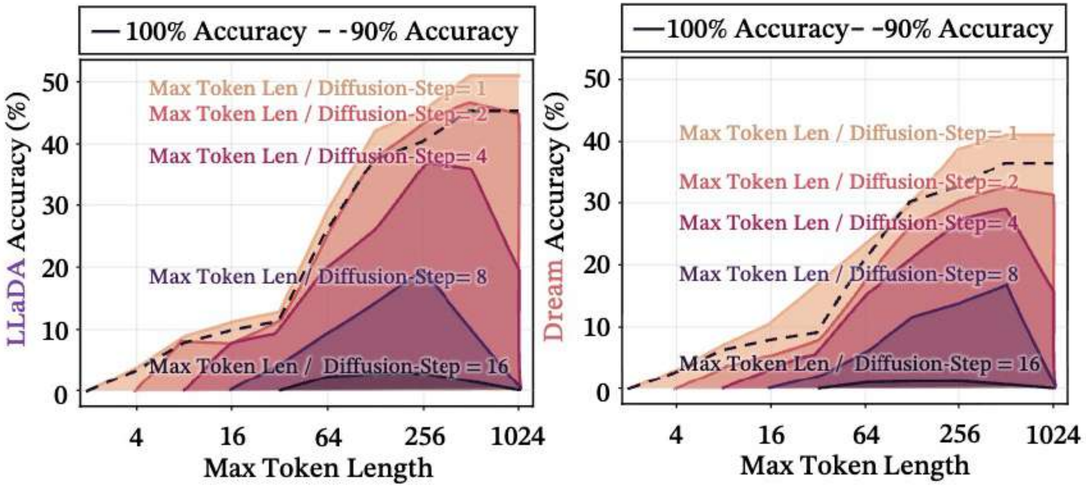
* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

扩散语言模型的并行推理——矛盾导致效率问题

- 如果要和自回归模型性能接近，则**需要双倍的解码步骤**（带Remask）
- 推理大部分**Token需要充分的diffusion**，每个token至少一次，否则性能下降10%以上



(a) The impact of diffusion/decoding steps and generated token length.



(b) Impact of the mismatch between diffusion steps and the multiple of max token length on model performance.

* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

扩散语言模型的并行推理——矛盾带来的影响

- 矛盾存在会，导致**序列型提示失效**，甚至会加重“并行-串行矛盾”
- **条件型提示**由于并行进行，与“并行-串行矛盾”并行不悖，**仍然有效**
- **鼓励并行推理会更有效**，缓解了“并行-串行矛盾”

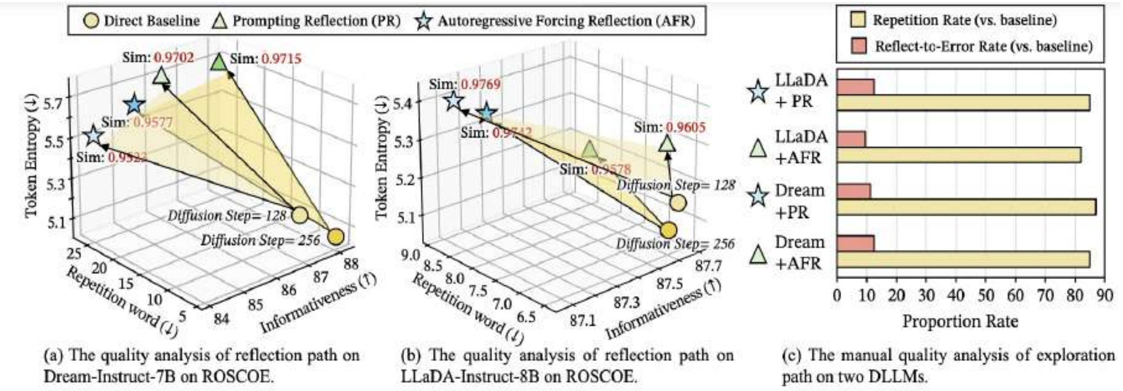
Model Name	BigGSM (Acc.)	GSM8K (Acc.)	Math-500 (Acc.)	HumanEval (Pass@1)	Average
Dream-7B-Instruct	41.15 (+0.00)	80.52 (+0.00)	37.00 (+0.00)	51.22 (+0.00)	52.47 (+0.00)
+Zero-CoT	41.15 (−0.00)	77.26 (−3.26)	34.00 (−3.00)	48.78 (−2.44)	50.30 (−2.18)
+Plan-and-Solve	34.75 (−6.40)	78.85 (−1.67)	20.20 (−16.80)	48.78 (−2.44)	45.65 (−6.83)
+Least-to-Most	35.25 (−5.90)	77.03 (−3.49)	16.20 (−20.80)	43.29 (−7.93)	42.94 (−9.53)
+Complex-CoT	41.80 (+0.65)	80.95 (+0.43)	37.40 (+0.40)	**52.44** (+1.22)	53.15 (+0.68)
+MARP	42.95 (+1.80)	80.52 (+0.00)	37.20 (+0.20)	51.22 (+0.00)	52.97 (+0.50)
+Diff-MARP	**47.21** (+6.06)	**82.64** (+2.21)	**43.60** (+6.60)	**52.44** (+1.22)	**56.47** (+4.00)
LLaDA-8B-Instruct	48.03 (+0.00)	75.36 (+0.00)	34.80 (+0.00)	32.32 (+0.00)	47.63 (+0.00)
+ Zero-CoT	35.57 (−12.46)	73.46 (−1.90)	32.40 (−2.40)	28.05 (−4.27)	42.37 (−5.26)
+ Plan-and-Solve	31.64 (−16.39)	72.33 (−3.03)	29.00 (−5.80)	27.44 (−4.88)	40.10 (−7.53)
+ Least-to-Most	34.75 (−13.28)	73.31 (−2.05)	30.80 (−4.00)	27.44 (−4.88)	41.58 (−6.05)
+ Complex-CoT	48.03 (+0.00)	76.50 (+1.14)	36.20 (+1.40)	36.59 (−4.27)	49.33 (+1.70)
+ MARP	48.20 (+0.17)	76.35 (+0.99)	34.40 (−0.40)	35.37 (−3.05)	48.58 (+0.95)
+Diff-MARP	**55.74** (+7.71)	**76.80** (+1.44)	**38.20** (+3.40)	**38.41** (+6.09)	**52.29** (+4.66)
LLaDA-v1.5	41.80 (+0.00)	74.98 (+0.00)	38.00 (+0.00)	36.59 (+0.00)	47.84 (+0.00)
+ Zero-CoT	36.39 (−5.41)	71.87 (−3.11)	37.20 (−0.80)	35.98 (−0.61)	45.36 (−2.48)
+ Plan-and-Solve	30.16 (−11.54)	74.37 (−0.61)	34.40 (−3.60)	35.98 (−0.61)	43.73 (−4.12)
+ Least-to-Most	35.90 (−5.90)	73.69 (−1.29)	34.60 (−3.40)	31.71 (−4.88)	43.98 (−3.87)
+ Complex-CoT	50.16 (+8.41)	75.51 (+0.53)	**39.40** (+1.40)	**39.02** (+2.43)	51.04 (+3.19)
+ MARP	42.13 (+0.33)	74.37 (0.61)	38.20 (+0.20)	37.20 (+0.61)	47.98 (+0.13)
+Diff-MARP	**54.49** (+12.79)	**76.50** (+1.52)	42.80 (+4.80)	38.41 (+1.82)	**53.08** (+5.23)
LLaDOU-Math	42.13 (+0.00)	81.88 (+0.00)	45.80 (+0.00)	39.02 (+0.00)	52.21 (+0.00)
+Zero-CoT	38.52(−3.61)	80.95(−0.93)	45.80(−0.00)	37.80(−1.22)	50.77(−1.44)
+Plan-and-Solve	40.82(−1.31)	81.12(−0.76)	43.20(−2.60)	38.41(−0.61)	50.89(−1.32)
+Least-to-Most	40.16(−1.97)	79.08(−2.80)	43.00(−2.80)	36.59(−2.43)	49.71(−2.50)
+ Complex-CoT	43.77(+1.64)	83.70(+1.82)	45.80(+0.00)	**42.07** (+3.05)	52.47(+0.26)
+ MARP	41.15(−0.98)	82.18(+0.30)	45.60(−0.20)	40.26(+1.24)	52.30(+0.09)
+Diff-MARP	**54.26** (+12.13)	**84.76** (+2.88)	**49.00** (+3.20)	40.85(+1.83)	**57.22** (+5.01)

* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

扩散语言模型的并行推理——矛盾带来的影响

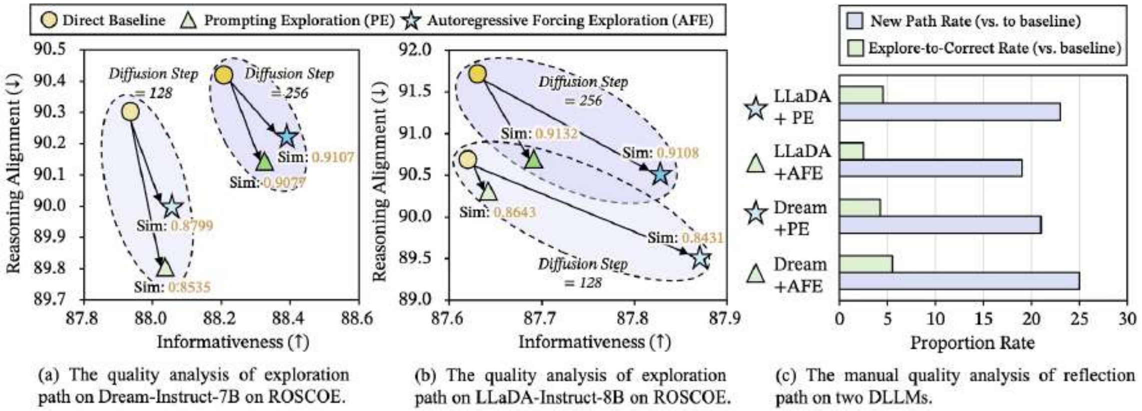
- **自我反思会失效**，提示型反思（不可并行）效果最差
- **自我探索效果一般**，提示型探索（可并行）效果相对效果较好
- **深度推理边界会大幅度下降**，甚至完全可行推理边界消失了

a



(a) The quality analysis of reflection path on Dream-Instruct-7B on ROSCOE.
(b) The quality analysis of reflection path on LLaDA-Instruct-8B on ROSCOE.
(c) The manual quality analysis of exploration path on two DLLMs.

b



(a) The quality analysis of exploration path on Dream-Instruct-7B on ROSCOE.
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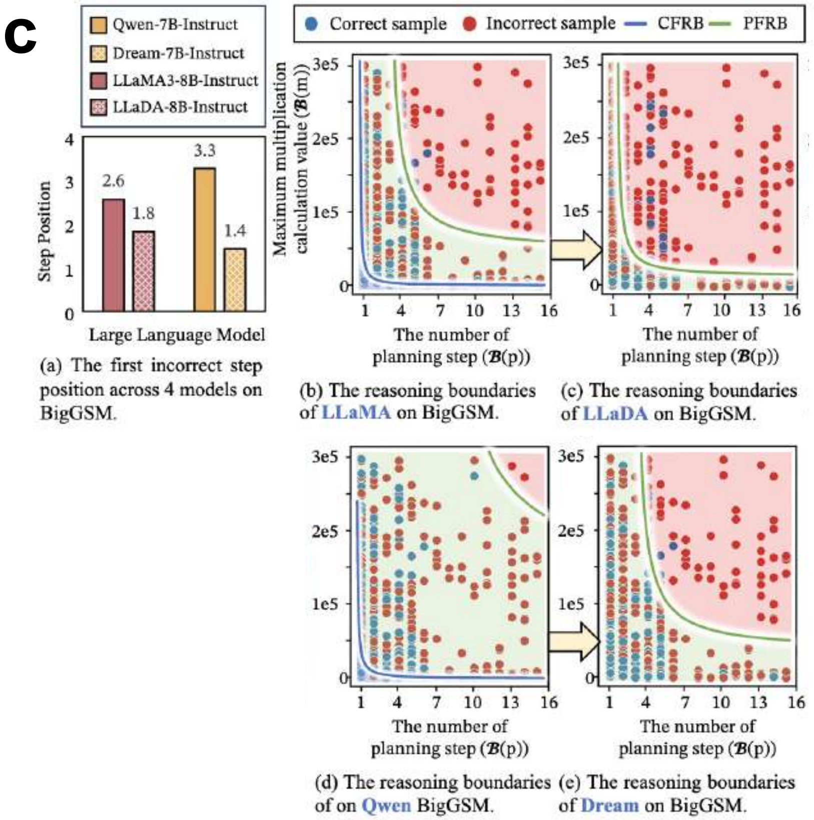
(c) The manual quality analysis of reflection path on two DLLMs.

c

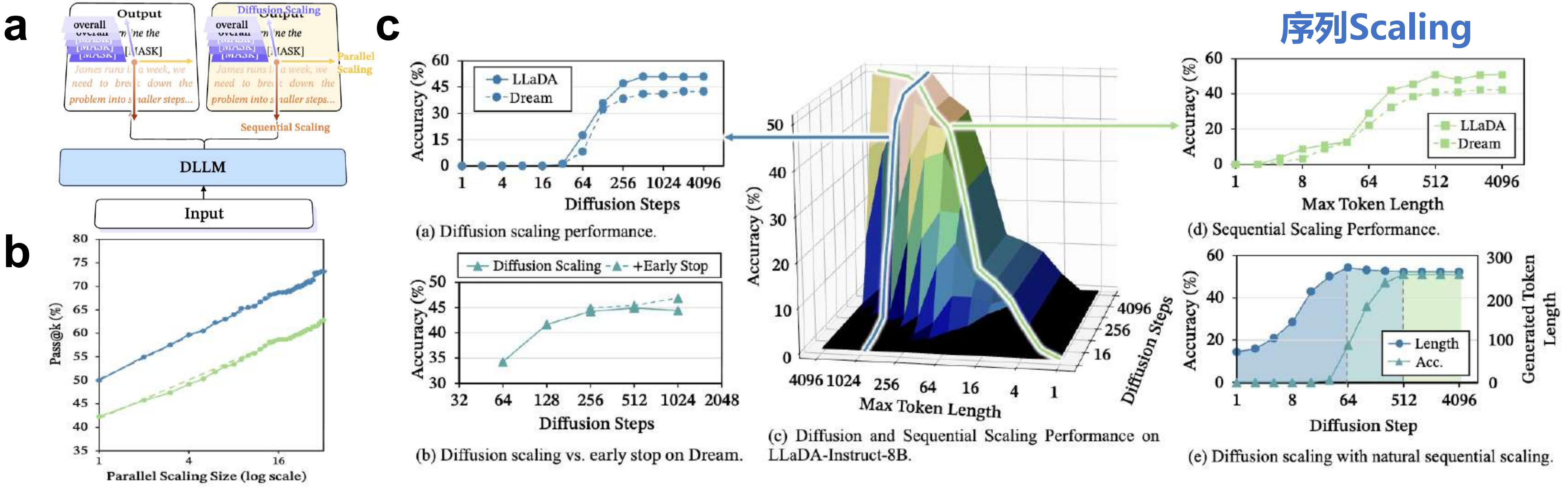


(a) The first incorrect step position across 4 models on BigGSM.
(b) The reasoning boundaries of LLaMA on BigGSM.
(c) The reasoning boundaries of LLaDA on BigGSM.
(d) The reasoning boundaries of Qwen on BigGSM.
(e) The reasoning boundaries of Dream on BigGSM.

* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

扩散语言模型的并行推理——矛盾限制了Scaling

- **并行Scaling**由于与“并行串行矛盾”正交，所以**仍然有效**
- **序列Scaling**由于与“并行串行矛盾”相关，效果会**存在上界**
- **扩散Scaling**由于与“并行串行矛盾”相关，效果也**有上界**
- 扩散Scaling会自然的进行序列扩展



并行Scaling　　扩散Scaling　　扩散Scaling-序列Scaling

* Chen et al. Beyond Surface Reasoning: Unveiling the True Long Chain-of-Thought Capacity of Diffusion Large Language Models. Arxiv 2025.

长思维链的科学应用

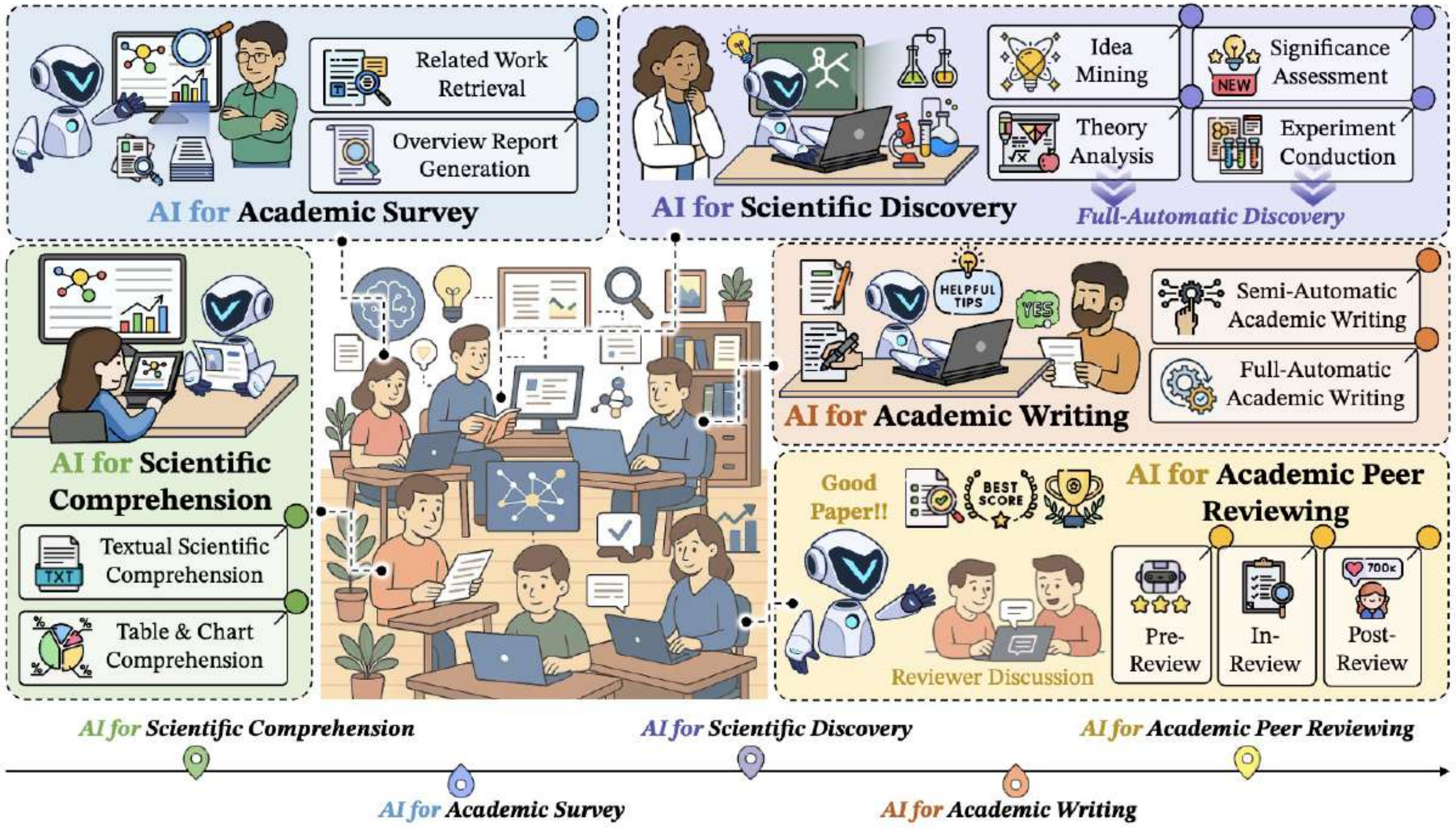
长思维链的科学应用

AI系统如何帮助与加速科学研究?

静态压缩技术——存在问题

🎯随着智能Agent系统越来越多地应用到科研过程中，开启了“AI创新” 新时代

✅ 创新、规划能力远超从前　✅ 思考探索深度远超从前　✅ 全链路深度参与

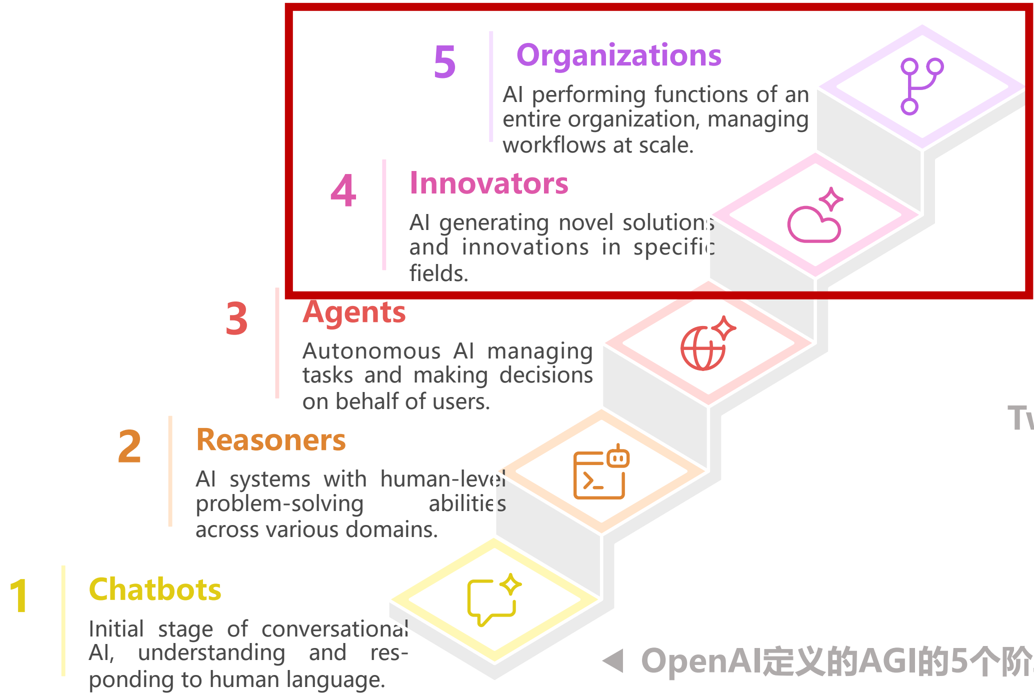


* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

动态压缩技术

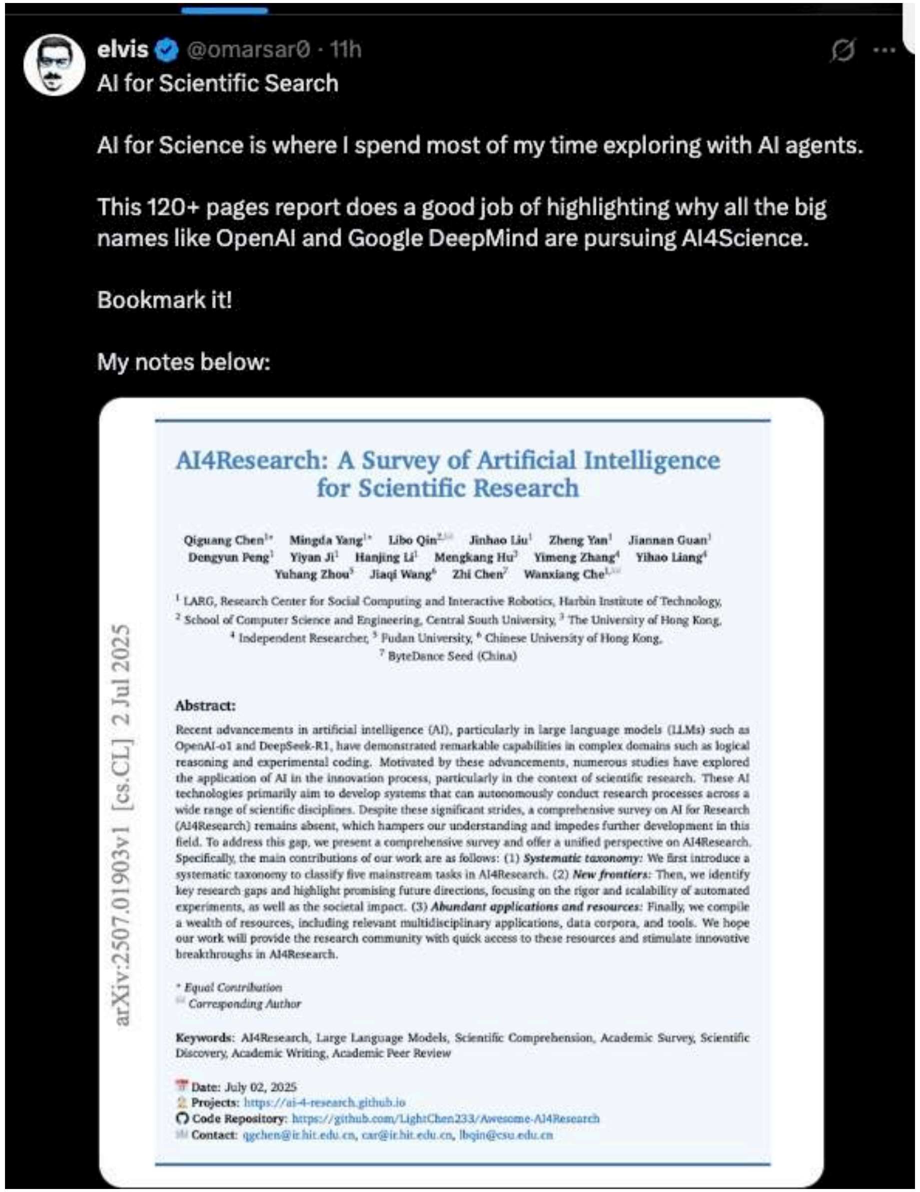
🎯 到达OpenAI定义的第4-5世代

- ✅ 让模型**创新**性的研究世界
- ✅ 让模型自发的**组织**研究流程



◀ OpenAI定义的AGI的5个阶段

Twitter大V的介绍 ▶



* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

动态压缩技术

🎯AI4Research和AI4Science有何区别?

- ✅ **覆盖面更大：** AI4Research涵盖更全面的研究流程，而AI4Science侧重科学发现/数据分析过程
- ✅ **目标更广：** AI4Research追求论文发表、科学发现、方法设计实施，而AI4Science更追求科学突破
- ✅ **应用目标不同：** AI4Research应用在更广的各个科研流程中，而AI4Science更应用在各个学科上
- ✅ **目标用户不同：** AI4Research目标为更广泛的科研人群，而AI4Science更依赖学科专家的运用

	AI4Science	**AI4Research**
Scope	🔬 Scientific Discovery, 📊 Data Analysis.	📚 Broader Research workflows.
Goal	🧬 Scientific Breakthroughs.	📝 Publications, 🔧 Methods, 📈 Overall Productivity.
Applications	🧪 Material Discovery, 💊 Drug Design, 🧬 Genomics, *etc.*	📖 Comprehension, Writing, 🧐 Peer Review, *etc.*
Target Users	🥇 Research Experts.	Both 🥇 Research Experts and 🥈 New Scientists.

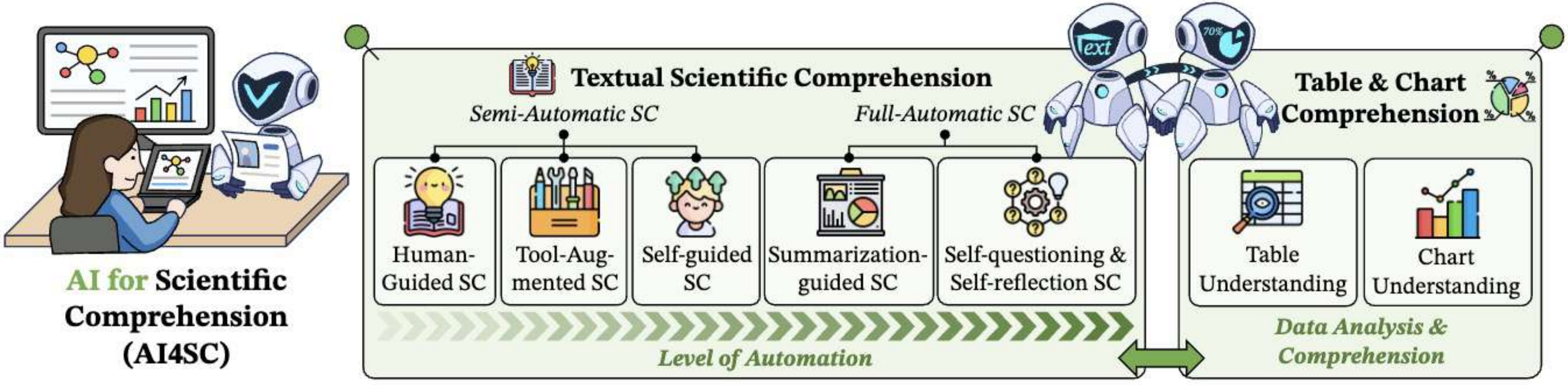
Table 1: Comparison and discussion between AI4Science and AI4Research, especially in terms of scope, goal, applications, and target users.

* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI for Scientific Comprehension

AI for Scientific Comprehension

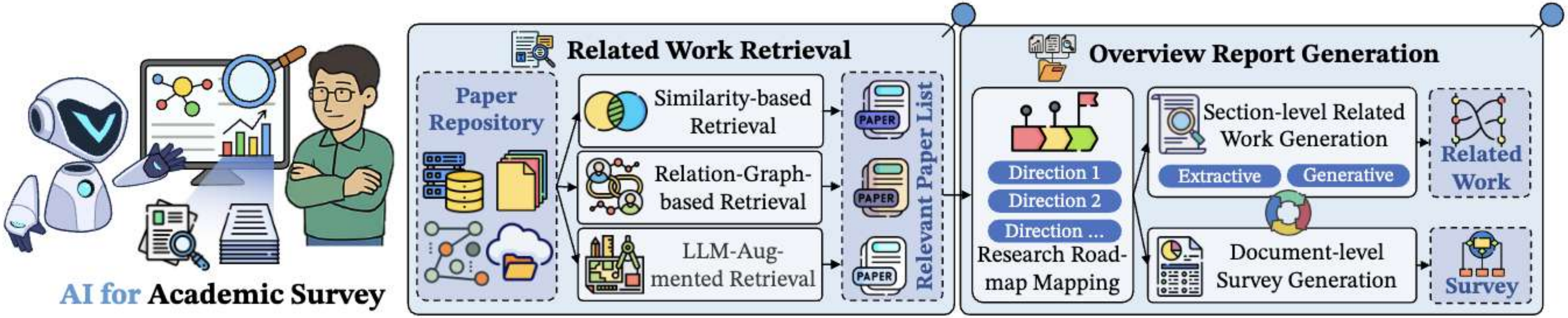
- **科学文本理解：** 1 半自动理解（需要人类+工具驱动） 2 全自动理解（摘要+自我提问）
- **科学图表理解：** 1 表格理解（数据增强/推理格式增强） 2 Chart理解（主要就是数据合成）



* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI for Academic Survey

- ◆ **相关文献检索：** 1 语义相似度检索 2 关系图检索 3 大模型增强的检索
- ◆ **相关报告生成：** 1 研究路线梳理 2 段落级相关工作（抽取式+生成式） 3 生成文章级综述生成

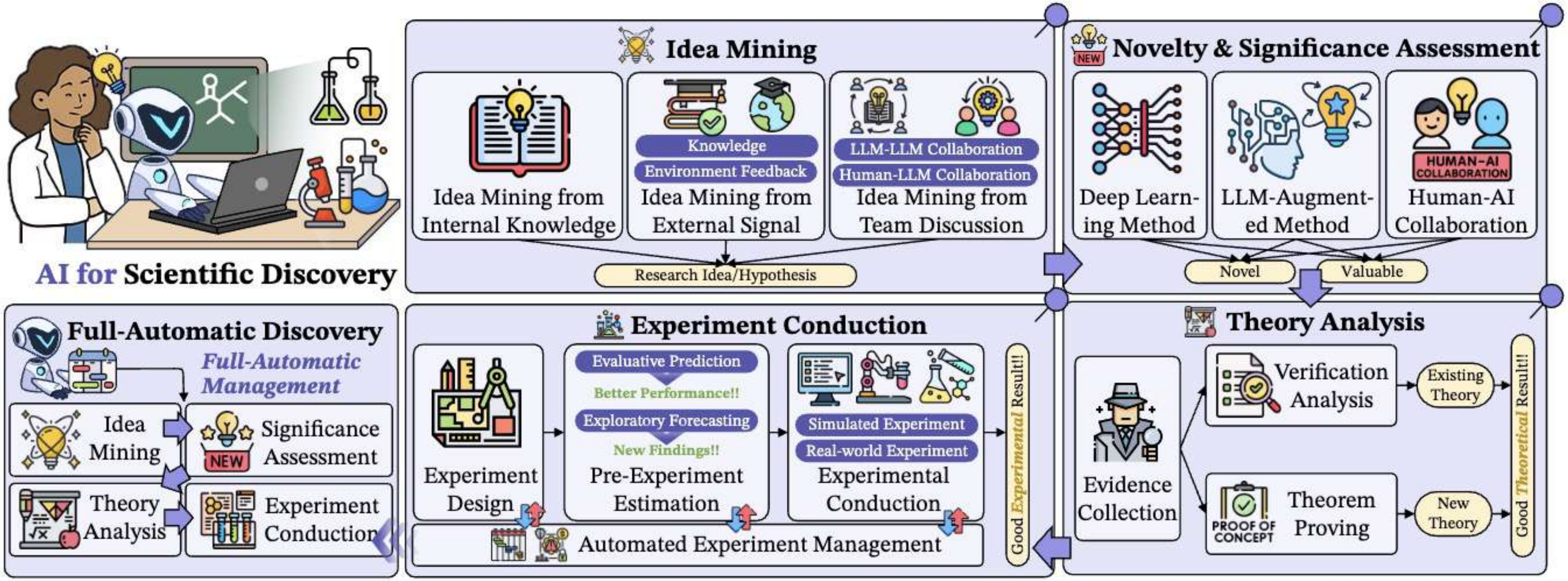


* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI for Scientific Discovery

AI for Scientific Discovery

- ✅ Idea挖掘
- ✅ 研究创新性/重要性评估
- ✅ 理论分析
- ✅ 实验实施
- ✅ 全自动科学发现

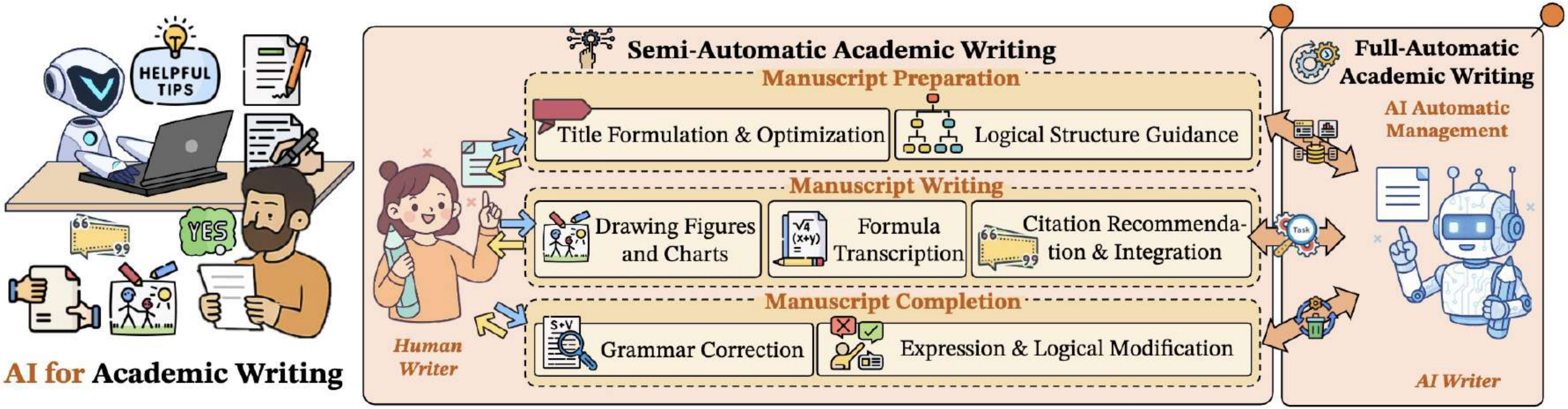


* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI for Academic Writing

AI for Academic Writing

✅ 手稿准备阶段辅助 ✅ 手稿写作阶段辅助 ✅ 手稿完成后辅助 ✅ 全自动写作

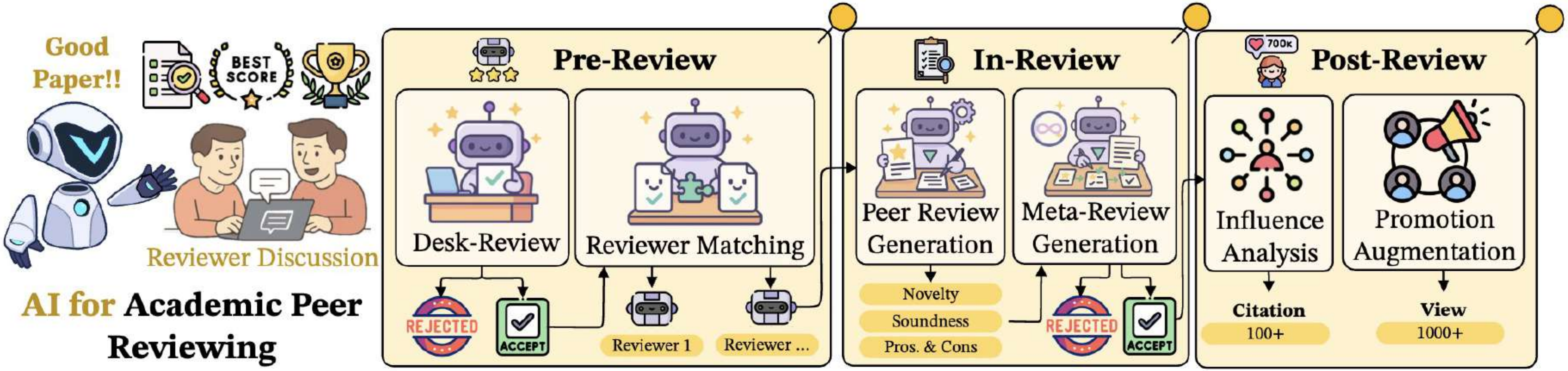


* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI for Academic Peer Reviewing

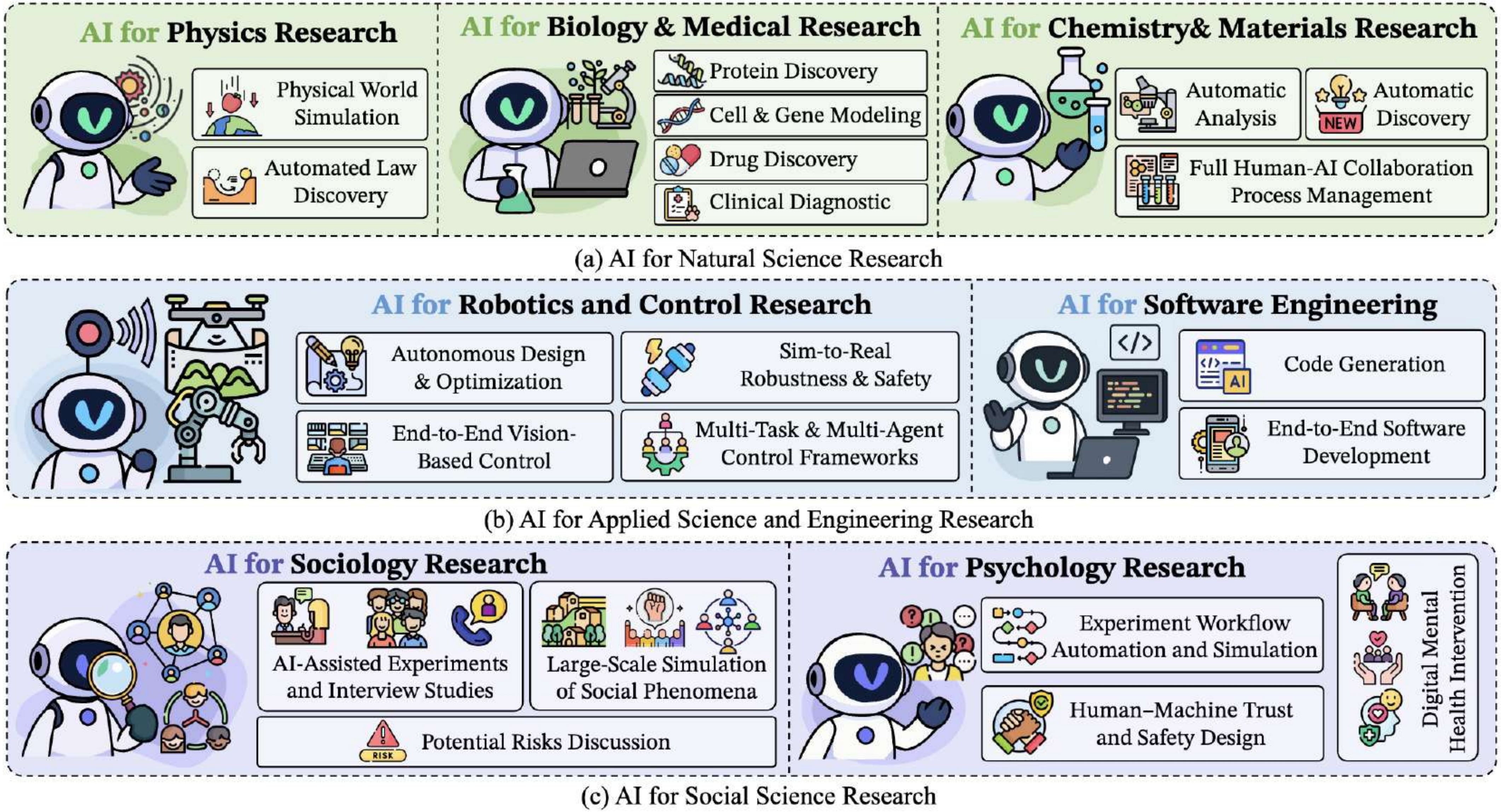
AI for Academic Peer Reviewing

- 预审稿阶段的AI系统
- 正式审稿的AI系统
- 后同行评议阶段的AI系统



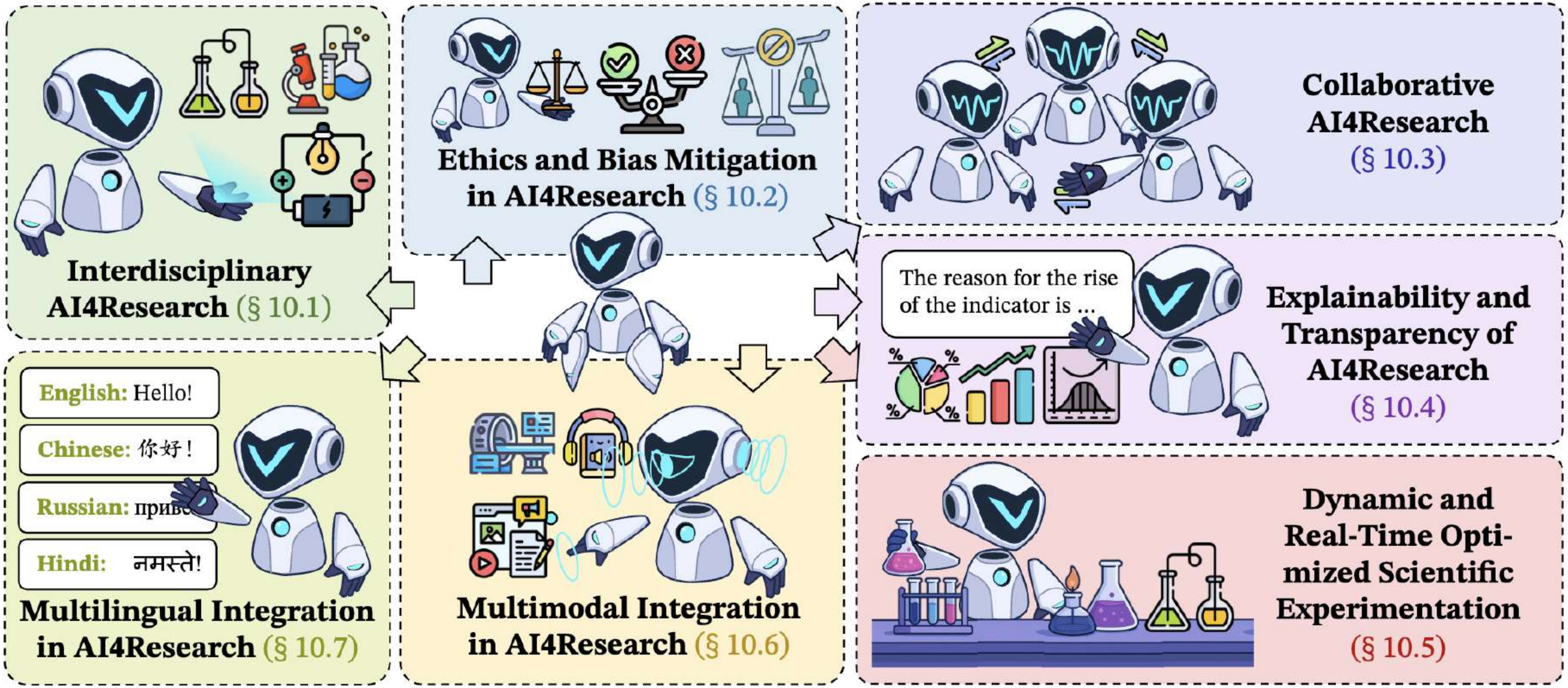
* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

AI4Research的应用



* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

未来工作



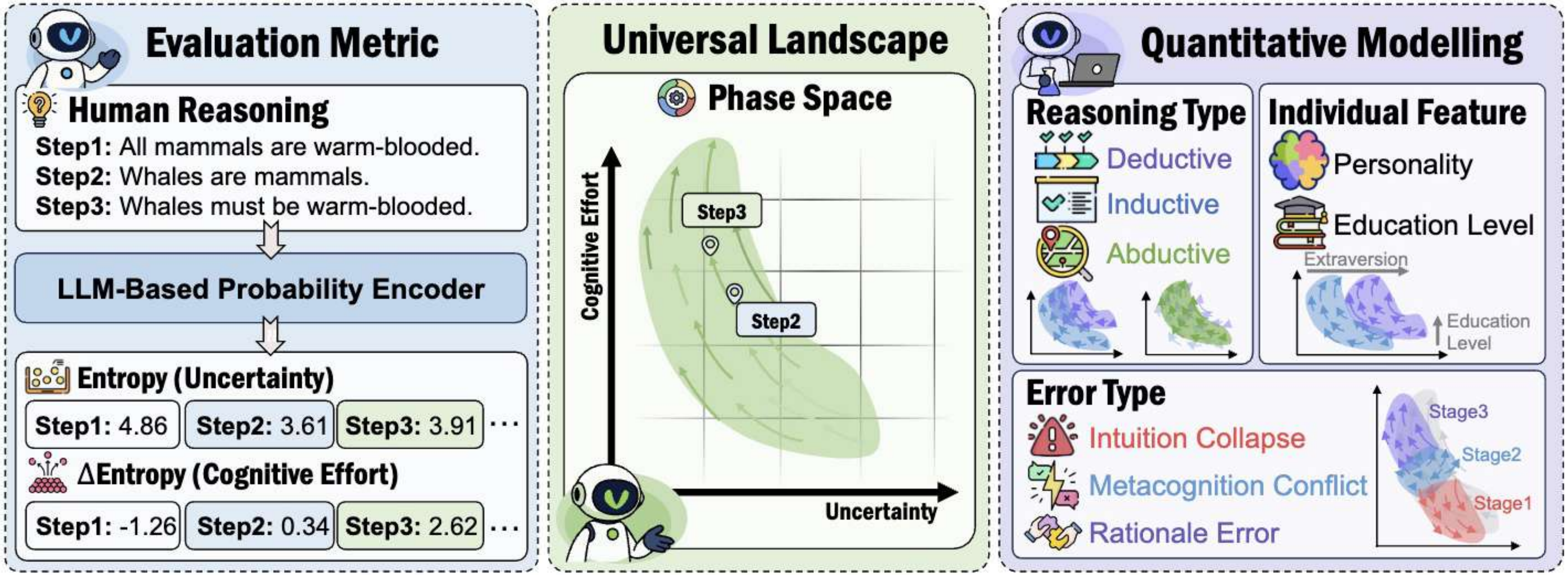
* Chen et al. AI4Research: A Survey of Artificial Intelligence for Scientific Research. Arxiv 2025.

长思维链的科学应用

人类推理信息相空间建模

人类通用推理建模

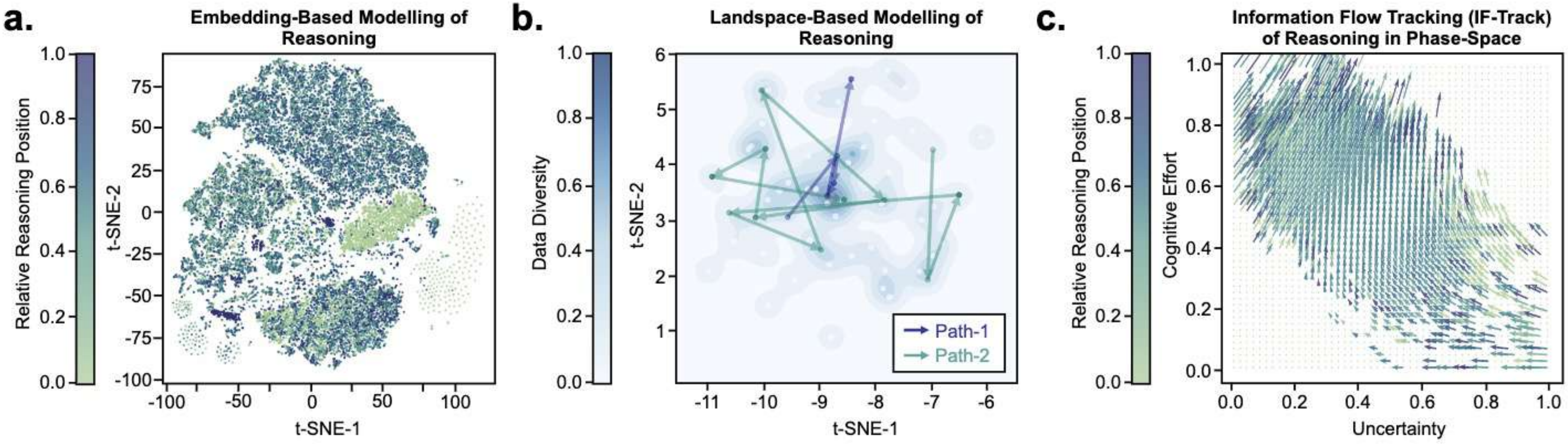
将**人类推理**视为**在信息相空间中连续演化的轨迹**。LLM-based 概率编码器计算每一步的**信息熵**（即该步骤的**不确定性**）和**信息增益**（即该步骤的**认知负担**）从而在二维相空间中重建推理的动态流向。



* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

- 现有Embedding/Landspace-of-Thought建模混乱
- 我们的方法能够**建立统一的信息流** （方差降低了4个数量级）

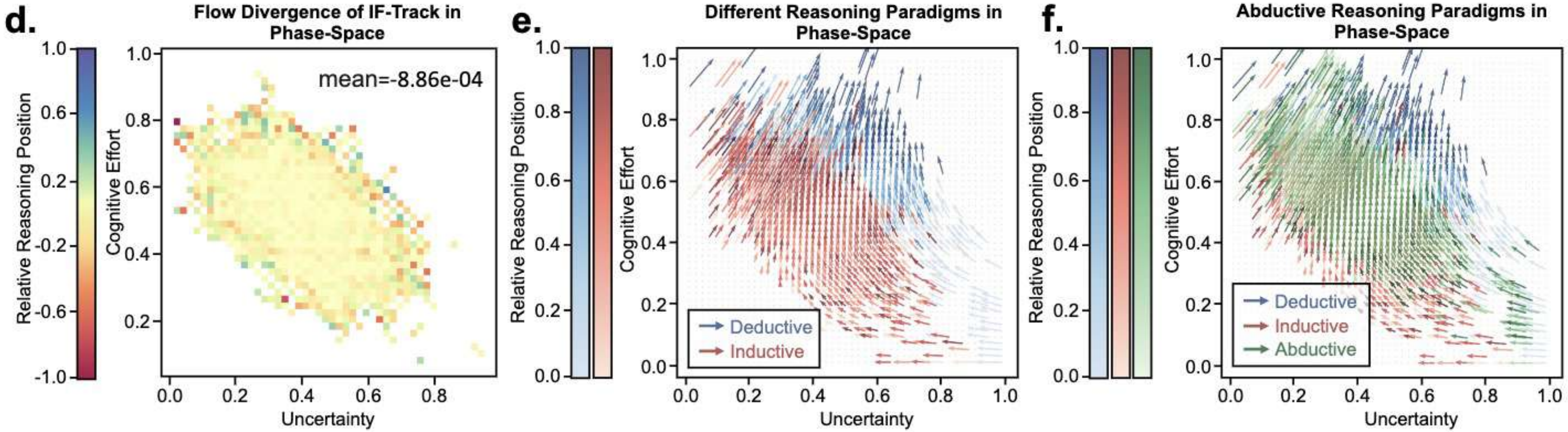


* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

🎯 IF-Track良好的建模了推理的固有属性

- ◆ 良好地建模不同**推理类型**（演绎，归纳，溯因）
- ◆ 演绎和归纳**整体流向一致**，**初期步骤不同。**
- ◆ 归纳早期的步骤在**猜测规律**，具有**较低认知负担和不确定性**
- ◆ 溯因推理被认为是演绎和归纳的**反复应用**，其流向在二者之间

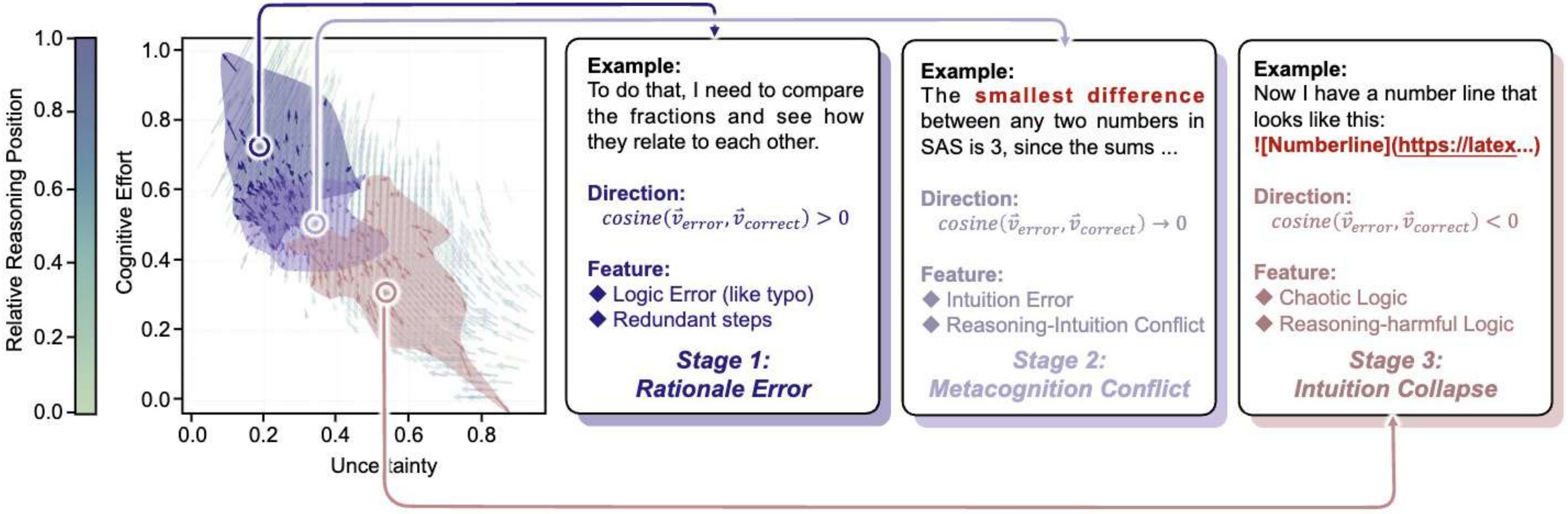


* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

🎯 IF-Track良好的建模了推理的固有属性

- 良好地建模了多种**推理错误类型**
- 根据和通用流向的差异，prm800k数据集中错误步骤被自然划分为三类
- 这三类和Gordon Pennycook的三阶段错误理论相对齐

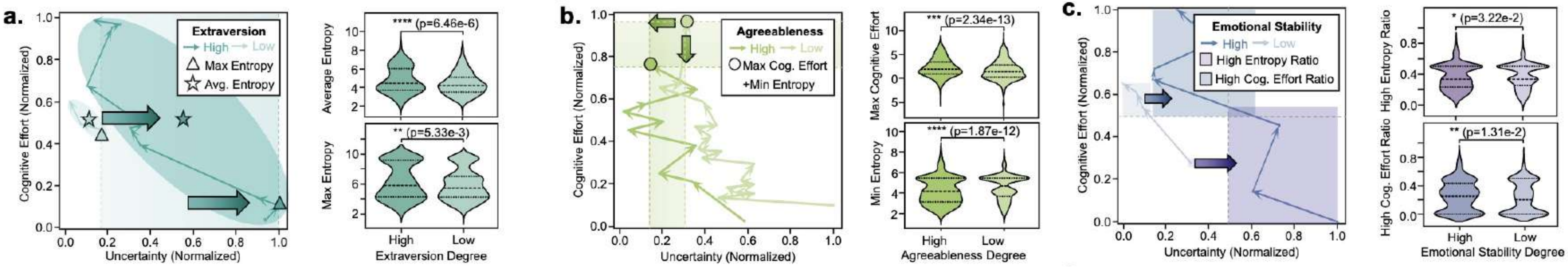


* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

🎯 IF-Track良好的建模了不同的推理个体属性

- ◆ **外向性：**外向程度更高的个体具有更高的平均与最大熵值，表明对不确定性更具容忍度
- ◆ **尽责性：**高尽责组表现出较低的平均熵和最大认知负担，与其明确目标导向的推理风格相一致
- ◆ **情绪稳定性：**高情绪稳定个体保持更均衡的状态分布，反映出更稳定的推理节奏与状态调控能力。

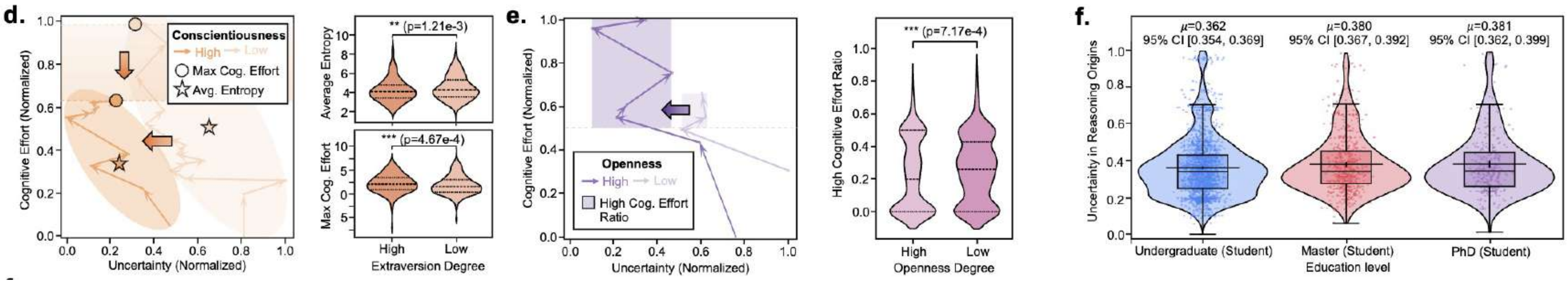


* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

IF-Track良好的建模了不同的推理个体属性

- **开放性：**高开放组呈现出更高比例的高负担推理阶段，更深入的认知投入
- **宜人性：**高宜人组拥有更高的最大认知负担与更低的最小熵值，表现出更专注、持续的推理特征
- **教育水平**：教育水平越高群体在推理起点呈现更高的不确定性，表明在早期推理中具备更广泛的假设

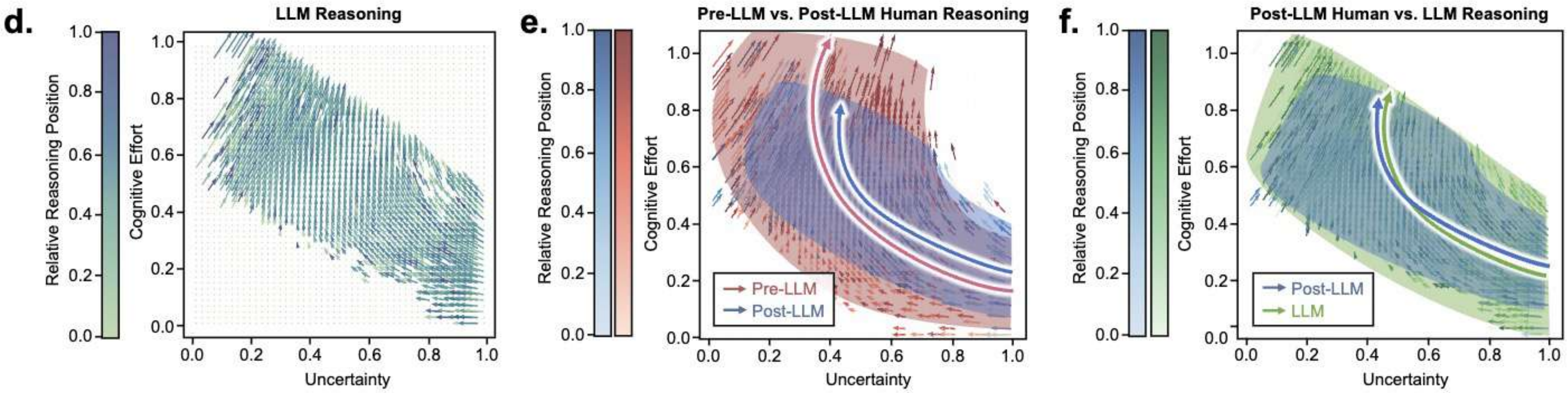


* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

人类通用推理建模

IF-Track揭示了AI时代对人类思维的重塑

- **前-LLM 时代的人类推理** 在 LLM 出现之前，人类推理通常从低认知负担、高不确定性开始，最终终点的认知负担也更高
- **后-LLM 时代的人类推理** 伴随对 GPT-4o 等模型的持续使用，现在往往从较高的认知负担起步
- **人机推理趋同** 如图所示，后-LLM 人类推理流与 GPT-4o 的推理轨迹高度重叠。



* Chen et al. The Universal Landscape of Human Reasoning. Arxiv 2025.

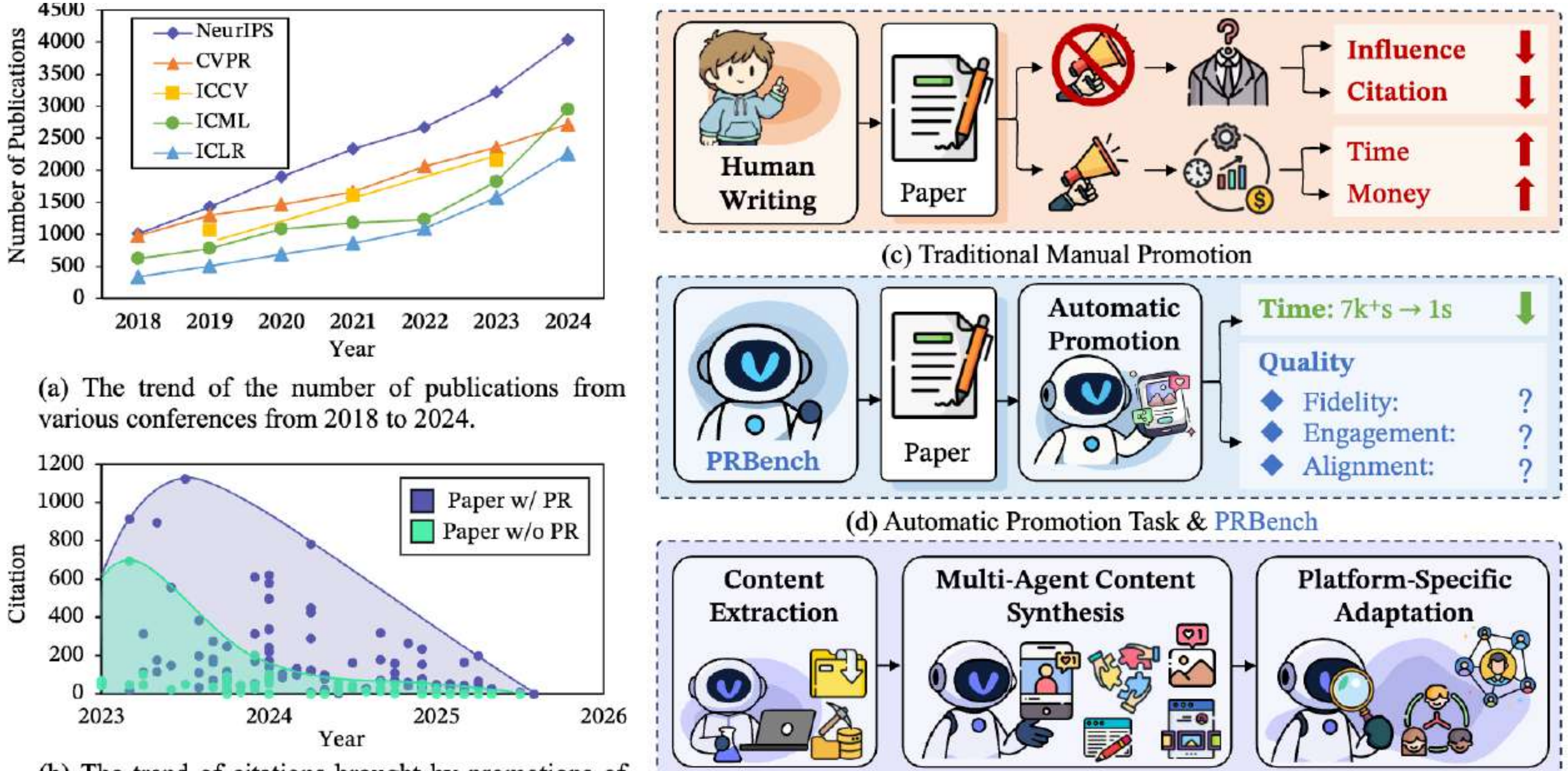
长思维链的科学应用

自动学术推广大模型

动机

🎯 学术论文发表量激增，科研人员面临信息过载的挑战

- 作者为了维持作品的可见度和引用量，需要投入**大量精力进行推广**



(a) The trend of the number of publications from various conferences from 2018 to 2024.

(b) The trend of citations brought by promotions of 20 researchers with 200 papers (2023-2025).

(c) Traditional Manual Promotion

(d) Automatic Promotion Task & PRBench

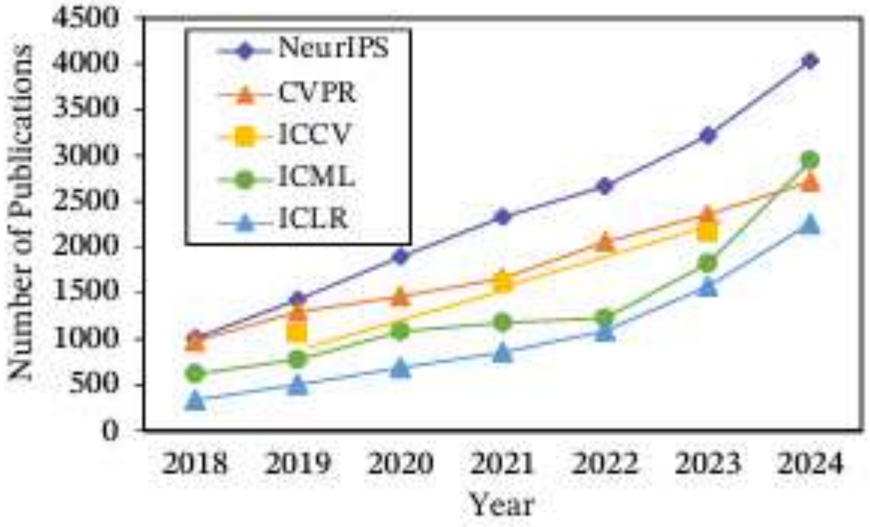
(e) PRAgent Framework

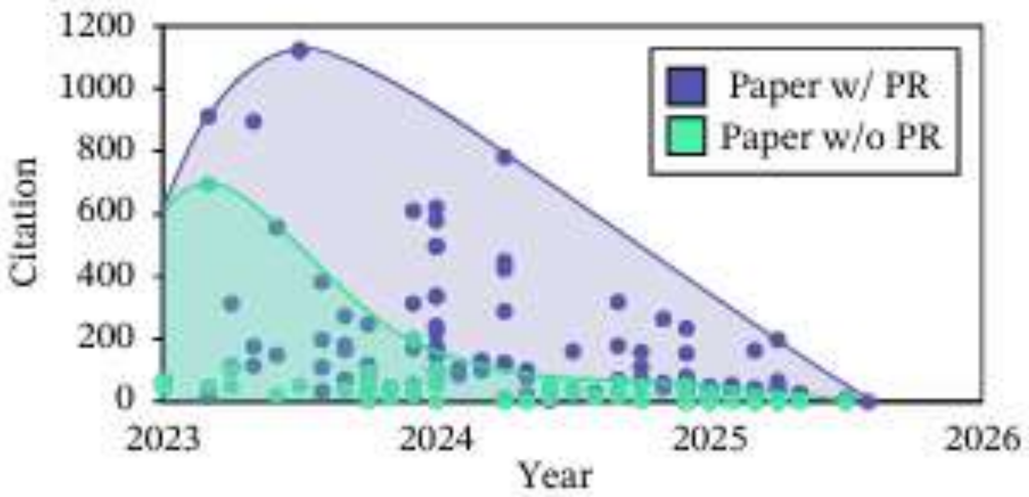
* Weissburg et al. Position: AI/ML influencers have a place in the academic process. ICML 2024.

动机

🎯 学术论文发表量激增，科研人员面临信息过载的挑战

- 有效的学术推广**能显著提升论文的影响力和引用次数**
- 社交媒体已成为学者获取前沿信息的重要渠道



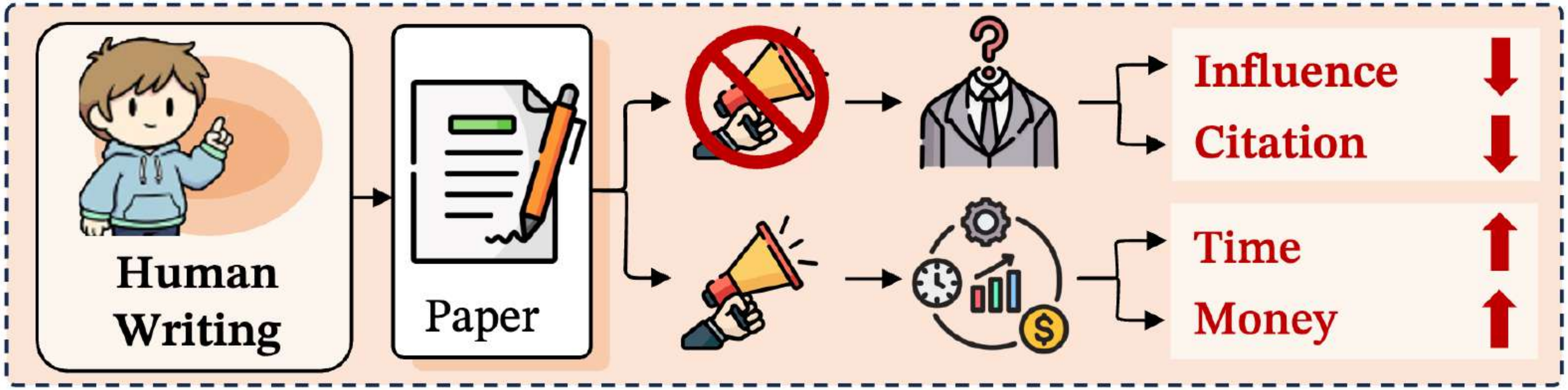


* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

动机

人工推广的挑战

- 撰写高质量的推广材料需要投入大量时间和**精力**。
- 设计图文内容需要专业的技能和不菲的**成本**。
- 推广效果**依赖个人经验**，缺乏系统性方法。



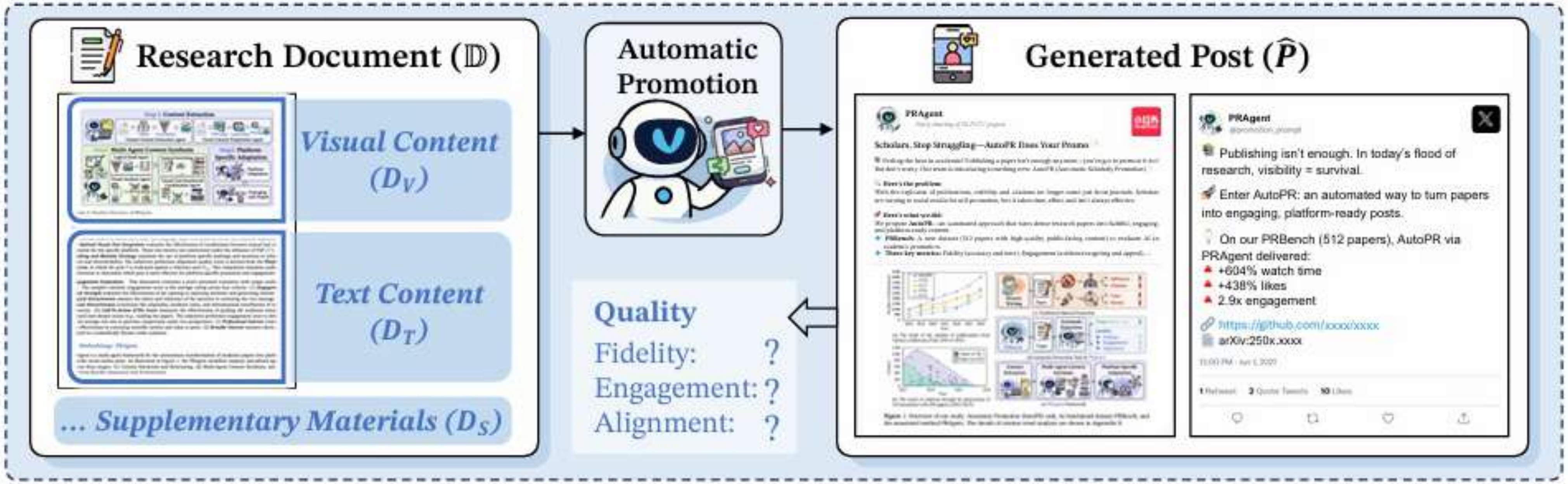
(c) Traditional Manual Promotion

* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

AutoPR任务定义与评测基准

🎯 AutoPR 任务详解

- **输入：** 研究文档（D），包括文本（D_T）、视觉内容（D_V）和补充材料（D_S）。
- **输出：** 针对特定平台（T_P）和受众（T_A）生成的推广帖子（ $\hat{P}$ ）
- **核心：** 一个多目标优化问题，旨在最大化忠实性，吸引力，平台适配度



* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

AutoPR任务定义与评测基准

AutoPR 评测的三大维度

- **忠实度 (Fidelity)**：内容准确性、信息完整性。
- **吸引力 (Engagement)**：能否有效吸引目标受众（同行、业余爱好者等）。
- **适配度 (Alignment)**：内容风格是否与目标平台的特点和用户行为习惯相符。

$$\hat{P} = \operatorname*{argmax}_{P} \mathbf{Pr}(P \mid \mathbb{D}, \mathbb{T}_P, \mathbb{T}_A)$$

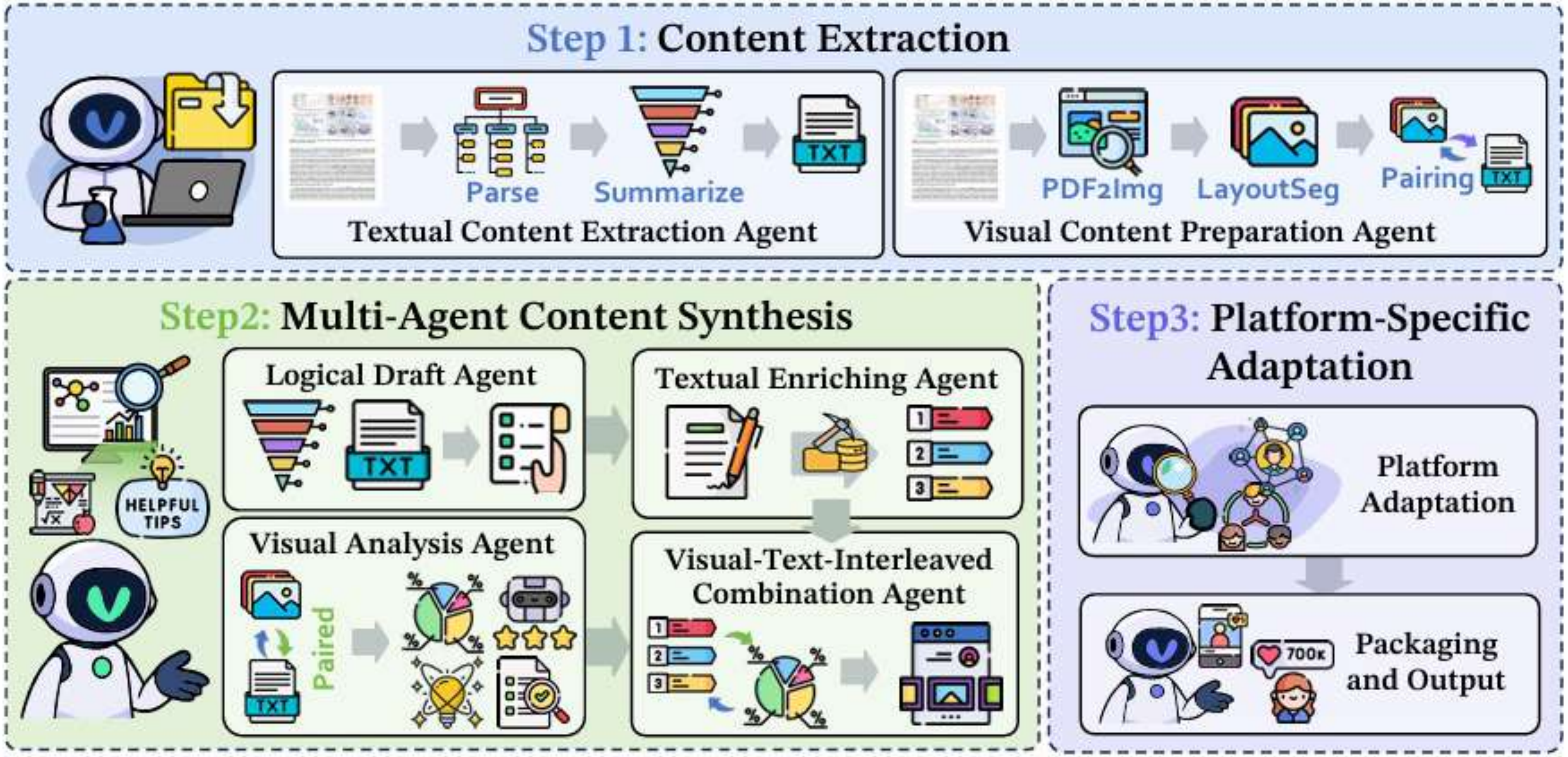
$$\max_{\hat{P}} \vec{F}(P) = \max_{\hat{P}} \left\{ \alpha_1 S_{\text{Fidelity}}(\hat{P} \mid \mathbb{D}) + \alpha_2 S_{\text{Align}}(\hat{P} \mid \mathbb{T}_P) + \alpha_3 S_{\text{Engage}}(\hat{P} \mid \mathbb{T}_A) \right\}$$

* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

PRAgent 框架

PRAgent 框架总览

模块化的多智能体框架。包括内容提取与结构化，多智能体内容合成，平台适配与发布三部分



* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

LLM是否能替代人类进行社交媒体推广贴子的评估?

- LLM评测员与人类的判断在大多数指标上呈现出强正相关（>0.5）。
- 开源模型（如Qwen）与人类判断的对齐度甚至优于部分闭源模型。
- LLM擅长评估客观、基于文本的指标，但在主观、多模态（如视觉美学）方面仍有挑战。

Metric	Qwen-2.5-VL-72B-Inst.		GPT-4o		Qwen-2.5-VL-32B-Inst.		GPT-5-mini	
	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman
Fidelity								
Authorship & Title Accuracy	**0.7511**	**0.6573**	0.5910	0.5176	0.3223	0.3543	0.5215	0.4202
Factual Checklist Score	**0.9811**	**0.9777**	0.9013	0.8470	0.9452	0.8968	0.9433	0.9208
Engagement								
Logical Attractiveness	**0.7414**	**0.7451**	0.5559	0.5579	0.5877	0.5581	0.5795	0.5509
Visual Attractiveness	0.4859	**0.5024**	0.0838*	0.0561*	**0.5156**	0.4827	0.4398	0.3255
Engagement Hook Strength	**0.7280**	**0.7204**	0.5784	0.5759	0.6099	0.6108	0.5817	0.5746
Call-To-Action Score	**0.8073**	**0.7762**	0.5393	0.5328	0.3095	0.3309	0.5994	0.5665
Alignment								
Contextual Relevance	**0.6799**	**0.6840**	0.5585	0.5526	0.5117	0.4729	0.4329	0.4531
Visual–Text Integration	**0.6266**	**0.6028**	0.3594	0.3065	0.5055	0.4728	0.3745	0.2855
Hashtag & Mention	0.7552	0.7849	0.4859	0.3894	0.5550	0.5741	**0.8473**	**0.8258**

* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

现有LLM在AutoPR任务上的局限

总体表现: 即使是SOTA模型，直接生成内容的得分也普遍不高。

Model Name	Fidelity		Engagement						Alignment				Avg.
	A&T Acc.	Factual Score	Hook	Logical Attr.	Visual Attr.	CTA	Prof. Pref.	Broad Pref.	Context Rel.	Vis-Txt Integ.	Hashtag	Plat. Pref.	
DeepSeek-R1-Distill-7B^{R,T}	43.25	21.45	33.07	45.04	-	15.34	37.70	43.25	31.28	-	17.13	23.02	31.05
Qwen-2.5-VL-7B-Ins	49.15	39.17	62.83	46.60	-	39.19	34.77	58.59	55.86	-	40.46	60.16	48.68
DeepSeek-R1-Distill-14B^{R,T}	51.37	43.57	69.14	54.92	-	29.56	60.16	75.78	64.23	-	50.13	81.64	58.05
DeepSeek-R1-Distill-32B^{R,T}	50.00	42.49	68.03	55.66	-	35.61	51.95	77.73	67.25	-	50.46	85.16	58.43
Qwen3-30B-A3B^{T}	51.11	43.03	71.68	51.69	-	35.22	47.66	74.61	67.84	-	60.16	83.59	58.66
InternVL3-38B	51.37	43.82	71.16	53.91	-	50.07	44.14	77.73	68.46	-	50.81	85.94	59.74
GPT-OSS-20B^{R,T}	52.30	56.11	69.34	40.62	-	44.21	73.44	74.22	71.52	-	54.88	90.62	62.73
InternVL3-8B	52.67	48.55	72.01	53.09	-	50.00	63.67	81.64	66.34	-	56.58	85.16	62.97
Qwen3-8B^{T}	51.76	45.09	73.83	51.69	-	44.27	62.50	78.91	72.10	-	61.46	91.41	63.30
InternVL3-14B	52.41	49.12	71.29	54.52	-	55.66	56.64	80.86	68.52	-	57.06	88.67	63.48
GPT-OSS-120B^{R,T}	52.67	59.85	68.55	41.02	-	43.29	76.17	78.91	73.86	-	67.45	92.19	65.40
Qwen-2.5-VL-72B-Ins	52.08	44.43	**74.41**	62.83	-	57.81	58.20	83.98	**74.67**	-	55.53	93.75	65.77
Qwen3-32B^{T}	52.73	52.56	72.98	54.04	-	47.27	79.30	80.08	70.41	-	61.98	92.97	66.43
Qwen3-235B-A22B^{T}	55.34	54.28	74.22	57.29	-	51.82	80.47	84.38	74.41	-	**69.99**	**96.09**	69.83
Qwen-2.5-VL-32B-Ins	**57.55**	**59.87**	70.90	**70.15**	-	**58.92**	**88.67**	**87.50**	67.68	-	53.32	91.02	**70.56**
GPT-4o	50.52	30.73	72.93	48.06	-	42.84	28.12	64.45	60.58	-	53.26	55.08	50.66
GPT-4.1	51.00	38.75	74.00	56.00	-	45.67	50.00	70.00	69.00	-	52.33	84.00	59.08
GPT-5-nano^{R}	49.80	57.91	51.56	37.34	-	34.31	58.59	51.95	52.51	-	49.28	73.05	51.63
GPT-5-mini^{R}	51.37	**61.80**	55.27	38.74	-	31.90	65.23	61.72	57.71	-	40.30	79.69	54.37
GPT-5^{R}	52.73	50.19	74.15	45.15	-	37.70	**74.61**	83.20	75.03	-	52.02	**94.92**	63.97
Gemini-2.5-Flash	**55.01**	45.10	**74.48**	**61.78**	-	**48.96**	39.06	**83.98**	**80.47**	-	**61.20**	93.75	**64.38**

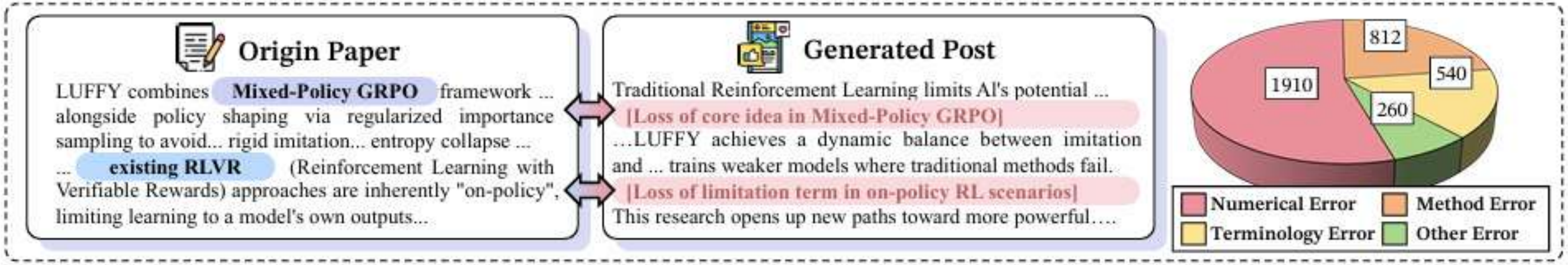
* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

局限一：忠实度瓶颈

问题：模型难以保持事实的精确性，常丢失核心技术细节。

答案：对Qwen2.5-VL-32B进行错误分析，超过92%的错误集中在“数值/方法/术语错误或遗漏”。



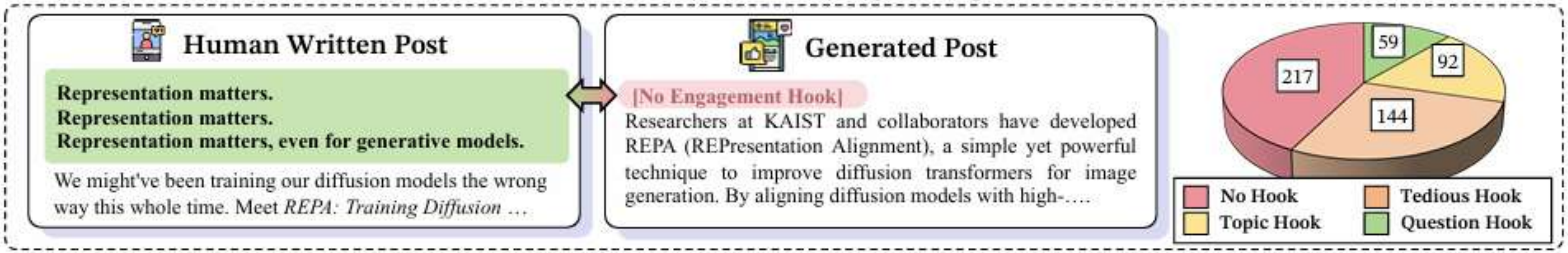
* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

局限二：缺乏真正的吸引力

? **问题：**生成内容风格刻板，缺乏能引发共鸣的叙事和情感钩子。

答案：42%的帖子完全没有使用任何吸引用户的开篇hook



* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

局限三：肤浅的平台适配

问题： 模型仅模仿表面风格，无法理解平台特性和用户生态。

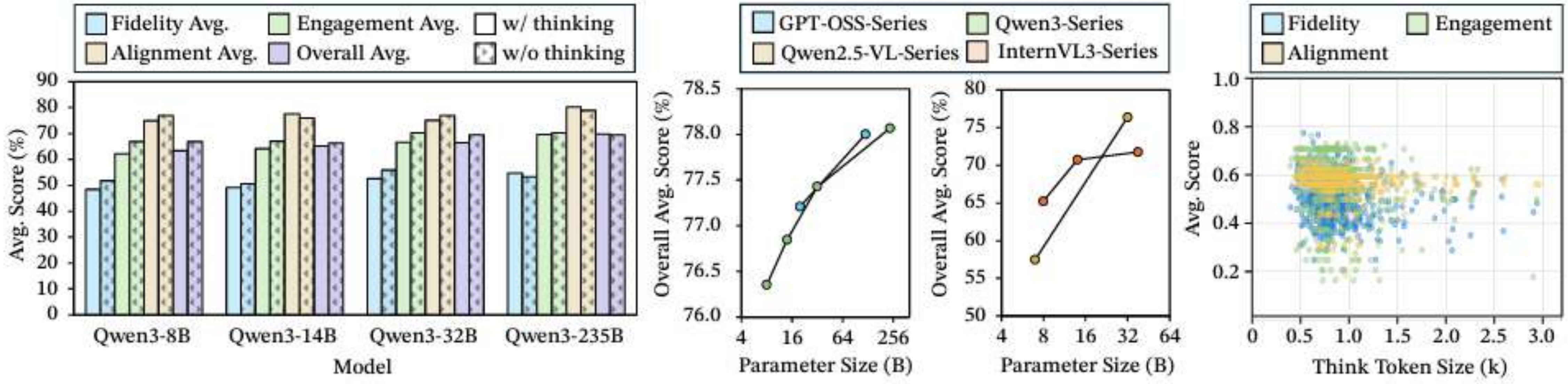
答案： 生成的标签过于通用，与人类作者的标签Jaccard相似度仅为0.0347。

* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

通用提升策略在AutoPR任务中的表现

- **长思维链 (Long CoT):** 效果不显著，甚至对大模型有负面影响。
- **参数扩展定律:** 依然有效，绝大多数情况下模型参数量越大，性能越好。
- **推理时间扩展:** 无效，过多的无引导推理反而导致性能下降。



* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

🎯 PRAgent 性能显著提升

- **PRAgent 框架在所有模型和指标上均显著优于直接生成的基线。**
- 平均提升超过7.15%，在GPT-5-mini上提升超过20%。

Model Name	Fidelity		Engagement						Alignment				Avg.
	A&T Acc.	Factual Score	Hook	Logical Attr.	Visual Attr.	CTA	Prof. Pref.	Broad Pref.	Context Rel.	Vis-Txt Integ.	Hashtag	Plat. Pref.	
Qwen2.5-VL-7B-Ins	49.15	39.17	62.83	46.60	-	39.19	34.77	58.59	55.86	-	40.46	60.16	48.68
+ PRAgent	62.17	57.89	62.57	58.33	59.32	15.62	66.41	74.61	57.40	60.61	50.26	70.31	57.96
InternVL3-14B	52.41	49.12	71.29	54.52	-	55.66	56.64	80.86	68.52	-	57.06	88.67	63.48
+ PRAgent	64.78	55.91	75.26	67.06	73.05	52.80	73.05	92.19	80.79	71.55	53.22	87.89	70.63
GPT-OSS-20B^{R,T}	52.30	56.11	69.34	40.62	-	44.21	73.44	74.22	71.52	-	54.88	90.62	62.73
+ PRAgent	70.12	75.28	75.00	64.84	72.46	47.33	99.22	98.05	83.59	73.63	62.76	99.22	76.79
Qwen2.5-VL-32B-Ins	57.55	59.87	70.90	70.15	-	58.92	88.67	87.50	67.68	-	53.32	91.02	70.56
+ PRAgent	72.85	72.49	74.80	82.03	75.33	51.69	98.05	100.00	83.82	75.03	61.65	96.48	78.69
Qwen3-32B^{T}	52.73	52.56	72.98	54.04	-	47.27	79.30	80.08	70.41	-	61.98	92.97	66.43
+ PRAgent	70.31	64.94	75.00	83.72	74.61	42.32	99.22	100.00	86.91	75.39	60.71	99.22	77.70
GPT-OSS-120B^{R,T}	52.67	59.85	68.55	41.02	-	43.29	76.17	78.91	73.86	-	67.45	92.19	65.40
+ PRAgent	69.34	79.42	75.00	66.37	72.79	46.94	100.00	98.05	81.74	74.12	60.61	100.00	77.03
Qwen3-235B-A22B^{T}	55.34	54.28	74.22	57.29	-	51.82	80.47	84.38	74.41	-	69.99	96.09	69.83
+ PRAgent	66.80	66.92	75.33	83.69	74.87	42.58	97.66	100.00	87.17	75.10	61.13	97.66	77.41
GPT-5^{R}	52.73	50.19	74.15	45.15	-	37.70	74.61	83.20	75.03	-	52.02	94.92	63.97
+ PRAgent	68.16	73.30	75.00	80.40	75.20	34.70	99.22	100.00	86.65	75.33	53.06	98.44	76.62
GPT-5-mini^{R}	51.37	61.80	55.27	38.74	-	31.90	65.23	61.72	57.71	-	40.30	79.69	54.37
+ PRAgent	72.33	83.61	74.61	68.07	74.61	43.49	99.22	99.61	81.97	73.83	52.60	96.48	76.70

* Chen et al. AutoPR: Let's Automate Your Academic Promotion! Arxiv 2025.

实验与结果分析

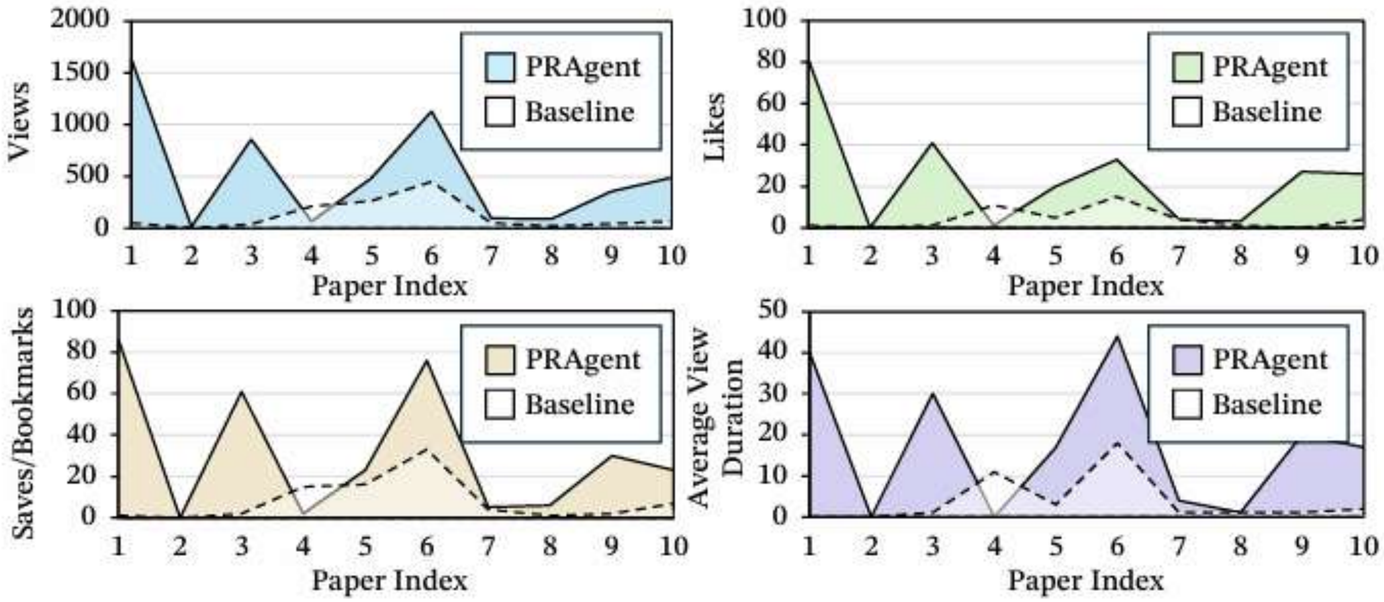
🎯 真实世界研究：PRAgent 在社交媒体上的表现

🧐 在RedNote（小红书）上进行了为期10天的真实世界对比实验。

🥥 **实验组:** PRAgent + GPT-5 生成内容。

🥥 **对照组:** Direct Prompting + GPT-5 生成内容。

🥥 **结果:** PRAgent 在所有互动指标上全面碾压基线。



	Baseline	PRAgent
Views	1178	5059 (+329%)
Total Watch Time	12555	88375 (+604%)
Profile Visitors	28	189 (+575%)
Likes	42	226 (+438%)
Saves/Bookmarks	78	308 (+294%)
Followers Gained	4	27 (+575%)
Shares	39	190 (+387%)

总结

- **推理已经成为新一代技术范式**
- **推理能力存在边界及组合现象**
  - 模型推理存在宏观逻辑电路
  - 推理过度自信现象分析
- **多模态推理**
  - 更符合人类推理方式
  - 基准 → 技术 → 机理
- **推理加速**
  - 推理思维链动态压缩
  - 扩散语言模型的并行推理
- **推理应用**
  - 长思维链科学应用
  - 通用人类推理路径建模
  - 自动科学宣传

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谢谢！

感谢聆听！

Q&A